Introduction

Electrical machines are used to convert energy:

- From electrical to electrical (transformers). Transformers are used to convert electrica energy at one voltage level to electrical enegry at another voltage level. Note that the frequency is the same at the two sides (primay & secondary) of the transformer.
If the transformer is used to produce a higher voltage level, it is called a step-up transformer. If it is used to produce a lower voltage level it is a step-down transformer.

- From electrical to mechanical, these are the motors. Motors are supplied by electrical energy and produce mechanical energy at their shafts.

- From mechanical to electrical, these are the generatos. Generators are supplied by mechanical enegry and produce electrical enrgy.

Note that the principle of operation of all electical machines is based on the action of the magnetic field.

Note that in all the types of enegry conversions the principle of energy conservation must be fullfilled. Or in other words allways the input enegry is equal to the output enegy plus the losses. The losses in the electric machines are dissipated in a form of heat.

Transformers are stationary machines (with no moving parts), while motors and generators have rotating shafts. Their operation is described by:

- angular displacement, θ, in radians: is the angle at which the shaft is rotated. It is measured from an arbitrary reference point. Usualy, the angle θ is positive if the rotation is counter clockwise (CCW) and is negative for clockwise (CW) rotation.

- angular velocity, ω, in rad/s: is the rate of change of the angular position with respect to time, or $\omega = \frac{d\theta}{dt}$.
Other units can also be used to describe the angular velocity (annular speed), these are:

- revolution per second (frequency), usually denoted by f, note that:

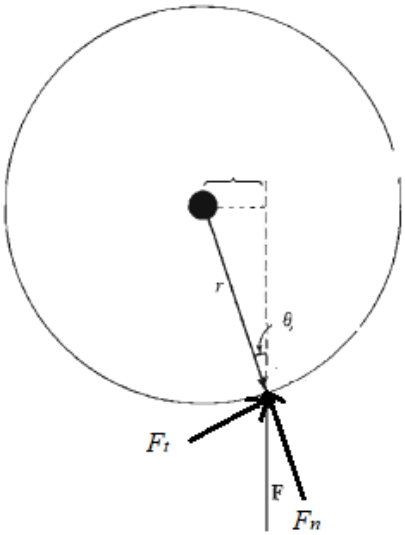
$f = \frac{\omega}{2\pi}$, or $\omega = 2\pi f$

- revolution per minute (RPM), usually denoted by n, note that:

$n = 60f$, and $n = 60f = 60\left(\frac{\omega}{2\pi}\right) = \frac{30\omega}{\pi}$

- angular acceleration, α, in rad/s^2: rad/s: is the rate of change of the angular speed with respect to time, or $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$.

- Torque, T , in $N.m$: it is the cause of changing of the angular speed, or it is the cause of the angular acceleration. It is simial to the force in linear motion.
The torque on an object is defined as the product of the force applied to the object and the distance between the line of action of the force and the object's axis of rotation.
The figure below shows a cross section of a cylinder that is free to rotate on its axis, with a force acting on it as shown.



Note that the acting force can be resolved into two components: the normal component (F_n), $F_n = F.\cos\theta$, which causes a pressure and the tangantial compenent (F_t), $F_n = F.\sin\theta$, which causes a rotation.
The tourq acting on the cylinder is:
$T = F_t.r = F.\sin\theta.r$
If vector notation is used,
$\vec{T} = \vec{F}X\vec{r}$
The torque is assumed in a positive direction if it causes a rotation in a counter clockwise (CCW) direction.

- Newtons law: in linear motion, Newtons law describe the relation between the acting force (F), the mass (m) and the acceleration (a), or:
$F = m.a$
For rotational motion, similar law describes the realtion between the torque (T), the momemt of inertia (J) and the angular acceleration (α), that is:
$T = J.\alpha$

- Work, in Joules: the work can be defined as the energy pefrormed by the force. For linear motion, the work is:
$W = \int \vec{F}.d\vec{r}$
Where $\vec{F}$ is the force and $\vec{r}$ is the displacement caused by the force action.
If the force is constant and colinear with the displacement, the work is:
$W = F.r$

For rotational motion, the work is:
$W = \int T.d\theta$

Where T is the torque and θ is the angular displacement caused by the torque action.
If the torque is constant, the work is:
$W = T.\theta$

- **Power**, in Watts: the power is defined as the rate of doing a work, or:

$$P = \frac{dW}{dt}$$

For linear motion:

$$P = \frac{dW}{dt} = \frac{d}{dt}\int \vec{F}.d\vec{r} = \int \vec{F}.d\left(\frac{d\vec{r}}{dt}\right) = \int \vec{F}.d\vec{v}$$

If the force is constant and colinear with the displacement, the power is:
$P = F.v$

For rotational motion:

$$P = \frac{dW}{dt} = \frac{d}{dt}\int T.d\theta = \int T.d\left(\frac{d\theta}{dt}\right) = \int T.d\omega$$

If the torque is constant, the power is:
$P = T.\omega$
This equation describes the mechanical power on the shaft of a motor or generator.

The followin is a comparison between the quantities and the laws describing linear and rotational motions:

	Linear motion	Rotational morion
displacement	x (in meters)	θ (in radians)
velocity (speed)	$v = \frac{dx}{dt}$ (in m/s)	$\omega = \frac{d\theta}{dt}$ (in rad/s)
acceleration	$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$ (in m/s^2)	$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ (in rad/s^2)
action	Force (in N)	Torque in ($N.m$)
Newton law of motion	$F = m.a$	$T = J.\alpha$
work	$W = \int \vec{F}.d\vec{r}$	$W = \int T.d\theta$
Power	$P = \int \vec{F}.d\vec{v}$	$P = \int T.d\omega$

The Manetic Field

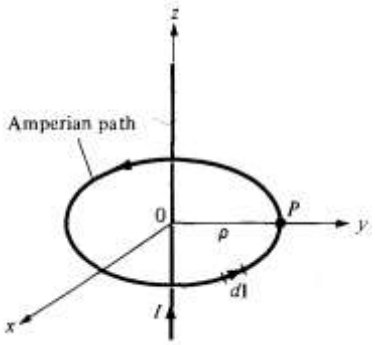
The magnetic fields is the fundamental mechanism by which energy is converted from one form to another in electrical machines. Four basic principles describe how the magnetic fields are used in electrical machines, these are:

1) A current carrying conductor produces a magnetic field around it, (Ampere's law).
2) A varying magnetic field through a coil induces a voltage through the coil (Faraday's law).
3) A force is acting on a current carrying conductor in a magnetic field (the basis of motor action).
4) a voltage is induced in a wire moving in a magnetic field.

Production of a Magnetic Field

The basic law describing the generation of a magnetic field around a current carring conductor is Ampere' law.

Consider a coductor with current I , as shown in the figure.



According to Ampere's law:

$$\oint \vec{H}.d\vec{l} = I_{enc} \qquad (1)$$

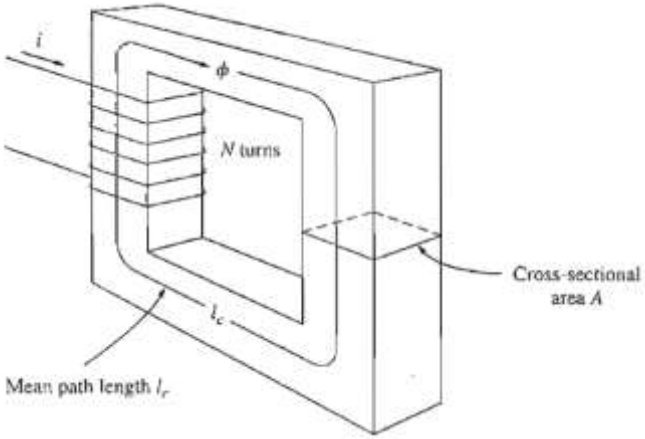
In this law:

$\vec{H}$ is the magnetic field intensity (strength) in A/m.

dl is the differential length of the closed path.

I_{enc} is the current in enclosed in the closed path.

Consider the following simple magnetic core with a winding having N number of turns carrying current I , as shown in the figure below.



According to Ampere's law:

$$\oint \vec{H}.d\vec{l} = I_{enc}$$

As H is constant along the path of integration, thus:

$$H.l_C = N.i \qquad (2)$$

Here l_C is the mean path of inegration and N is the number of turns of the winding.

Note that $N.i$ is usually called the magnetomotive force (MMF).

According to the above equation:

$$H = \frac{N.i}{l_C} \qquad (3)$$

The magnetic field intensity, H , is producing a magnetic flux, the relation between the magnetic flux density B and the magnetic field intensity H is:

$$B = \mu H \qquad (4)$$

Where μ is the permeability of the magnetic material of the core.

Note that B is measured in tesla (T) or (web/m^2) .

For the magnetic material:

$$\mu = \mu_r.\mu_0 \qquad (5)$$

Here:

$\mu_0 = 4\pi x10^{-7} H/m$ is the permeability of free space.

μ_r is the relative permeability of the magnetic material. For the ferromagnetic material usually used in electric machines μ_r is between 2000 and 6000 or more.

Substituting (3) in (4), we find:

$$B = \frac{\mu.N.i}{l_C} \qquad (6)$$

The flux through a given area is given by:

$$\phi = \int_{(A)} BdA \qquad (7)$$

If the flux density is constant through the area, thus:

$$\phi = B.A \qquad (8)$$

Substituting (6) in (8), we find:

$$\phi = \frac{\mu.N.i.A}{l_C} \qquad (9)$$

<u>Magnetic Circuits</u>

Equation (9) shows that the magnetic flux generated in magnetic core of material haing permeability μ with a cross section A and mean length of the path l_C around which a coil having N turns in which current i is passing, as shown in the figure below, is:

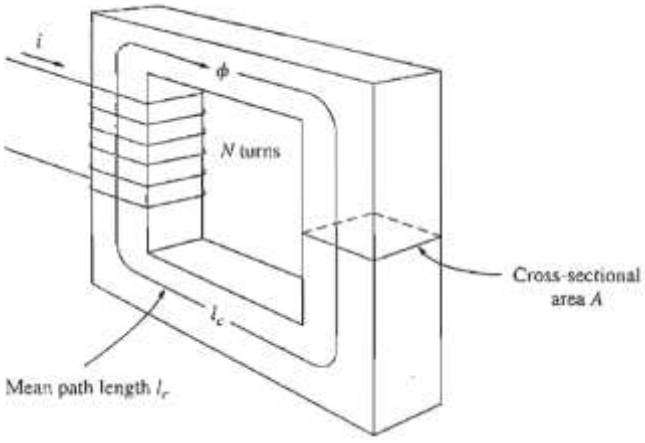
$$\phi = \frac{\mu.A}{l_C}(.N.i) \qquad (1)$$

As said before, the product of the current in the coil (i) and the number of the coil turns (N) is called the magnetomotive force (MMF), or:

$$MMF = N.i \qquad (2)$$

Thus equation (1) can be written as:

$$\phi = \frac{\mu.A}{l_C}(MMF) \qquad (3)$$



This equation shows that the magnetic flux is generated by the magneto motive force (MMF).
thus equation (3) can be written as:

$$MMF = \frac{l_C}{\mu.A}\phi \quad (4)$$

If we assume that the magnetomotive force is similar to the voltage in the electric circuit and the magnetic flux ϕ is similar to the electric current, thus above equation can be wriiten in a form similar to that of Ohm's law ($V = RI$)
Or equation (4) can be written as:

$$MMF = R_m.\phi \quad (5)$$

Where:

$$R_m = \frac{l_C}{\mu.A} \quad (6)$$

and is called the magnetic relactance.

Notes:
1- The polarity of the MMF is determined by the right-hand rule: If the fingers of the right hand curl in the direction of the current flow in the coil, the thumb will point in the direction of the positive MMF.

2- The reciprocal of the magnetic reluctance is called the magnetic permeance and is denoted by G_m. Note that:

$$G_m = \frac{1}{R_m} = \frac{\mu.A}{l_C} \quad (7)$$

As in equation (5), we can write:

$$\phi = G_m.MMF \quad (8)$$

Reluctances in a magnetic circuit obey the same rules as resistances in an electric circuit. The equivalent reluctance of a number of reluctances in series is just the sum of the individual reluctances:

$$R_{m,eq} = R_{m1} + R_{m2} + R_{m2} + ... + R_{mn} = \sum_{i=1}^{n} R_{mi} \quad (9)$$

Similarly, reluctances in parallel combine according to the equation:

$$\frac{1}{R_{m,eq}} = \frac{1}{R_{m1}} + \frac{1}{R_{m2}} + \frac{1}{R_{m3}} + ... + \frac{1}{R_{mn}} = \sum_{i=1}^{n} \frac{1}{R_{mi}} \quad (10)$$

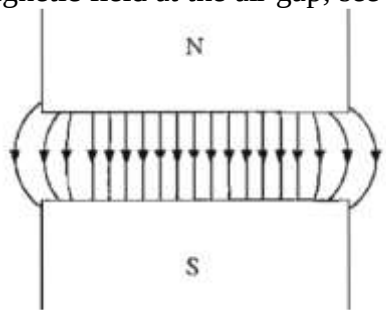
It has to be noted that using the conceopt of the magnetic circuit to calculate the magnetic flux in a core is always approximate. (usyally are accurate to within about 5% of the real answer). This because of the following reasons:

1) The magnetic flux is assumed to pass only through the magnetic core, which is not quite true. A small fraction of the flux escapes from the core into the surrounding low-permeability air. This flux outside the core is called the leakage flux.

2) The actual cross section and the mean path of the magnetic material in which the flux passes can't be measured accuretly, specially at the corners.

3) The permeability of the ferromagnetic material is not constant but varies with the amount of flux passing in the material.

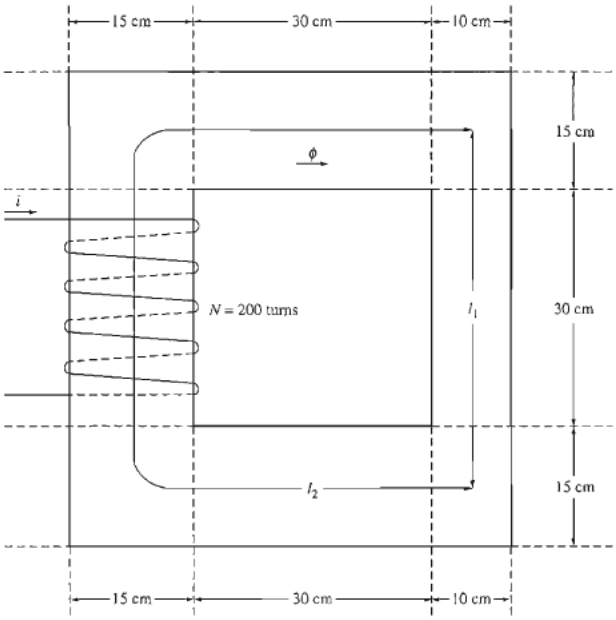
4) If air gaps are existing in the flux path, the effective cross-sectional area of the air gap will be larger than the cross-sectional area of the iron core. This is caused by the "fringing effect" of the magnetic field at the air gap, see the figure below.



The errors caused by the above reasons are compensated by using the effective mean path length and the effective cross-sectional area instead of the actual length and area.

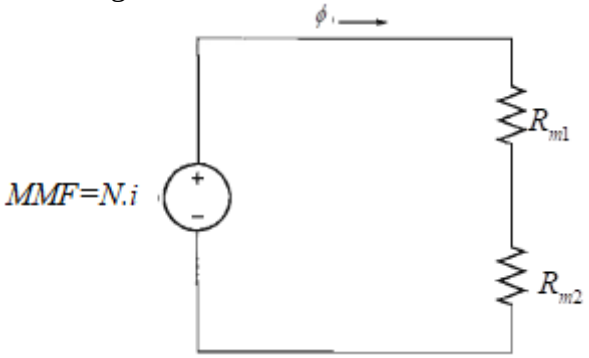
Example 1:

A ferromagnetic core is shown in the figure below. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200 turn coil wrapped around the left side of the core. Assuming relative permeability μ_r of 2500, how much flux will be produced by a 1-A input current?



Solution:

Three sides of the core have the same cross-sectional areas, while the fourth side has a different area. Thus, the core can be divided into two regions: (1) the single thinner side and (2) the other three sides taken together. The magnetic circuit corresponding to this core is shown in the figure below.



$$R_{m1} = \frac{l_1}{\mu.A_1} = \frac{l_1}{\mu_r.\mu_0..A_1} = \frac{0.45}{2500x4\pi x10^{-7}\,x0.01} = 14300 A.turns/Wb$$

$$R_{m2} = \frac{l_2}{\mu.A_2} = \frac{l_2}{\mu_r.\mu_0..A_2} = \frac{1.3}{2500x4\pi x10^{-7}\,x0.015} = 27600 A.turns/Wb$$

$$R_{m,eq} = R_{m1} + R_{m2} = 14300 + 27600 = 41900 A.turns/Wb$$

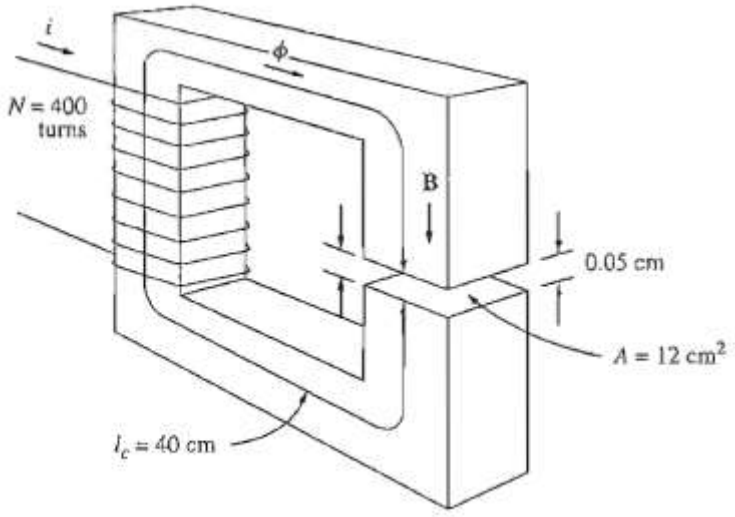
$$MMF = N.i = 200 A.turns$$

$$\phi = \frac{MMF}{R_{m,eq}} = \frac{200}{41900} = 0.0048 Wb$$

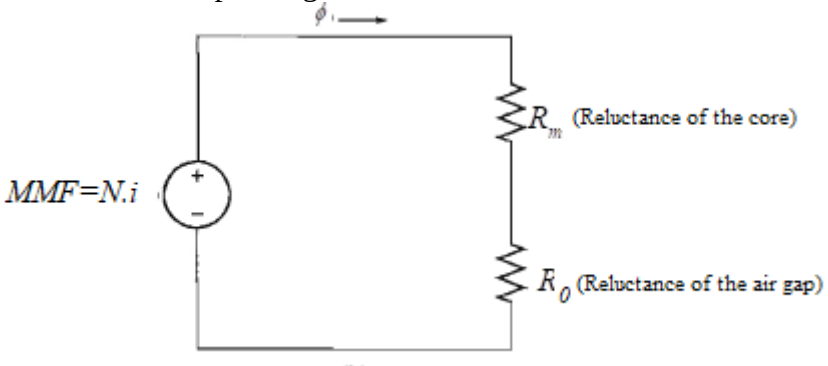
Example 2:

The figure shows a ferromagnetic core whose mean path length is 40 cm. A small air gap of 0.05 cm is exising in the strcture. The cross-sectional area of the core is $12cm^2$, the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Find:

a) the total reluctance of the flux path (iron plus air gap).
b) the current required to produce a flux density of 0.5 T in the air gap.



Solution:
The magnetic circuit corresponding to this core is shown below.



Here, the reluctance of the core is:

$$R_m = \frac{l_c}{\mu.A_c} = \frac{l_c}{\mu_r.\mu_0.A_c} = \frac{0.4}{4000x4\pi x10^{-7}\,x0.0012} = 66300 A.turns/Wb$$

The effective area of the air gap is $1.05x12 = 12.6cm^2$, so the reluctance of the air gap is:

$$R_0 = \frac{l_0}{\mu_0.A_0} = \frac{0.05x10^{-2}}{4\pi x10^{-7}\,x12.6x10^{-4}} = 316000 A.turns/Wb$$

Therefore, the total reluctance of the flux path is:

$$R_{m,eq} = R_m + R_0 = 382300 A.turns/web$$

(Note that the air gap reluctance is much greater than the reluctance of the magnetic core, even it has 800 times shorter length).
We know that:

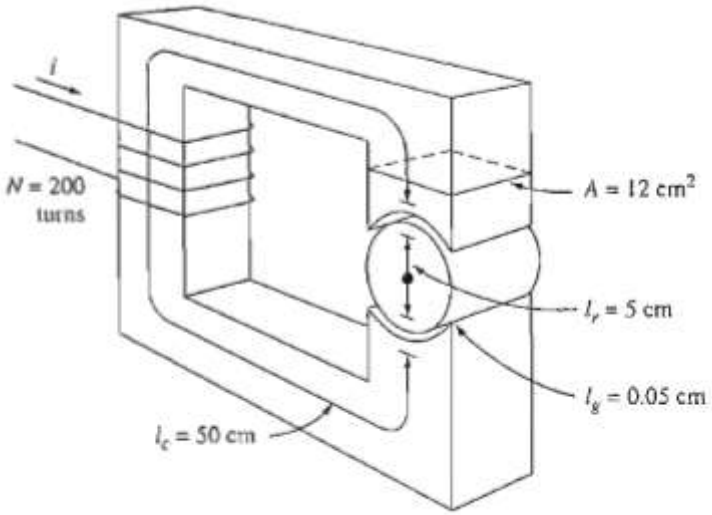
$$MMF = N.i = \phi.R_{m,eq}$$

Thus:

$$i = \frac{\phi.R_{m,eq}}{N} = \frac{B.A.R_{m,eq}}{N} = \frac{0.5x12.6x10^{-4}\,x382300}{400} = 0.602A$$

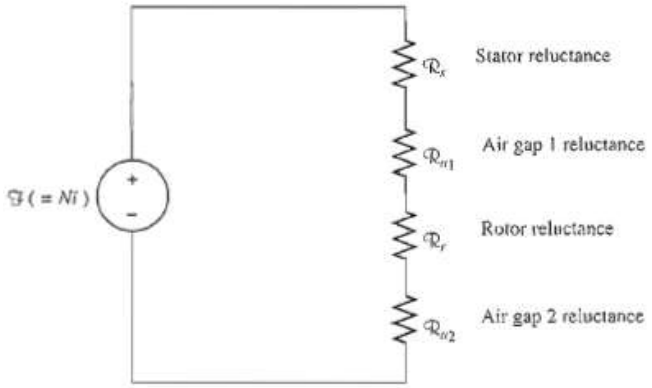
Example 3:

The figure shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross-sectional area is $12cm^2$, The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be $12cm^2$, Each air gap between the rotor and the stator is 0.05 cm wide, and the cross-sectional area of each air gap (including fringing) is $14cm^2$, The iron of the core has a relative permeability of 2000, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be?



Solution:

The magnetic circuit of the motor is as below.



The reluctance of the stator is:

$$R_s = \frac{l_s}{\mu_r.\mu_0.A_s} = \frac{0.5}{2000x4\pi x10^{-7}\,x0.0012} = 166000 A.turns/Wb$$

The reluctance of the rotr is:

$$R_r = \frac{l_r}{\mu_r.\mu_0.A_r} = \frac{0.05}{2000x4\pi x10^{-7}\,x0.0012} = 16600 A.turns/Wb$$

The reluctance of the air gaps is:

$$R_0 = \frac{l_0}{\mu_r.\mu_0.A_0} = \frac{0.0005}{1x4\pi x10^{-7}\,x0.0014} = 284000 A.turns/Wb$$

The total reluctance of the flux path is:

$$R_{eq} = R_s + R_r + 2R_0 = 751000 A.turns/Wb$$

The net magneto motive force applied to the core is:

$$MMF = N.i = 200x1 = 200A.turns$$

The total flux in the core is:

$$\phi = \frac{MMF}{R_{eq}} = \frac{200}{751000} = 0.00266Wb$$

The magnetic flux density in the motor's air gap is:

$$B = \frac{\phi}{A} = \frac{0.00266}{0.0014} = 0.19T$$

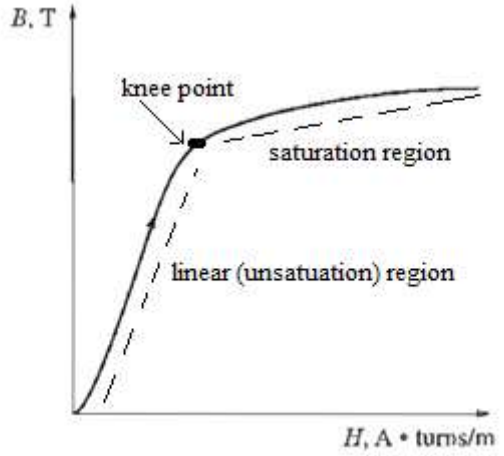
Magnetic Behavior of Ferromagnetic Materials

Previously, it has been stated that the relation between the magnetic flux density (B) and the magnetic field intensity (H) is linear, or:

$B = \mu.H$

Where μ is the permeability of the magnetic material, and is assumed to be constant.

For all ferromagnetic materials used in electric machines the realtion between the magnetic flux density (B) and the magnetic field intensity (H) is nonlinear, and is as shown in the figure below.



This type of plot is called the magnetization curve. Note that up to the knee point the relation between the magnetic flux density (B) and the magnetic field intensity (H) is linear and the magnetic permeability of the magnetic material μ is constant and doesn't depend on the field intensity (H).

After the knee point, an increase in the field intensity (H) produces relatively a small increases in the the magnetic flux density (B). And the curve nearly flattens out.

Note that a similar plot between the magnetic flux (ϕ) and the magnetomotive force ($MMF = N.i$) can be found using the relations:

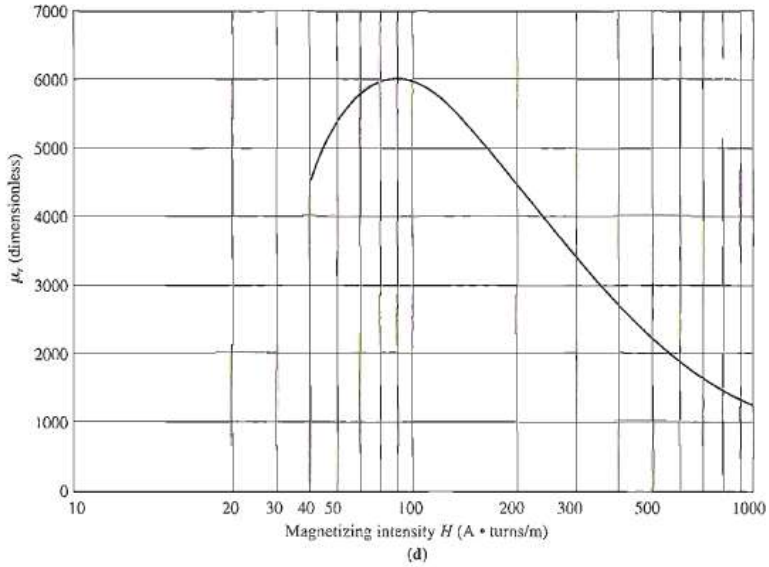
$\phi = B.A$

$MMF = H.l_c$

Where A is the cross section and l_c is the mean path of of the magnetic material.

The slope of the magnetization curve, shown in the previous figure, at any point, defines the value of the permeability of the material (μ) at this point.

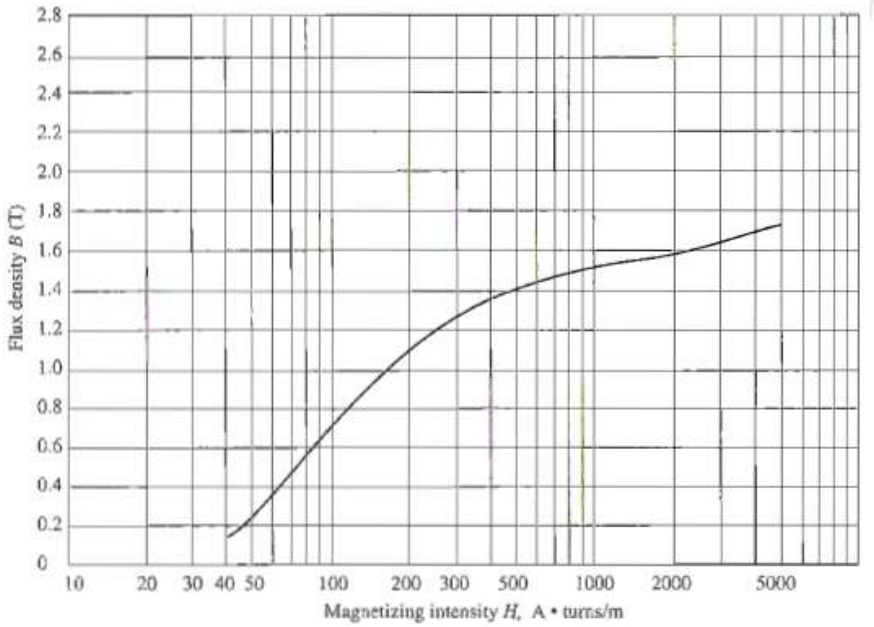
The figure below shows the permeability of the magnetic material (μ) as a function of the mahnetic field intensity (H).



It can be seen from this curve that the permeability is large and relatively constant in the unsaturated region and then gradually drops to a very low value as the core becomes heavily saturated.

It has to be noted that, for reducing the amount of the magnetic material used to build the electic machine, and thus reducing the size and the cost of the machine, most real electic machines operate near the knee of the magnetization curve, and the flux in their cores is not linearly related to the magnetomotive force producing it. This nonlinearity accounts for many of the nonliear characheristics of electric machines.

Example 1:
Find the relative permeability of the typical ferromagnetic material whose magnetization curve is shown in the figure below, at:
a) $H = 50 A.turns/m$, (b) $H = 100 A.turns/m$, (c) $H = 500 A.turns/m$, aud (d) $H = 1000 A.turns/m$



Solution:

$\mu = \frac{B}{H}$ and $\mu_r = \frac{\mu}{\mu_0}$

a) from the figure, at $H = 50 A.turns/m$, $B = 0.25T$

$$\mu = \frac{B}{H} = \frac{0.25}{50} = 0.005 H/m$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.005}{4\pi x 10^{-7}} = 3980$$

b) at $H = 100 A.turns/m$, $B = 0.72T$

$$\mu = \frac{B}{H} = \frac{0.72}{100} = 0.0072 H/m$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.0072}{4\pi x 10^{-7}} = 5730$$

c) at $H = 500 A.turns/m$, $B = 1.4T$

$$\mu = \frac{B}{H} = \frac{1.4}{500} = 0.0028 H/m$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.0028}{4\pi x 10^{-7}} = 2230$$

d) at $H = 1000 A.turns/m$, $B = 1.51T$

$$\mu = \frac{B}{H} = \frac{1.51}{1000} = 0.00151 H/m$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00151}{4\pi x 10^{-7}} = 1200$$

Notice that as the magnetizing intensity is increased, the relative permeability first increases and then starts to drop off.

Example 2:

A square magnetic core has a mean path length of $55cm$ cm and a cross sectional area of $150cm^2$. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve as shown in the previous figure.

a) How much current is required to produce $0.012Wb$ of flux in the core.

b) What is the core's relative permeability at that current level.

c) What is its reluctance.

Solution:

a)

The flux density is:

$$B = \frac{\phi}{A} = \frac{0.012}{150x10^{-4}} = 0.8T$$

From the figure, the required magnetizing intensity is $H = 115 A.turns/m$

The required magnetomotivc force is:

$$MMF = N.i = H.l_C = 115x0.55 = 63.25 A.turns$$

The current is:

$$i = \frac{MMF}{N} = \frac{63.25}{200} = 0.316A$$

b) The core permeability is:

$$\mu = \frac{B}{H} = \frac{0.8}{115} = 0.00696 H/m$$

The relative permeability is:

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696}{4\pi x 10^{-7}} = 5540$$

c) The reluctance of the core is:

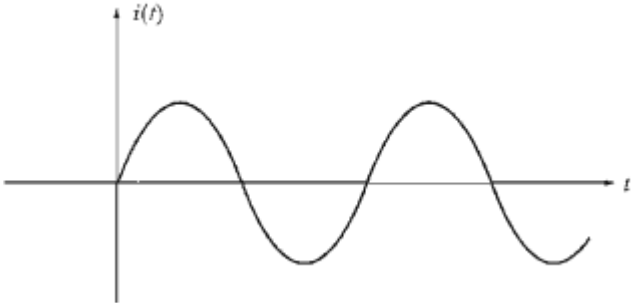
$$R_m = \frac{l}{\mu.A} = \frac{0.55}{0.00696 x 150 x 10^{-4}} = 5268 A.turns/Wb$$

Or:

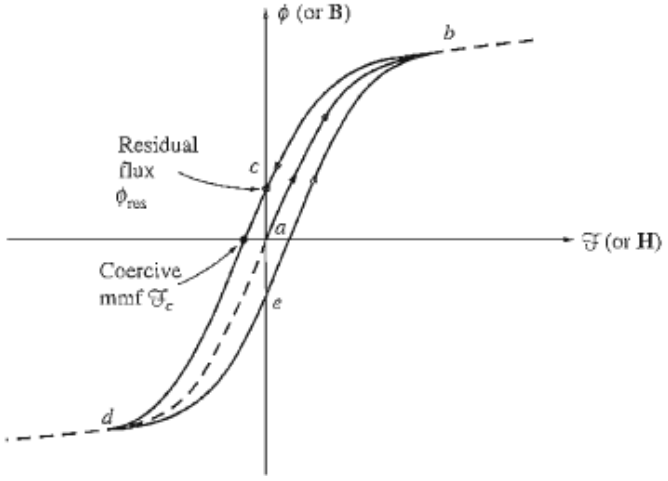
$$R_m = \frac{MMF}{\phi} = \frac{N.i}{\phi} = \frac{200 x 0.316}{0.012} 5268 A.turns/Wb$$

Energy Losses in a Ferromagnetic Core

Assume that a varying current, as shown below, is applied to the winding of a magnetic core.



Assume that the flux in the core is initially zero. As the current increases for the first time, the flux in the core traces out path *ab* in the figure below.



However, when the current falls again, the flux traces out the path *bcd*, and later when the current increases again, the flux traces out path *deb*. Notice that the amount of flux present in the core depends not only on the amount of current applied to the windings of the core, but also on the previous history of the flux in the core. This dependence on the preceding flux history and the resulting failure to retrace flux paths is called hysteresis. Path *bcdeb* traced out in the previous figure as the applied current changes is called a hysteresis loop.

Notice that if a large magnetomotive force is first applied to the core and then removed, the flux path in the core will be *abc*. When the magnetomotive force is removed, the flux in the core does not go to zero. Instead, a magnetic field is left in the core. This magnetic field is called the residual flux in the core. It is in precisely this manner that permanent magnets are produced. To force the flux to zero, an amount of magnetomotive force known as the coercive magnetomotive force F_C must be applied to the core in the opposite direction.

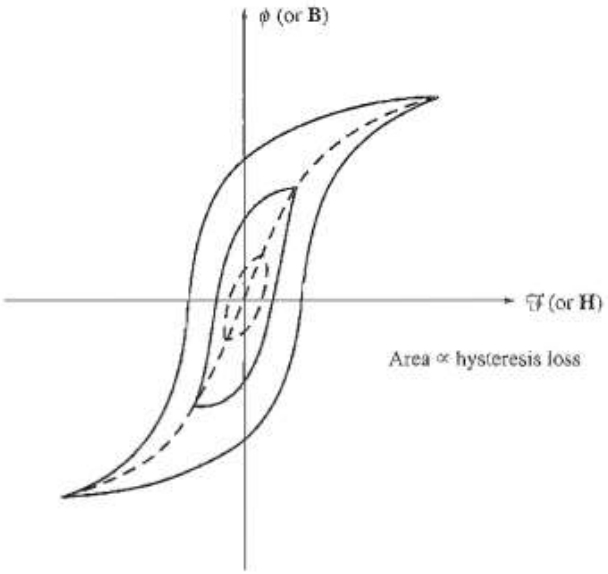
The reason for this behavior of the ferromagnetic materials is that the atoms of iron and similar metals (cobalt, nickel, and some of their alloys) have their magnetic fields (caused by the rotating electrons). The whole block of iron appear to have no flux because the fields of the different attoms are oriented randomly.

When an external magnetic field is applied to this block of iron, it causes the atoms to align themselves with the applied magnetic field. The extra atoms aligned with the field increase the magnetic flux in the iron, which in turn causes more atoms to switch orientation, further increasing the strength of the magnetic field. It is this positive feedback effect that causes iron to have a permeability much higher than air.

As the strength of the external magnetic field continues to increase, whole atoms are aligned up with the field and any further increase in the magnetomotive force can cause only the same flux increase that it would in free space. At this point, the iron is saturated with flux.

The key to hysteresis is that when the external magnetic field is removed, the atoms do not completely randomize again. This is because turning the atoms in requires energy. Originally, energy was provided by the external magnetic field to accomplish the alignment; when the field is removed, there is no source of energy to cause all the atoms to rotate back. The piece of iron is now a permanent magnet.

The fact that turning atoms in the iron requires energy leads to a common type of energy loss in all machines and transformers. This energy loss is called the hysterisis loss. The hysteresis loss in an iron core is the energy required to accomplish the reorientation of atoms during each cycle of the alternating current applied to the core. It can be shown that the area enclosed in the hysteresis loop formed by applying an alternating current to the core is directly proportional to the energy lost in a given ac cycle. The smaller the applied magnetomotive force excursions on the core, the smaller the area of the resulting hysteresis loop and so the smaller the resulting losses. The figure blow illustrates this point.



Another type of losses existing in the magnetic material is is the eddy current losses, which are caused by the currents circulating within the magnetic material. These currents are generated because of the induced voltages within the magnetic material, caused by the varying magnetic field.

Both hysteresis and eddy current losses cause heating in the core material, they are usually called core losses.

FARADAY'S LAW- INDUCED VOLTAGE FROM A TIME-CHANGING MAGNETIC FIELD

Faraday's law is the basis of transformer operation. It states that if a flux passes through a turn of a coil of wire, a voltage will be induced in the turn of wire that is directly proportional to the rate of change in the flux with respect to time. Or:

$$e_{ind} = -\frac{d\phi}{dt} \qquad (1)$$

If a coil has N turns and if the same flux passes through all of them, then the voltage induced across the whole coil is given by:

$$e_{ind} = -N\frac{d\phi}{dt} \qquad (2)$$

The minus sign in the equations is an expression of Lenz's law. Lenz's law states that the direction of the voltage builds up in the coil is such that if the coil ends were short circuited, it would produce a current that would cause a flux opposing the original flux change.

Equation (2) can be written as:

$$e_{ind} = -N\frac{d\phi}{dt} = -\frac{d}{dt}(N.\phi) = -\frac{d\lambda}{dt}$$

Where:

$\lambda = N.\phi$ and is called the flux linkage.

Faraday's law explains the eddy current losses in the magnetic material. A time-changing flux induces voltage within a ferromagnetic core. These voltages cause swirls of current to flow within the core. These eddy currents are flowing in a resistive material (the iron of the core), so energy is dissipated by them. The lost energy goes into heating the iron core.

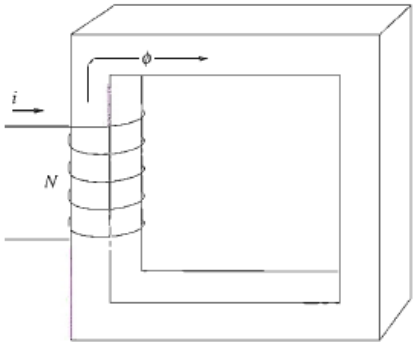
Two approaches can be used to reduce the eddy current losses in electrical machines:

1) The ferromagnetic core is broken up into many small strips. or laminations, to reduce the size of a current swirls.
2) Increase the resistivity of the core material, by adding some silicon to the steel of the core.

Example:

The figure below shows a coil of wire wrapped around an iron core, The flux in the core is given by the equation $\phi = 0.05\sin(314t)Wb$. If there are 100 turns on the core, what voltage is produced at the terminals of the coil?

Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).



Solution:

$$e_{ind} = N\frac{d\phi}{dt} = (100)(0.05)\frac{d}{dt}\sin(314t) = 5x314\cos(314t) = 1570\cos(314t)$$

PRODUCTION OF INDUCED FORCE ON AWIRE

If a conductor of length l and current i is exising in a magnetic field of uniform flux density B, a force F will act on the conductor, where:

$$\vec{F} = i\vec{l}X\vec{B}$$

The direction of the force is given by the right-hand rule: If the index finger of the right hand points in the direction of the vector $\vec{l}$ and the middle finger points in the direction of the flux density vector $\vec{B}$, then the thumb points in the direction of the resultant force on the conductor. The magnitude of the force is given by:

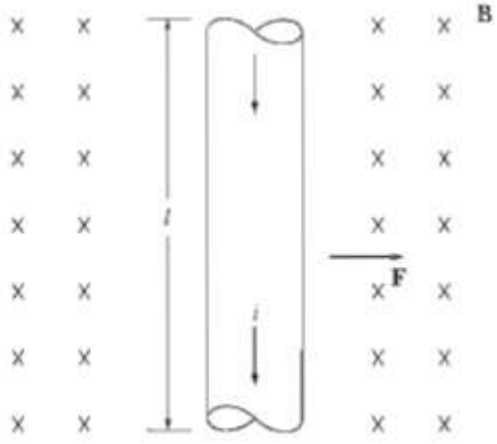
$$F = i.l.B.\sin\theta$$

where θe is the angle between the conductor and the flux density vector.

The induction of a force in a wire by a current in the presence of a magnetic field is the basis of the motor action.

Example:

The figure below shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?



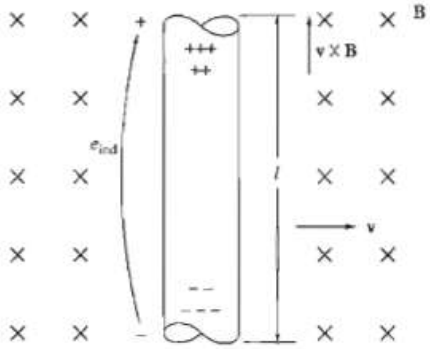
Solution:

$$F = i.l.B.\sin\theta = 0.5x1x0.25x\sin(90^0) = 0.125N$$

The direction of the force is given by the right-hand rule as being to the right.

INDUCED VOLTAGE ON A CONDUCTOR MOVING IN A MAGNETIC FIELD

If a wire moves through a magnetic field, a voltage is induced in it. See the figure below.



The voltage induced in the wire is given by:

$e_{ind} = (\vec{V}X\vec{B}).\vec{l}$

Where:

$\vec{V}$: the velocity of the wire

$\vec{B}$: is the magnetic flux density vector

$\vec{l}$: is the length of conductor in the magnetic field

Vector $\vec{l}$ points along the direction of the wire toward the end making the smallest angle with respect to the vector $\vec{V}X\vec{B}$.

The voltage in the wire will be built up so that the positive end is in the direction of the vector $\vec{V}X\vec{B}$.

Example 1:

The previous figure shows a conductor moving with a velocity of $5.0m/s$ to the right in the presence of a magnetic field. The flux density is $0.5T$ into the page, and the wire is $1.0m$ in length, oriented as shown. What are the magnitude and the polarity of the resulting induced voltage?

Solution:

The direction of the quantity $\vec{V}X\vec{B}$ is up. Therefore, the voltage on the conductor have a positive at the top with respect to the bottom of the wire. The direction of vector $\vec{l}$ is up, so that it makes the smallest angle with respect to the vector $\vec{V}X\vec{B}$.

The magnitude of the induced voltage is:

$e_{ind} = \left|(\vec{V}X\vec{B}).\vec{l}\right|$

Where:

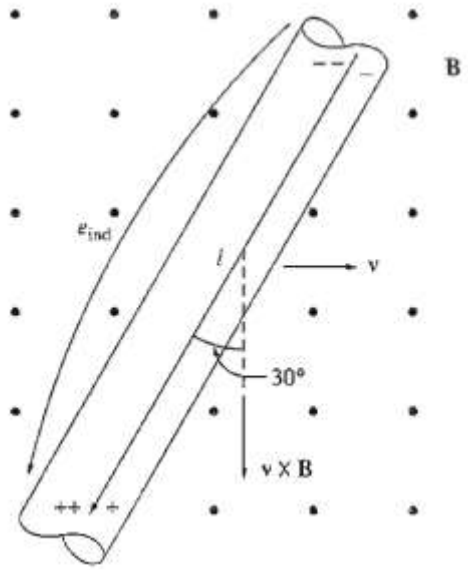
$\left|\vec{V}X\vec{B}\right| = \left|\vec{V}\right|.\left|\vec{B}\right|\sin\theta = 5x0.5x\sin 90^0 = 2.5$

$e_{ind} = \left|(\vec{V}X\vec{B}).\vec{l}\right| = \left|\vec{V}X\vec{B}\right|.\left|\vec{l}\right|.\cos\theta = 2.5x1x\cos 0^0 = 2.5V$

Positive at the top of the wire.

Example 2: The figure below shows a conductor moving with a velocity of $10m/s$ to the right in a magnetic field. The flux density is $0.5T$, out of the page, and the wire is

1.0*m* in length, oriented as shown. What are the magnitude and the polarity of the resulting induced voltage?



Solution:

$e_{ind} = \left|\left(\vec{V}X\vec{B}\right).\vec{l}\right|$

Where:

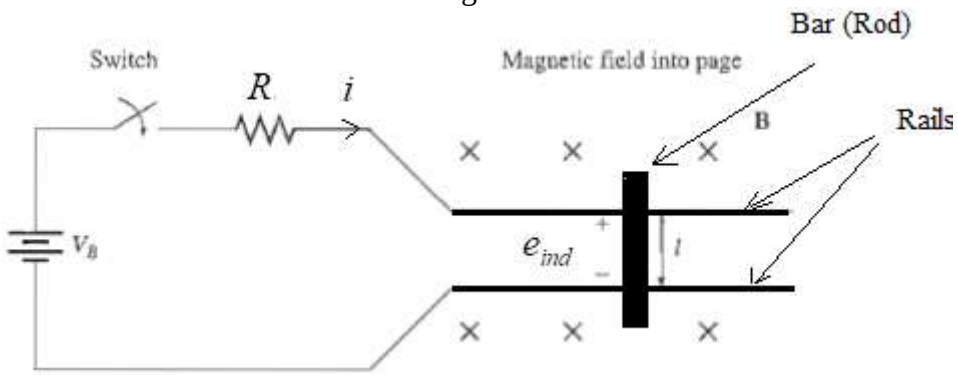
$\left|\vec{V}X\vec{B}\right| = \left|\vec{V}\right|.\left|\vec{B}\right|\sin\theta = 10x0.5x\sin 90^0 = 5V/m$ directed down

$e_{ind} = \left|\left(\vec{V}X\vec{B}\right).\vec{l}\right| = \left|\vec{V}X\vec{B}\right|.\left|\vec{l}\right|.\cos\theta = 5x1x\cos 30^0 = \frac{5\sqrt{3}}{2}V$

Note that the direction of $\vec{l}$ is to make the smallest possible angle with the direction of $\vec{V}X\vec{B}$.

THE LINEAR DC MACHINE

The Linear DC machine is shown in the figure below.



It consists of two smooth frictionless rails of conducting material with a rod of conducting material which can freely move along the rails. This structure provides a linear motion (not a rotational motion as a real machine) but can be simply used to explain all the principles describing the behavior of of the real generator and the real motor.

The operation of this simple machine is based on 4 basic laws:

1) The law for the force on a wire in the presence of a magnetic field, as mentioned in the previous lectures:

$$\vec{F} = i\vec{l}X\vec{B} \qquad (1)$$

2) The law for the voltage induced on a wire moving in a magnetic fiel, as stated before:

$$e_{ind} = \left(\vec{V}X\vec{B}\right)\vec{l} \qquad (2)$$

3) Kirchhoff's voltage law, or:

$$V_B = R.i + e_{ind} \qquad (3)$$

4) Newton's law for the force acting on the freely moving bar:

$$F_{net} = m.a \qquad (4)$$

Starting the Linear DC Machine

When the switch closes, a current flows in the bar, which is given by:

$$i = \frac{V_B - e_{ind}}{R} \qquad (5)$$

Since the bar is initially at rest, $e_{ind} = 0$, so $i = \frac{V_B}{R}$.

This current, in the presence of a magnetic field induces a force on the wire, which magnitude is:

$$F = iLB$$

And direction to the right.

Because of this force, the bar accelerates and its velocity begins to increase. An induced voltage, e_{ind}, will appear across the bar and its magnitudes continuously increasing as the velocity is increasing. This is because:

$$e_{ind} = V.B.l$$

As e_{ind} increases the current in the bar decreases, as by equation (5).

As the current in the bar decreases, the force decrease and the acceleation decreases, till it becomes zero, and the speed becomes constant, or the speed has reached a

constant steady-state value V_{SS}, where the net force on the bar is zero. This will occur when e_{ind} has risen to a value equals to the voltage V_B. At that time:

$$e_{ind} = V_B$$

Or:

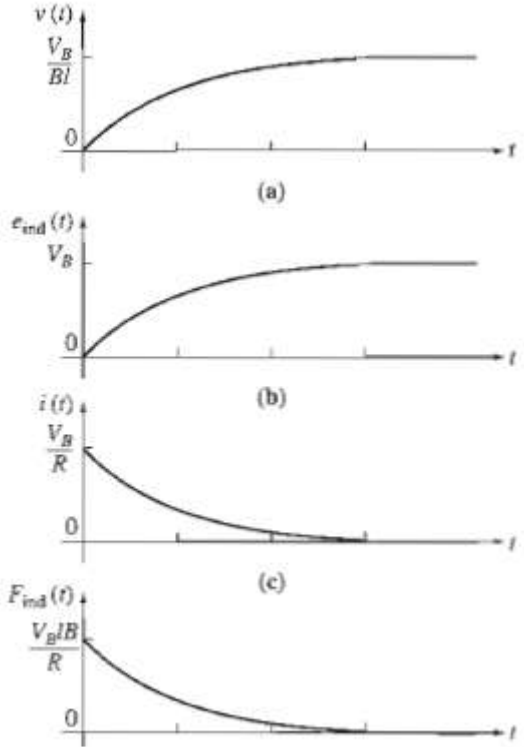
$$V_{SS}.B.l = V_B$$

Or:

$$V_{SS} = \frac{V_B}{B.l}$$

The bar will continue to coast along at this no-load speed forever unless some external force disturbs it.

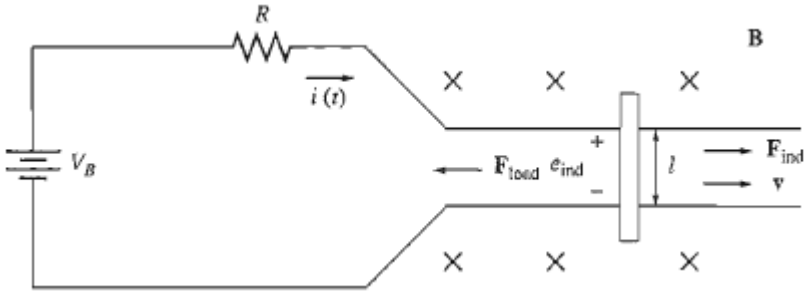
The figure below shows the speed, the induced voltage and the force on the rod, as functions of time, after closing the switch.



This desribed behavior of the linear machine, is similar to that of real motors.

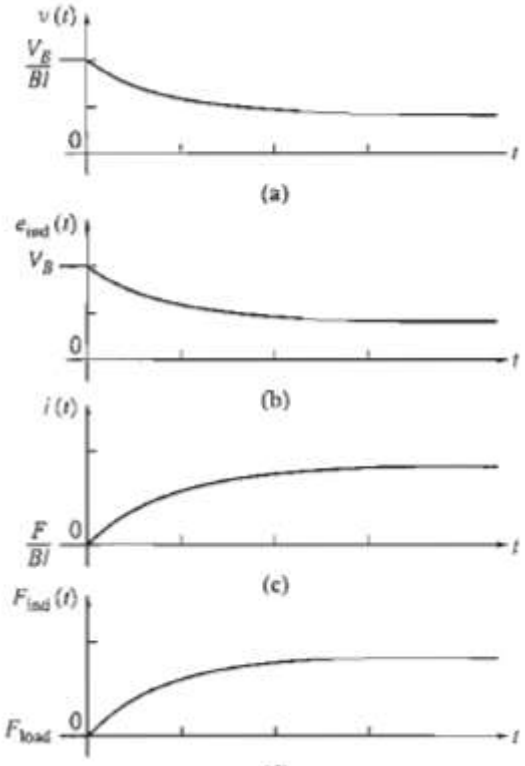
The Linear DC Machine as a Motor

The figure below shows a linear DC machine that is initially nunning at no-load steady-state speed. A load with a force F_{load}, acting in a direction opposite the direction of movement, is applied to the bar.



The effect of F_{load} will be to slow the bar. But just as soon as the bar begins to slow down, the induced voltage on the bar drops ($e_{ind} = V.B.l$). As the induced voltage decreases, the current flow in the bar rises ($i = \frac{V_B - e_{ind}}{R}$). Therefore, the induced force rises too ($F = iLB$). The induced force rises until it is equal and opposite to the load force , and the bar again travels in steady state, but at a lower speed.

The figure below shows the speed, the induced voltage, the current and the induced for as a function of time, when a load is applied to the linear DC machine.



As an induced force, to keep the bar moving, appears, this indicate that a power is being converted from electrical form to mechanical form. The power being converted is:

$$P_{conv} = e_{ind}.i = F.V$$

Or an electric power equals to $P_{elect.} = e_{ind}.i$ is being converted to an equivalent mechanical power equals to $P_{mech} = F.V$ and the machine is operating as a motor.

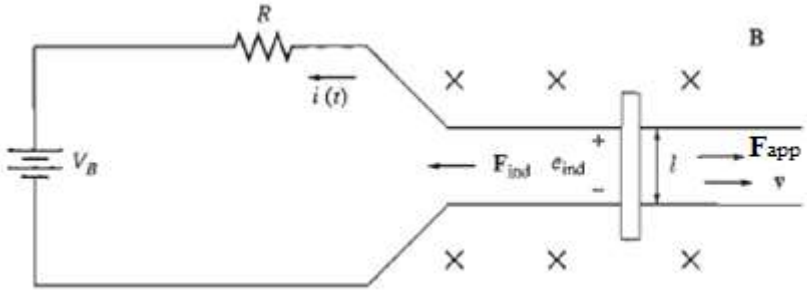
A real DC motor behaves in a precisely analogous fashion when it is loaded: As a load is added to its shaft, the motor begins to slow down, which reduces its internal voltage, increasing its current flow. The increased current flow increases its induced torque, and the induced torque will equal the load torque of the motor at a new, slower speed.
For the motor with rotational motion:

$$P_{conv} = e_{ind}.i = T.\omega$$

Where T is the torque and ω is the angular speed of the rotor shaft.

The Linear DC Machine as a Generator

Consider again, a linear DC machine that is initially nunning at no-load steady-state speed. A force F_{app}, in the direction of motion, is being applied to the rod.



The applied force will cause the bar to accelerate in the direction of motion, and the velocity v of the bar will increase. As the velocity increases, $e_{ind} = V.B.l$ will increase and will be larger than the battery voltage V_B.
With $e_{ind} > V_B$, the current reverses direction and is now given by the equation:

$$i = \frac{e_{ind} - V_B}{R}$$

This current now is charging the battery.

Since this current now flows up through the bar, it induces a force in the bar given by:

$$F = iLB$$

Acting to the left and opposing the applied force on the bar, F_{app}.

Note also that, the induced force will be equal and opposite to the applied force, and the bar will be moving at a higher steady-state speed than before. Notice that now the battery is charging. The linear machine is now serving as a generator, converting mechanical power into electric power, where:

$$F.V = e_{ind}.i$$

Note that the real DC generator behaves in the same manner: A torque is applied to the shaft in the direction of motion, the speed of the shaft increases, the internal voltage increases, and current flows out of the generator to the loads. The amount of mechanical power converted to electrical form in the real rotating generator is given by:

$$P_{conv} = T.\omega$$

Where T is the torque and ω is the angular speed of the rotor shaft.

It has to be noted that the same machine can acts as both a motor or a generator. The only difference between the two is whether the externally applied forces are in the

direction of motion (generator) or opposite to the direction of motion (motor). Electrically, when $e_{ind} > V_B$ the machine acts as a generator, and when $e_{ind} < V_B$ the machine acts as a motor.

Note also that when the machine acts as a motor or as a generator, it moves in the same direction, but to act as a generator an external force is applied to increase the speed, and when acts as a motor a force is applied on the load and the speed decreases.

Starting Problems with the Linear Machine

Because of the small internal resistance in the real machine circuit (around 0.1Ω for real medium size DC motor) and as the induced voltage (e_{ind}) is zero at starting, the starting current of the machine becomes very high (and be more than 10 times the rated value), and may damage the machine winding.
Note that:

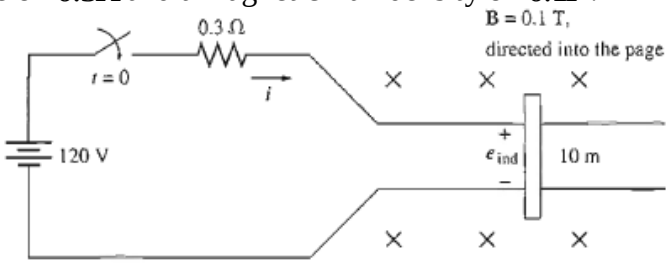
$$I_{Starting} = \frac{V_B}{R}$$

The high starting current problem can be solved by:

1) inserting an extra resistance into the circuit during starting to limit the current flow until e_{ind} builds up enough to limit it.
2) Applying a lower supply voltage during starting.

Example:

The linear dc machine, shown in the figure below, has a battery voltage of $120V$, an internal resistance of 0.3Ω and a magnetic flux density of $0.1T$.



a) What is this machine's maximum starting current? What is its steady-state velocity at no load?

b) Suppose that a $30N$ force pointing to the right were applied to the bar. What would the steady-state speed be? How much power would the bar be producing or consuming? How much power would the battery be producing or consuming? Explain the difference between these two figures (the power of the bar and the power of the battery). Is this machine acting as a motor or as a generator?

c) Suppose a $30N$ force pointing to the left were applied to the bar. What would the new steady-state speed be? Is this machine acts as a motor or as a generator?

d) Assume that a force pointing to the left is applied to the bar. Calculate speed of the bar as a function of the force for values from $0N$ to $50N$ in $10N$ steps. Plot the velocity of the bar versus the applied force.

e) Assume that the bar is unloaded and that it suddenly runs into a region where the magnetic field is weakened to 0.08 T. How fast will the bar go now?

Solution:

a) The current is:

$$i = \frac{V_B - e_{ind}}{R}$$

at starting $e_{ind} = 0$ and the starting current is:

$$i_{starting} = \frac{120 - 0}{0.3} = 400A$$

At no-load $i = 0$ and $e_{ind} = V_B = 120V$

But:

$$e_{ind} = V.B.l$$

Or:

$$V = \frac{e_{ind}}{B.l} = \frac{120}{0.1x10} = 120m/s$$

b) At steady state, $F_{ind} = F_{app} = 30N$

but:

$$F_{ind} = iBl$$

From this equation:

$$i = \frac{F_{ind}}{Bl} = \frac{30}{0.1x10} = 30A$$

But as the machine operates as a generator:

$$i = \frac{e_{ind} - V_B}{R}$$

Or:

$$e_{ind} = V_B + R.i = 120 + 0.3x30 = 129V$$

But:

$$e_{ind} = V.B.l$$

Thus:

$$V = \frac{e_{ind}}{B.l} = \frac{129}{0.1x10} = 129m/s$$

The producing power by the machine (the machine acts as a generator) is:

$$P_m = e_{ind}.i = 129x30 = 3870W$$

The power consumed by the battery (the machine is charging the battery) is:

$$P_B = V_B.i = 120x30 = 3600W$$

The difference between the power produced by the machine and the power consumed by the battery is:

$P_{loss} = P_m - P_B = 3870 - 3600 = 270W$ and is losses on the machine resistance R, which is dissipated as heat.

c) as the force is pointing to the left, the machine is acting as a motor.

Also, at steady state, $F_{ind} = F_{app} = 30N$

but:

$$F_{ind} = iBl$$

Or:

$$i = \frac{F_{ind}}{Bl} = \frac{30}{0.1x10} = 30A$$

But as the machine operates as a motor:

$$i = \frac{V_B - e_{ind}}{R}$$

Or:

$$e_{ind} = V_B - R.i = 120 - 0.3x30 = 111V$$

But:

$$e_{ind} = V.B.l$$

Thus:

$$V = \frac{e_{ind}}{B.l} = \frac{111}{0.1x10} = 111m/s$$

d) For F_{app} from $0N$ to $50N$ in $10N$ steps, the same procedure as in part (c) is applied and the characteristic is plotted .
A MATLAB program can be built.

e) When the bar is not loaded $e_{ind} = \mathrm{V_B} = 120V$
where:

$e_{ind} = V.B.l$, Or: $V = \frac{e_{ind}}{B.l}$

Initially, $B = 0.1T$, thus $V = \frac{e_{ind}}{B.l} = \frac{120}{0.1x10} = 120m/s$

When B is decreased to $0.08T$, the speed will increase and becomes:

$$V = \frac{e_{ind}}{B.l} = \frac{120}{0.08x10} = 150m/s$$

Real, Reactive, and Apparent Powers in Single-Phase AC Circuits

Consider the case with a sinusoidal voltage on a load $v(t) = V_m.\sin(\omega.t)$ and a current lagging shifted by ϕ, $i(t) = I_m.\sin(\omega.t - \phi)$, as shown.

The instantaneous power will be:

$$\begin{aligned}
p = v.i &= V_m.I_m.\sin(\omega.t).\sin(\omega t - \phi) \\
&= V_m.I_m.\sin(\omega.t).(\sin\omega t.\cos\phi - \cos\omega t.\sin\phi) \\
&= V_m.I_m.\sin^2(\omega.t).\cos\phi - V_m.I_m.\sin\omega t.\cos\omega t.\sin\phi \\
&= \frac{V_m.I_m}{2}.\cos\phi.(1-\cos 2\omega t) - \frac{V_m.I_m}{2}.\sin\phi.\sin 2\omega t \\
&= \frac{V_m.I_m}{2}.\cos\phi - \frac{V_m.I_m}{2}.\cos\phi.\cos 2\omega t - \frac{V_m.I_m}{2}.\sin\phi.\sin 2\omega t \\
&= \frac{V_m.I_m}{2}.\cos\phi - \frac{V_m.I_m}{2}(\cos\phi.\cos 2\omega t + \sin\phi.\sin 2\omega t) \\
&= \frac{V_m.I_m}{2}.\cos\phi - \frac{V_m.I_m}{2}\cos(2\omega t - \phi) \\
&= V.I\cos\phi - V.I\cos(2\omega t - \phi)
\end{aligned}$$

or the instantaneous power is pulsating with a double frequency and have an average value of $V.I\cos\phi$. وتسمى القدرة الفعالة أو القدرة الحقيقية أو القدرة

The figure below shows the wave forms of the voltage, the current and the instantaneous power.



note that at some instances the instantaneous power is negative which means that the power is transferring from the load to the source.

Active and reactive power

It has been found that:

$$\begin{aligned}
p = v.i &= V_m.I_m.\sin(\omega.t).\sin(\omega t - \phi) \\
&= V_m.I_m.\sin(\omega.t).(\sin\omega t.\cos\phi - \cos\omega t.\sin\phi) \\
&= V_m.I_m.\sin^2(\omega.t).\cos\phi - V_m.I_m.\sin\omega t.\cos\omega t.\sin\phi \\
&= \frac{V_m.I_m}{2}.\cos\phi.(1-\cos 2\omega t) - \frac{V_m.I_m}{2}.\sin\phi.\sin 2\omega t \\
&= V.I.\cos\phi.(1-\cos 2\omega t) - V.I.\sin\phi.\sin 2\omega t
\end{aligned}$$

from this we note that the instantaneous power has two components:

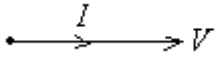
- the first component, $V.I.\cos\phi.(1-\cos 2\omega t)$ never goes negative and pulsates around the average value $V.I.\cos\phi$. This average value $V.I.\cos\phi$ is denoted by P and is called the real or active

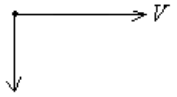
power, as it never goes negative this means that this component always goes from the source to the load.

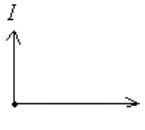
- the second component, $V.I.\sin\phi.\sin 2\omega t$ has an average value equals to zero. It has a maximum value of $V.I.\sin\phi$ and denoted by Q. This maximum value is called the reactive power. as this power pulsates negative and positive, this means that it is transferred from the load to the source and vice versa.
- As the real power is defined as $P = V.I.\cos\phi$, this can be interpreted as the product of the magnitude of the component of the current vector on the voltage vector (usually called active current) and the magnitude of the voltage.
- As the reactive power is defined as $Q = V.I.\sin\phi$, this can be interpreted as the product of the magnitude of the normal component of the current vector on the voltage vector (usually called reactive current) and the magnitude of the voltage.

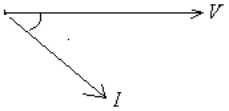
note that $\cos\phi$ is called the power factor.
note that P has the unit W, kW, MW,..etc.
and Q has a unit kvar, Mvar…etc.

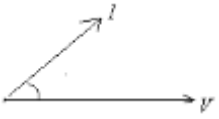
Note that in case of pure resistive load, $\phi = 0$, thus $P = V.I$ and $Q = 0$
The following shows the voltage and current phasors as well as the active power and the reactive power for different load types.

- pure resistive load: 

, $\phi = 0$, $P > 0$, $Q = 0$
- pure inductive load: 

, $\phi = -90^0$, $P = 0$, $Q < 0$
- pure capacitive load: 

, $\phi = 90^0$, $P = 0$, $Q > 0$
- resistive-inductive load: 

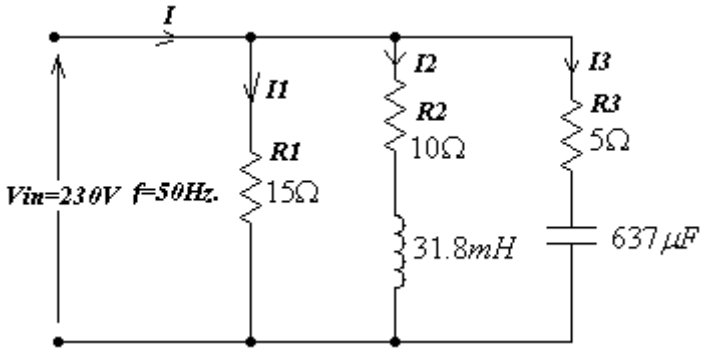
, $\phi = 0 to -90^0$, $P > 0$, $Q < 0$
- resistive-capacitive load: 

, $\phi = 0 to 90^0$, $P > 0$, $Q > 0$

Apparent Power
The apparent power is defined as $V.I$ and is measured in VA. As it was find that the active power is $P = V.I.\cos\phi$ and the reactive power is $Q = V.I.\sin\phi$, it is found that

$$\sqrt{P^2 + Q^2} = \sqrt{(V.I.\cos\phi)^2 + (V.I.\sin\phi)^2} = VA$$

Example:
Consider the following circuit.



compute the various powers absorbed in the three branches:
solution:

$$I_1 = \frac{V_{in}}{R_1} = \frac{230}{15} = 15.33\angle 0^0\, A$$

for the RL branch:

$$Z_2 = R_2 + j2\pi f.L = 10 + j10$$

$$I_2 = \frac{V_{in}}{Z_2} = \frac{230}{10 + j10} = 11.5 - j11.5 = 16.26\angle -45^0$$

for the RC branch:

$$Z_3 = R_3 + \frac{1}{j2\pi f.C} = 5 - j5$$

$$I_3 = \frac{V_{in}}{Z_3} = \frac{230}{5 - j5} = 23 + j23 = 32.53\angle 45^0$$

for the resistive branch:

$$P_1 = V_{in}xI_1x\cos\varphi_1 = 230x15.33x\cos(0) = 3527W = 3.527kW$$

$$Q_1 = V_{in}xI_1x\sin\varphi_1 = 230x15.33x\sin(0) = 0\,\text{var}.$$

for the RL branch:

$$P_2 = V_{in}xI_2x\cos\varphi_2 = 230x16.26x\cos(-45^0) = 2644W = 2.644kW$$

$$Q_2 = V_{in}xI_2x\sin\varphi_2 = 230x16.26x\sin(-45^0) = -2644\,\text{var} = -2.644k\,\text{var (inductive)}$$

for the RC branch:

$$P_3 = V_{in}xI_3x\cos\varphi_3 = 230x32.53x\cos(45^0) = 5291W = 5.291kW$$

$$Q_3 = V_{in}xI_3x\sin\varphi_3 = 230x32.53x\sin(45^0) = 5291\,\text{var} = 5.291k\,\text{var (capacitive)}$$

Note that:
- the total active power is:

$$P = P_1 + P_2 + P_3 = 11.462kW$$

- the total reactive power is:

$$Q = Q_1 + Q_2 + Q_3 = 0 - 2.644 + 5.291 = 2.647k\,\text{var}\ \text{(capacitive)}.$$

these total active and reactive power can be found as:

$$I = I_1 + I_2 + I_3 = 48.83 + j11.5 = 51.14\angle 13^0$$

thus: the active power is:

$P = V_{in} x I x \cos\varphi = 230 x 51.14 x \cos(13^0) = 11.462 kW$

the reactive power is:

$Q = V_{in} x I x \sin\varphi = 230 x 51.14 x \sin(13^0) = 2.647 k \text{ var}$

note that these values are the same as before.

The apparent powers are as follows:

The apparent power of the resistive branch is:

$V_{in} x I_1 = 230 x 15.33 = 3.526 kVA$

The apparent power of the RL branch is:

$V_{in} x I_2 = 230 x 16.26 = 3.74 kVA$

The apparent power of the RC branch is:

$V_{in} x I_3 = 230 x 32.53 = 7.48 kVA$

The total apparent power as computed from the input current is:

$V_{in} x I = 230 x 51.14 = 11.76 kVA$

not that the total apparent power is less than the sum of apparent powers for all branches which is equal to: $3.526 + 3.74 + 7.48 = 14.746 kVA$

or the algebraic summation rule doesn't apply for the apparent powers.

The concept of complex power:

if the voltage is $V = |V| \angle \theta_V$ and the current is $I = |I| \angle \theta_I$, the conjugate value of the current will be: $I^* = |I| \angle -\theta_I$

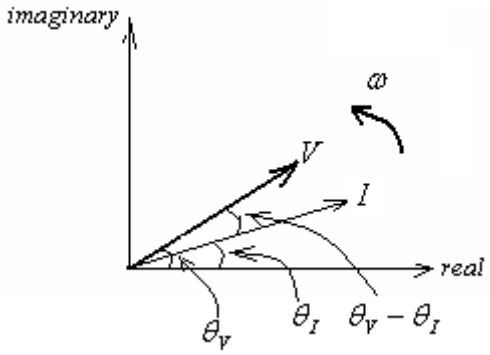
the complex power is defined as:

$$S = V.I^* = |V| \angle \theta_V . |I| \angle -\theta_I = |V|.|I| . \angle(\theta_V - \theta_I)$$
$$= |V|.|I| . \angle \phi = |V|.|I| . \cos\phi + j|V|.|I| . \sin\phi = P + jQ$$

note that by using in the definition the conjugate value of the current, the produced results complies with our definition.

Note:
$$V.I = |V| \angle \theta_V . |I| \angle \theta_I = |V|.|I| . \angle(\theta_V + \theta_I)$$
$$= |V|.|I| . \cos(\theta_V + \theta_I) + j|V|.|I| . \sin(\theta_V + \theta_I)$$

and doesn't comply with the definition of the real and reactive powers stated before, and the definition of the angle between the current and the voltage shown below.



if in the equation for the complex $S = V.I^*$, we replace $V = Z.I$, we find that:

$S = V.I^* = Z.I.I^* = Z.|I|^2$

if in the equation for the complex $S = V.I^*$, we replace $I = Y.V$,or $I^* = Y^*.V^*$ we find that:

$S = V.I^* = V.Y^*.V^* = Y^*.|V|^2$.

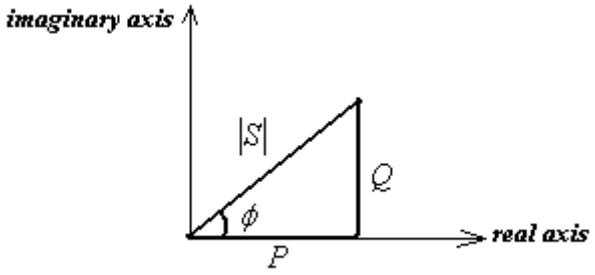
all these equations can be used to compute the complex power.

P and Q can be obtained as the real and the imaginary parts of S, respectively.

Note that the magnitude of the complex power is:

$|S| = \sqrt{P^2 + Q^2}$ which the apparent power.

As S is a complex number, it can be drawn in the complex plane as below.



this triangle is sometimes called the power triangle, from it, we find:

$$|S| = \sqrt{P^2 + Q^2}\ ,\ \phi = \tan^{-1}\left(\frac{Q}{P}\right) = \sin^{-1}\left(\frac{Q}{\sqrt{P^2 + Q^2}}\right) = \cos^{-1}\left(\frac{P}{\sqrt{P^2 + Q^2}}\right)$$

Example: (the same previous example, solved using the complex power).

or the algebraic summation rule apply for the complex powers.

TRANSFORMERS

A transformer is a device that converts ac electric power at one voltage level to ac electric power at another voltage level, at the same frequency, through the action of a magnetic field. The transformer has two or more coils of conductor wire wrapped around a common ferromagnetic core.
One of the transformer windings is connected to a source of ac electric power, and is called the primary winding. The second transformer winding (called the secondary winding) supplies electric power to loads. If there is a third winding on the transformer, it is called the tertiary winding.

In the power system, electric power is generated at voltages of 11 to 30 kV. Transformers step up the voltage to between 132 kV and nearly 1000 kV for transmission over long distances at very low losses. Transformers then step down the voltage to the 11- to 33-kV range for local distribution and finally permit the power to be used safely in homes, offices, and factories at voltages as low as 400/230V.
Depending on their use, power transformers have different names:
- Unit transformer: is the transformer connected to the output of a generator and used to step the generated voltage up to transmission levels (132 or 400 kV or higher).

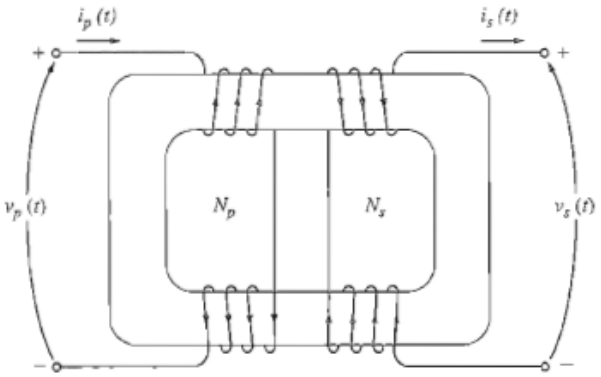
- Substation transformer: is the transformer at the other end of the transmission line and is used to step the transmission voltage down to distribution levels (33 kV or 11 kV).

- Distribution transformer: is the transformer that takes the distribution voltage and steps it down to the final voltage at which the power is actually used (400/230 V).

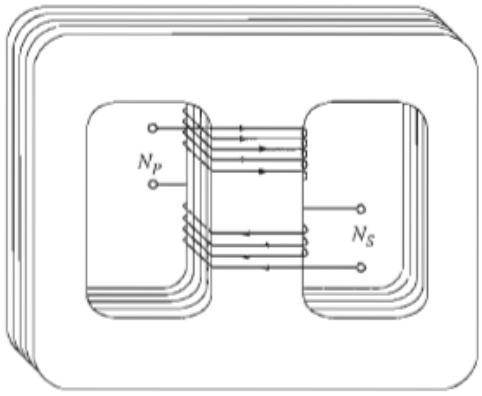
In addition to the various power transformers, two special-purpose transformers are used to measure voltage and current in electric machinery and power systems. The first of these special transformers is a device specially designed to sample a high voltage and produce a low secondary voltage directly proportional to it. Such a transformer is called a potential (or voltage) transformel, (VT).
The second type of special transformer is a device designed to provide a secondary current much smaller than but directly proportional to its primary current. This device is called a current transformer (CT).

TYPES AND CONSTRUCTION OF TRANSFORMERS

Power transformers generally are constructed on one of two types of cores:
1) core form: The core in this construction is of rectangular laminated piece of steel with the transformer windings wrapped around two sides of the rectangle, as shown below.



2) shell form: this type consists of a three-legged laminated core with the windings wrapped around the center leg, as shown below.



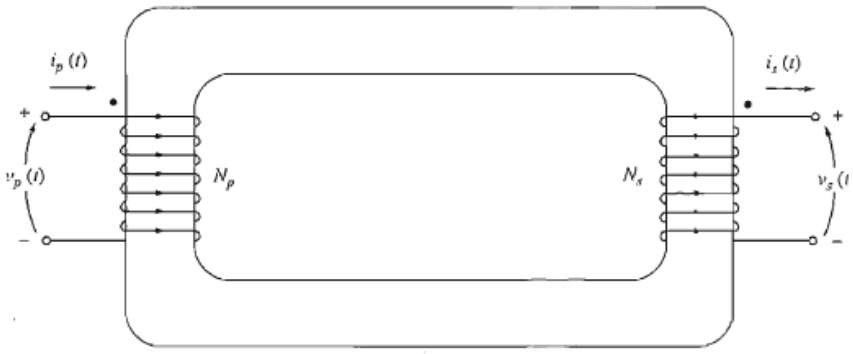
In each of these two constructions, the core is constructed of thin laminations electrically isolated from each other in order to minimize eddy currents.

In practice, the primary and secondary windings of the transformer are wrapped one on top of the other with the low-voltage winding innermost. Such an arrangement serves two purposes:

1) It simplifies the problem of insulating the high-voltage winding from the core.
2) It results in much less leakage flux than would be the case if the two windings were separated by a distance on the core.

THE IDEAL TRANSFORMER

The ideal transformer is a lossless device with an input winding and an output winding, as shown below.



The relationship between the voltage $v_p(t)$ applied to the primary side of the transformer and the voltage $v_S(t)$ produced on the secondary side is:

$$\frac{v_p(t)}{v_S(t)} = \frac{N_p}{N_S} = a \qquad (1)$$

where a is defined to be the turns ratio of the transformer:

$$a = \frac{N_p}{N_S}$$

As the transformer is lossless, the input and the output powers are equal, or:

$$v_p(t).i_p(t) = v_S(t).i_S(t)$$

And from it:

$$\frac{i_p(t)}{i_S(t)} = \frac{v_S(t)}{v_p(t)} = \frac{1}{a} \qquad (2)$$

In terms of phasor quantities, these equations are:

$$\frac{\vec{V}_P}{\vec{V}_S} = \frac{\vec{I}_S}{\vec{I}_P} = a \qquad (3)$$

Notice that the phase angle of $\vec{V}_P$ is the same as the angle of $\vec{V}_S$ and the phase angIe of $\vec{I}_P$ is the same as the phase angle of $\vec{I}_S$, or the turns ratio of the ideal transformer affects the magnitudes of the voltages and currents, but not their angles.

The polarity of the voltage and current on the primary and the secondary sides of the transformer utilizes the dot convention. The relationship is as follows:

1) If the primary voltage is positive at the dotted end of the winding with respect to the undotted end, then the secondary voltage will be positive at the dotted end also, or voltage polarities are the same with respect to the dots on each side of the core.
2) If the primary current of the transformer flows into the dotted end of the primary winding, the secondary current will flow out of the dotted end of the secondary winding.

Power in an Ideal Transformer

The real power P_{in} supplied to the transformer by the primary circuit is given by:

$$P_{in} = V_P.I_P.\cos\theta_P$$

The real power P_{out} supplied by the transformer secondary circuit to its loads is:

$$P_{out} = V_S.I_S.\cos\theta_S$$

Since for the indeal transformer;

$P_{in} = P_{out}$, or:

$$\frac{P_{in}}{P_{out}} = \frac{V_P.I_P.\cos\theta_P}{V_S.I_S.\cos\theta_S} = 1$$

Or:

$$\left(\frac{V_P}{V_S}\right).\left(\frac{I_P}{I_S}\right).\left(\frac{\cos\theta_P}{\cos\theta_S}\right) = 1$$

But: $\frac{V_P}{V_S} = a$ and $\frac{I_P}{I_S} = \frac{1}{a}$, thus: $\frac{\cos\theta_P}{\cos\theta_S} = 1$,or: $\cos\theta_P = \cos\theta_S$

Or, the primary and secondary windings of an ideal transfonner have the same power factor.

As the reactive power is concerner, the input reactive power is:

$Q_{in} = V_P.I_P.\sin\theta_P$

The output reactive power is:

$Q_{out} = V_S.I_S.\sin\theta_S$

Thus:

$$\frac{Q_{in}}{Q_{out}} = \frac{V_P.I_P.\sin\theta_P}{V_S.I_S.\sin\theta_S} = \left(\frac{V_P}{V_S}\right).\left(\frac{I_P}{I_S}\right).\left(\frac{\sin\theta_P}{\sin\theta_S}\right)$$

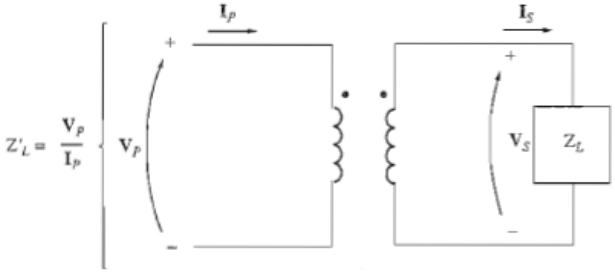
As: $\frac{V_P}{V_S} = a$ and $\frac{I_P}{I_S} = \frac{1}{a}$ and $\theta_P = \theta_S$, thus:

$\frac{Q_{in}}{Q_{out}} = 1$, or $Q_{in} = Q_{out}$

As $P_{in} = P_{out}$ and $Q_{in} = Q_{out}$, thus the input and output complex powers (as well as the input and output apparent power) of the ideal transformer are equal.

<u>Impedance Transformation through a Transformer</u>

Consider an ideal transformer with a load impedance Z_L, as shown below.



One of the interesting properties of a transformer is that, since it changes voltage and cunent levels, it changes the ratio between voltage and current and hence the apparent impedance of an element.

The impedance of the load is:

$$Z_L = \frac{V_S}{I_S} \qquad (1)$$

The apparent impedance of the primary circuit of the transfonner is:

$$Z_L' = \frac{V_P}{I_P} \qquad (2)$$

As:

$\frac{V_P}{V_S} = a$, thus:

$V_P = aV_S$ (3)

And as:

$\frac{I_P}{I_S} = \frac{1}{a}$, thus:

$$I_P = \frac{I_S}{a} \qquad (4)$$

Substituting (3) and (4) in (2), we find:

$$Z_L' = \frac{aV_S}{\frac{I_S}{a}} = a^2 \frac{V_S}{I_S} = a^2 Z_L \qquad (5)$$

Or an impedance Z_L on the secondary will be seen as $a^2 Z_L$ from the primary.

It can be seen that, with a transformer, it is possible to match the magnitude of a load impedance to a source impedance simply by picking the proper turns ratio.

Analysis of Circuits Containing Ideal Transformers

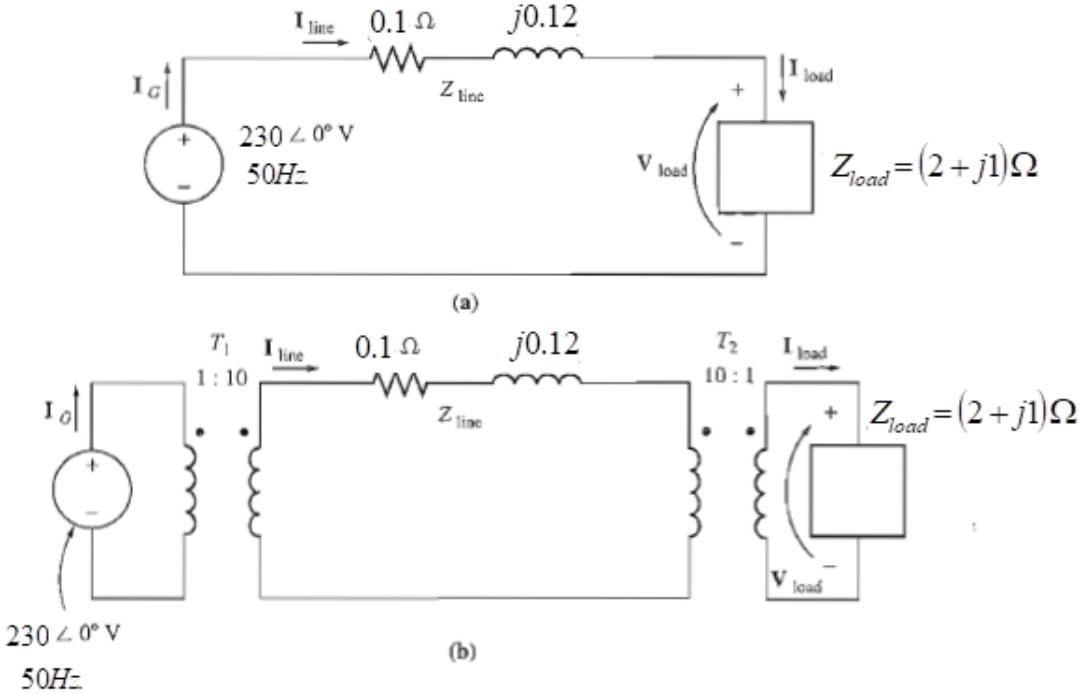
If a circuit contains an ideal transformer, then the easiest way to analyze the circuit is to replace the portion of the circuit on one side of the transformer by its equivalent on the other side, and the circuit will be solved for one voltage level, (without a transformer present). The turns ratio of the transformer can be used to determine the voltages and currents on the other side of the transformer. The process of replacing one side of a transformer with its equivalent at the other side's voltage level is known as referring the first side of the transformer to the second side.
Consider the following example.

Example:

A single-phase power system consists of a $230V$, $50Hz.$ generator supplying a load $Z_{load} = (2 + j1)\Omega$ through a transmission line of impedance $Z_{line} = (0.1 + j0.12)\Omega$.

a) Find the load voltage and the transmission line losses.

b) If a $1:10$ step-up transformer is placed at the generator end of the transmission line and a $10:1$ step-down transformer is placed at the load end of the line, as shown below, find the load voltage and the transmission line losses.



a)

$$I_G = I_{line} = I_{load} = \frac{V}{Z_{line} + Z_{load}} = \frac{230\angle 0^0}{(0.1 + j0.12) + (2 + j1)} = \frac{230\angle 0^0}{2.1 + j1.12}$$

$$= \frac{230\angle 0^0}{2.38\angle 28.07^0} = 96.64\angle -28.07^0$$

The load voltage is:

$$V_{load} = I_{line}.Z_{load} = 96.64\angle -28.07^0 (2 + j1) = 96.64\angle -28.07^0 \, x2.236\angle 26.57^0$$

$$= 216.09\angle -1.5^0$$

The line losses are:

$$P_{loss} = (I_{line})^2.R_{line} = (96.64)^2 \, x0.1 = 933.93W$$

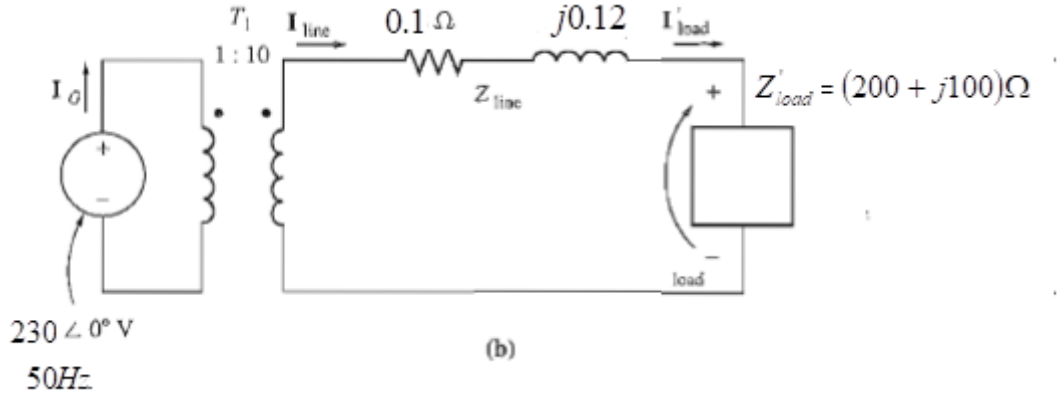
b)

To analyze the system with transformers, it is necessary to convert all the sections to one voltage level (the generator voltage level).

- To eliminale transformer T_2, the load is reffered to the transmission line's voltage level

$$Z'_{load} = a^2.Z_{load} = \left(\frac{10}{1}\right)^2 (2 + j1) = (200 + j100)\Omega$$

The equivalent circuit becomes as follows.



The total impedance at the transmission line level is now:

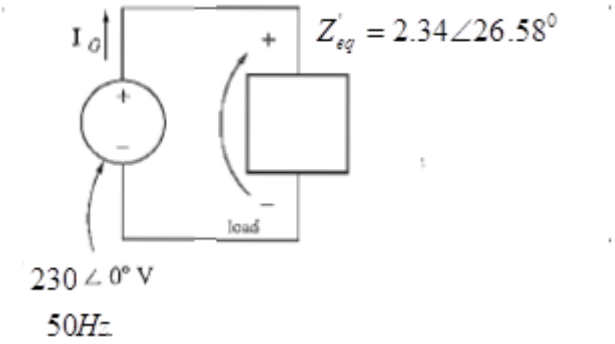
$$Z_{eq} = Z_{line} + Z'_{load} = (0.1 + j0.12) + (200 + j100) = 200.1 + j100.12$$

$$= 233.75\angle 26.58 \quad \Omega$$

- To eliminate transfonner T_1, The total impedance at the tra nsmission line level ($Z_{line} + Z'_{load}$) is now referred to the source voltage level, or:

$$Z'_{eq} = a^2 (Z_{line} + Z'_{load}) = \left(\frac{1}{10}\right)^2 (233.57\angle 26.58^0) = 2.34\angle 26.58^0$$

The resulting equivalent circuit is as below.



The generator's current is:

$$I_G = \frac{230\angle 0^0}{2.34\angle 26.58^0} = 98.29\angle -26.58^0\ A$$

Knowing the current I_G, we can work back and find I_{line} and I_{load}, or:

$$I_{line} = \frac{1}{10} x I_G = \frac{1}{10}\left(98.29\angle -26.58^0\right) = 9.829\angle -26.58^0\ A$$

And:

$$I_{load} = 10 x I_{line} = 10x\left(9.829\angle -26.58^0\right) = 98.29\angle -26.58^0\ A$$

The load voltage is:

$$V_{load} = Z_{load} x I_{load} = (2 + j1)\left(98.29\angle -26.58^0\right) = \left(2.236\angle 26.57^0\right)\left(98.29\angle -26.58^0\right)$$
$$= 219.777\angle -0.01^0 V$$

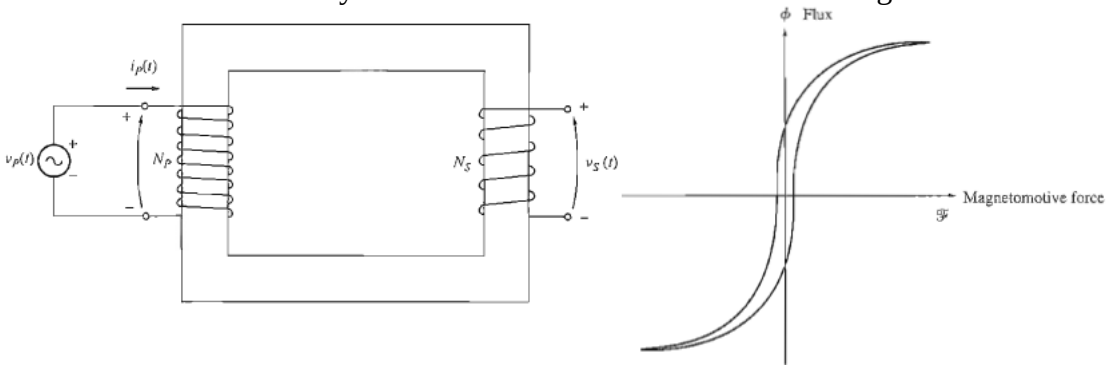
and the line losses are:

$$P_{loss} = \left(I_{line}\right)^2 . R_{line} = \left(9.829\right)^2 x0.1 = 9.66W$$

Notice that raising the transmission voltage of the power system reduced transmission losses. Also, the voltage at the load dropped much less in the system with transformers compared to the system without transformers. This example illustrates the advantages of using higher-voltage transmission lines as well as the extreme importance of transformers in modern power systems.

THEORY OF OPERATION OF REAL SINGLE· PHASE TRANSFORMERS

The real transformer is two or more coils of wire physically wrapped around a ferromagnetic core.
The figure shows a transformer consisting of two coils of wire wrapped around a transformer core and the hysteresis curve of the transformer ferromagnetic material.



It is known that:

$$e_{ind} = \frac{d\lambda}{dt}$$

Where:

$$\lambda = \sum_{i=1}^{N} \phi_i$$

Because the flux passing through each turn of a coil is different from the flux in the other turns (because of leakage), thus:

$\lambda = \sum_{i=1}^{N} \phi_i \neq N.\phi$, where N is the number of turns in the coil.

However, it is possible to define an average flux per turn in a coil.

$$\bar{\phi} = \frac{\lambda}{N}$$

and Faraday's law can be written as:

$$e_{ind} = N\frac{d\bar{\phi}}{dt}$$

The Voltage Ratio across a Transformer

The flux in the primary coil of the transformer is divided into two components:

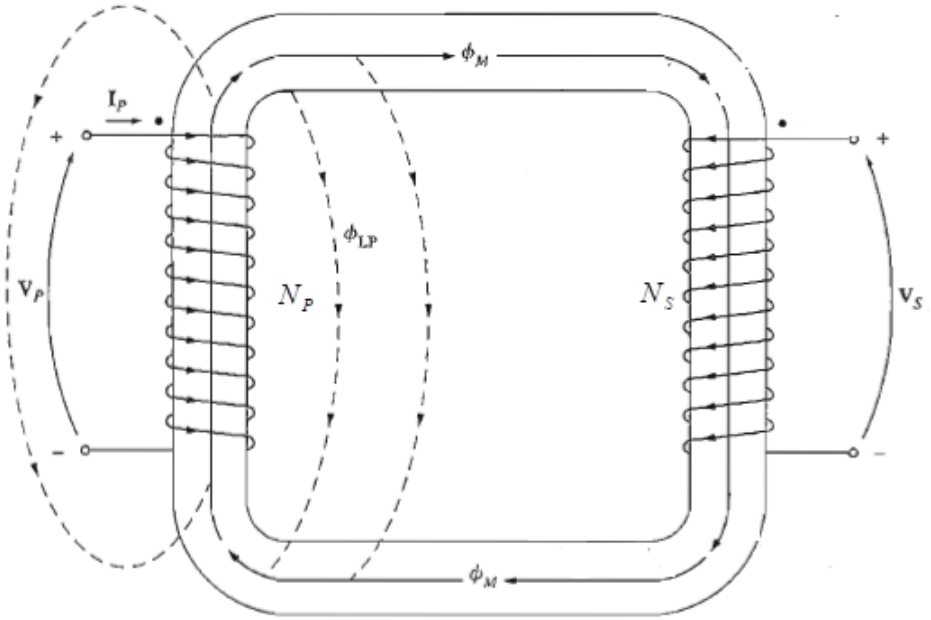
- Mutual flux (ϕ_M), which remains in the core and links both windings, (called also magneizing flux).

- Leakage flux (ϕ_{LP}), which passes through the primary winding but returns through the air, bypassing the secondary winding.

Or:

$$\bar{\phi}_P = \phi_M + \phi_{LP}$$

See the figure below.



A similar division of flux in the secondary winding can be done, or the flux in the secondary winding has two components:

- The mutual flux (ϕ_M)
- The leakage flux (ϕ_{LS}), which passes through the secondary winding but returns through the air and bypassing the primary winding.

Or:

$$\bar{\phi}_S = \phi_M + \phi_{LS}$$

Applying Faraday's law to the primary circuit:

$$v_P(t) = N_P \frac{d\bar{\phi}_P}{dt} = N_P \frac{d\phi_M}{dt} + N_P \frac{d\phi_{LP}}{dt}$$

Or:

$$v_P(t) = e_P(t) + e_{LP}(t)$$

Or the voltage of the primary winding is the sum of the induced voltage by the mutual flux and the induced voltage by the leakage flux.

Applying Faraday's law to the secondary circuit:

$$v_S(t) = N_S \frac{d\bar{\phi}_S}{dt} = N_S \frac{d\phi_M}{dt} + N_S \frac{d\phi_{LS}}{dt}$$

Or:

$$v_S(t) = e_S(t) + e_{LS}(t)$$

Or the voltage of the secondary winding is also the sum of the induced voltage by the mutual flux and the induced voltage by the leakage flux.

The primary voltage due to the mutual flux is:

$$e_P(t) = N_P \frac{d\phi_M}{dt}$$

The secondary voltage due to the mutual flux is:

$$e_S(t) = N_S \frac{d\phi_M}{dt}$$

From these two equations:

$$\frac{e_P(t)}{e_S(t)} = \frac{N_P \dfrac{d\phi_M}{dt}}{N_S \dfrac{d\phi_M}{dt}} = \frac{N_P}{N_S} = a$$

in a well-designed transformer $\phi_M >> \phi_{LP}$ and $\phi_M >> \phi_{LS}$, or:

$$v_P(t) = N_P \frac{d\bar{\phi}_P}{dt} \approx N_P \frac{d\phi_M}{dt} \approx e_P(t)$$

$$v_S(t) = N_S \frac{d\bar{\phi}_S}{dt} \approx N_S \frac{d\phi_M}{dt} \approx e_S(t)$$

Or:

$$\frac{v_P(t)}{v_S(t)} \approx \frac{e_P(t)}{e_S(t)} = a$$

The Magnetization Current in a Real Transforme

If the transformer is supplied by an AC source, a current flows in its primary circuit, even when the secondary circuit is open-circuited. This current has two components:

1) The magnetization component, i_M , which is required to produce the flux in the transformer core.

2) The core-loss current component, i_{h+e} which is required to make up for hysteresis and eddy current losses in the core.

The magnetization component, i_M

We know that the relation between the flux, ϕ , through a given coil and the voltage, v produced between its terminals is described by Faraday's law, or:

$$v(t) = \frac{d\lambda(t)}{dt} = N \frac{d\bar{\phi}(t)}{dt}$$

Or,

$$\bar{\phi}(t) = \frac{1}{N}\int v(t).dt$$

Thus, if the voltage applied to the primary winding of a transformer is given by:

$$v_P(t) = V_m \cos \omega t$$

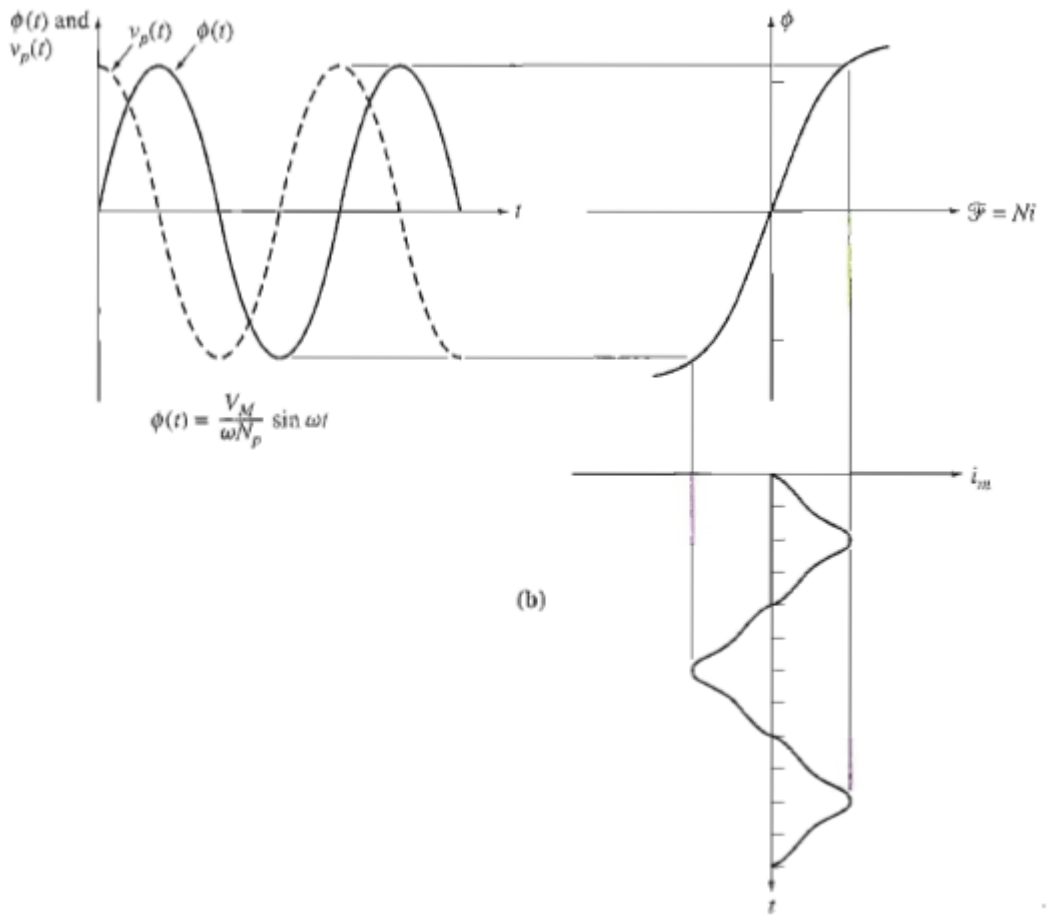
then the resulting flux will be:

$$\bar{\phi}(t) = \frac{1}{N}\int v(t).dt = \frac{1}{N}\int V_m \cos \omega t.dt = \frac{V_m}{N\omega} \sin \omega t$$

The current required to produce this flux can be determined from the magnetization curve of the core material.

Or the steps for drawing the magnetizing current passing in the primary winding when is supplied by a given voltage waveform is as follows:

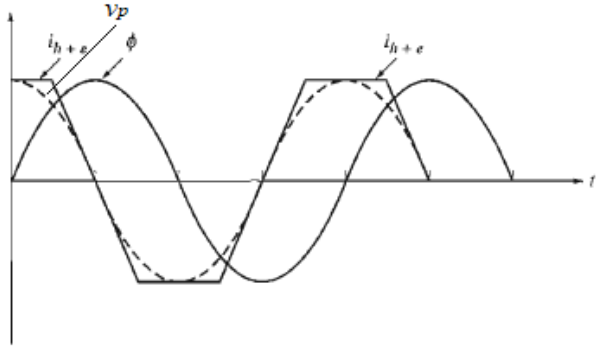
From the voltage v, find $\bar{\phi}(t)$, then from the magnetization curve of the core material find the required current i_M, as shown below:



Notice the following:
1) The magnetization current in the transformer is not sinusoidal. It has higher frequency components, which are due to saturation in the transfonner core.
2) Once the peak flu x reaches the saturation point in the core, a small increase in peak flux requires a very large increase in the peak magnetization current.
3) The fundamental component of the magnetization current lags the voltage applied to the core winding by 90°.

The core-loss current component, i_{h+e}

This component is required to supply power to make up the hysteresis and eddy current losses in the core.
As this current component is producing active power losses, thus its in phase with the primary supply voltage, as shown below.



Notice the following:
1) The core-loss current is nonlinear because of the nonlinear effects of hysteresis.
2) The fundamental component of the core-loss current is in phase with the voltage applied to the core.

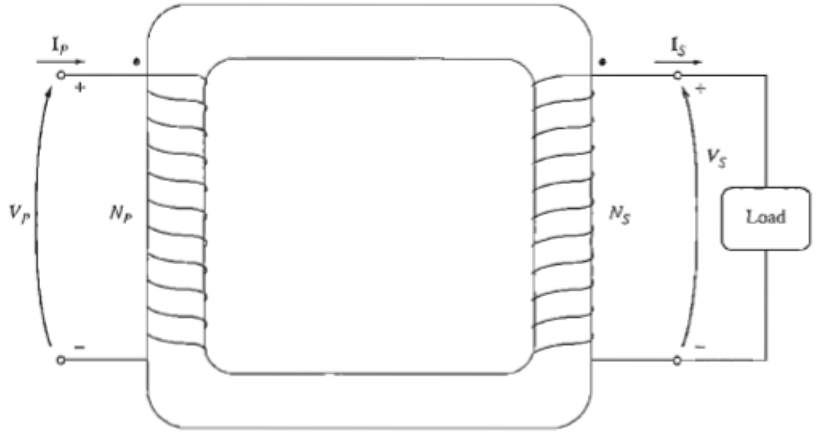
The total no-load current in the core is called the excitation current of the transformer. It is just the sum of the magnetization current and the core-loss current in the core:

$$i_{ex} = i_M + i_{h+e}$$

In a well-designed power transformer, the excitation current is much smaller than the full-load current of the transformer.

The Cnrrent Ratio on a Transformer and the Dot Convention

When a load is connected to the secondary of the transformer, a current will flow out of the secondary winding to supply the load, as shown below.



Notice the dots on the windings of the transformer which help to determine the polarity of the voltages and currents in the core. The physical significance of the dot convention is that a current flowing into the dotted end of a winding, I_P, produces a positive magnetomotiveforce while a current flowing out of the dotted end of a winding, I_S, produces a negative magnetomotive force.

The net magnetomotive force on the core will be:

$$F_{M,net} = N_P.i_P - N_S.i_S$$

This net magnetomotive force will produce the net fl ux in the core, or:

$$F_{M,net} = N_P.i_P - N_S.i_S = \phi.R_m$$

Where R_m is the magnetic reluctance of the core.

As the reluctance of a well-designed transformer core is very small (nearly zero) until the core is saturated, the relationship between the primary and secondary currents is approximately:

$$F_{M,net} = N_P.i_P - N_S.i_S \approx 0$$

Or:

$$N_P.i_P \approx N_S.i_S$$

Or:

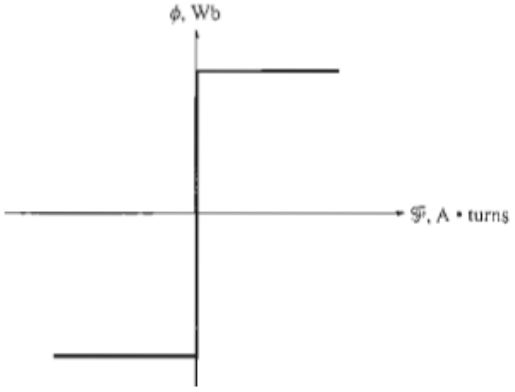
$$\frac{i_P}{i_S} \approx \frac{N_S}{N_P} = \frac{1}{a}$$

THE EQUIVALENT CIRCUIT OF A TRANSFORMER

As can be found from the previous descussion, the ideal transformer has the following assumptions:

1) The core have no hysteresis or eddy currents.

2) The magnetization curve must have the shape shown below.



Notice that for an unsaturated core the net magnetomotive force $F_{M,net} = 0$ implying that:

$N_P.i_P = N_S.i_S$

3) The leakage flux in the core is zero, implying that all the flux in the core couples both windings.

4) The resistances of the transformer windings are zeros.

The following items have to be taken into consideration in studying real transformers:

1) Copper (I^2R) losses. Copper losses are the resistive heating losses in the primary and secondary windings of the transformer. They are proportional to the square of the current in the windings.

2) Eddy current losses. Eddy current losses are resistive heating losses in the core of the transformer. They are proportional to the square of the voltage applied to the transformer.

3) Hysteresis Losses. Hysteresis losses are associated with the rearrangement of the magnetic domains in the core during each half-cycle. They are a complex, nonlinear function of the voltage applied to the transformer.

4) Leakage flux. The fluxes ϕ_{LP} and ϕ_{LS} which escape the core and pass through only one of the transformer windings are leakage fluxes. These escaped fluxes produce a leakage inductance in the primary and secondary coils.

The Exact Equivalent Circuit of a Real Transformer

The equivalent circuit of the transformer (transformer model) must take into account all the major imperfections in real transformers.

- Copper losses are resistive losses in the primary and secondary windings of the transformer core. They are modeled by placing a resistor R_P in the primary circuit of the transformer and a resistor R_S in the secondary circuit.

- The leakage flux in the primary windings ϕ_{LP} produces a voltage e_{LP} given by:

$$e_{LP}(t) = N_P \frac{d\phi_{LP}}{dt} \qquad (1)$$

The flux ϕ_{LP} is directly proportional to the primary circuit current i_P, or:

$$\phi_{LP} = \frac{N_P.i_P}{R_m} = N_P.i_P.G_m \qquad (2)$$

Where R_m is the magnetic reluctance and G_m is the magnetic permeance of the leakage flux path, which is mostly in air.

Substituting (2) in (1), we have:

$$e_{LP}(t) = N_P \frac{d}{dt}\left(N_P.i_P.G_m\right) = \left(N_P\right)^2 G_m \frac{di_P}{dt} = L_P \frac{di_P}{dt}$$

Here:

$L_P = \left(N_P\right)^2 G_m = \frac{\left(N_P\right)^2}{R_m}$ is the leakage inductance of the primary winding.

- In the same way, the leakage flux in the secondary windings ϕ_{LS} produces a voltage e_{LS} given by:

$$e_{LS}(t) = N_S \frac{d\phi_{LS}}{dt} \qquad (1)$$

The flux ϕ_{LS} is directly proportional to the secondary circuit current i_S, or:

$$\phi_{LS} = \frac{N_S.i_S}{R_m} = N_S.i_S.G_m \qquad (2)$$

Where R_m is the magnetic reluctance and G_m is the manetic permeance of the leakage flux path, which is mostly in air.

Substituting (2) in (1), we have:

$$e_{LS}(t) = N_S \frac{d}{dt}\left(N_S.i_S.G_m\right) = \left(N_S\right)^2 G_m \frac{di_S}{dt} = L_S \frac{di_S}{dt}$$

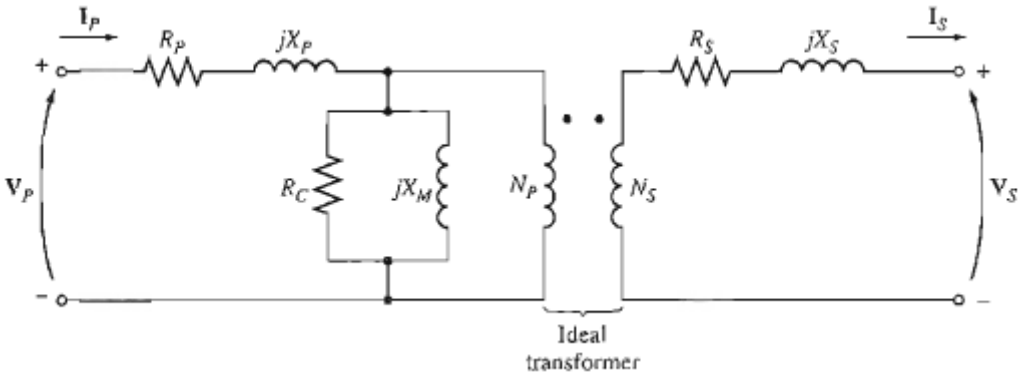
Here:

$L_S = \left(N_S\right)^2 G_m = \frac{\left(N_S\right)^2}{R_m}$ is the leakage inductance of the secondary winding.

- The magnetization current i_m is proportional (in the unsaturated region) to the voltage applied to the primary winding and lagging the applied voltage by 90^0, so it can be modeled by a reactance X_M connected across the primary voltage source.

- The core-loss current i_{h+e} is a current proportional to the voltage applied to the primary winding and is in phase with it, so it can be modeled by a resistance R_C connected across the primary voltage source.

After taking all the previous items into consideration, the resulting equivalent circuit is shown in the figure below.



In this figure:

R_P is the resistance of the primary winding.

$X_P = \omega L_P$ is the reactance due to the primary leakage inductance

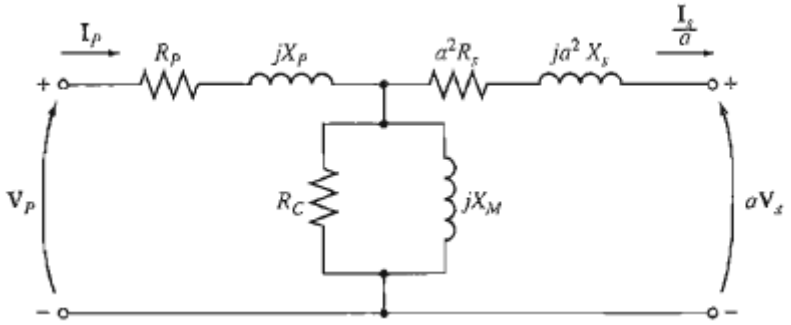
R_S is the resistance of the secondary winding,

$X_S = \omega L_S$ is the reactance due to the secondary leakage inductance.

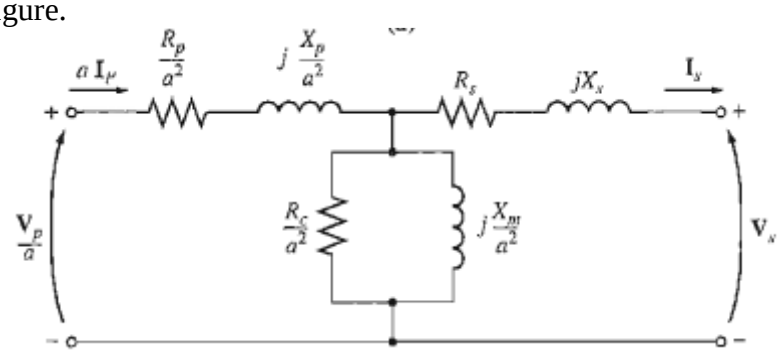
R_C reflects hysteresis and eddy current losses in the core.

X_M is the magnetizing reactance of the transformer.

Usually to analyze practical circuits containing transformers, the entire transformer circuit is refered to a single voltage level. The equivalent transformer circuit refered to the primary side is shown below.



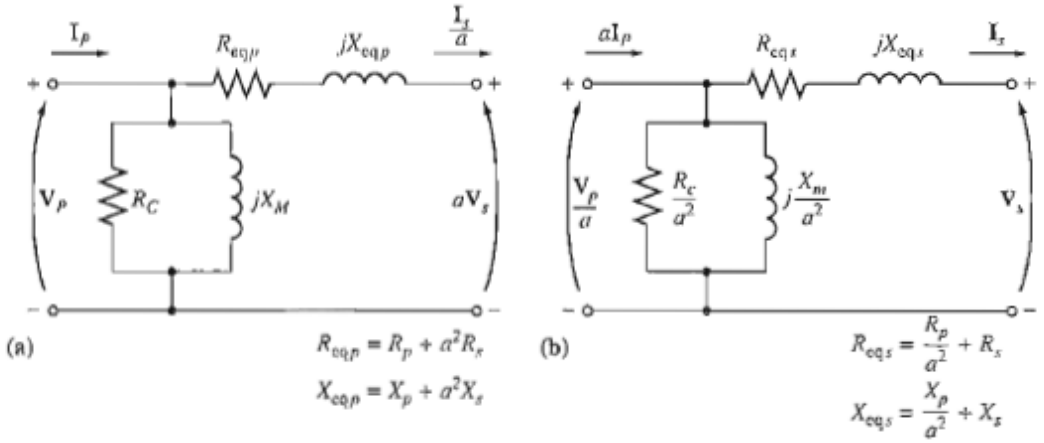
While the equivalent transformer circuit refered to the secondary side is shown in the following figure.



Approximate Equivalent Circuits of a Transformer

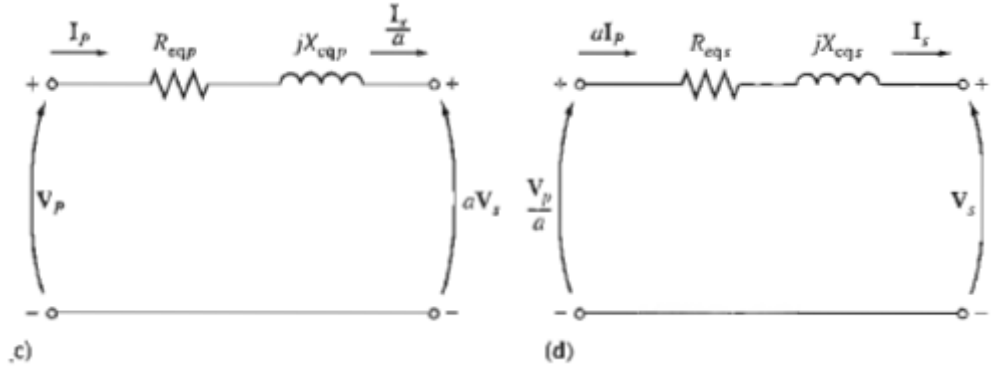
The transformer models shown before are relatively exact by are little complex for analysis.

For practical engineering applications, this circuit can be simplified by removing the excitation branch to the front of the transformer, as shown in the figure below.



As seen from the figure, the primary and secondary impedances becom in series with each other and their equivalent is found by just adding their impedances.
This modification is justified because the excitation branch has a very small current compared to the load current of the transformers. In fact, the excitation current is only about 2-3% of the full load current for typical power transformers.

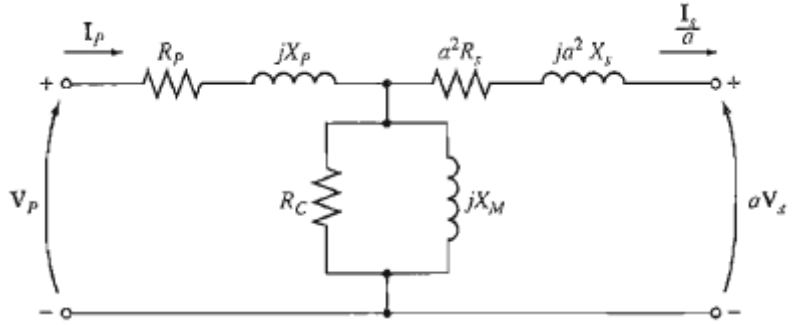
Further simplification can be done by neglecting entirely the excitation branch as in the simplified e circuits, shown below.



Determining the Valnes of Components in the Transformer Model

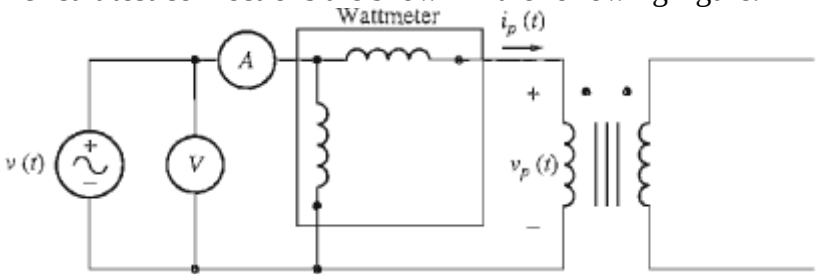
The values of the resistances and inductances in the transformer equivalent circuis can be found by the open-circuit test and the short-circuit test.

The open-circuit test: in this test, one of the transformer winding is open-circuited, and the other winding is connected to full rated supply voltage, see the circuit below.



As seen from the transformer equivalent circuit, the input current flows through the excitation branch of the transformer. The series elements, R_P and X_P are too small in comparison to R_C and X_M, so essentially all the input voltage is dropped across the excitation branch.

The open-circuit test connections are shown in the following figure.



Usually, full line voltage is applied to one side of the transformer, and the input voltage, input current, and input power to the transformer are measured.

From the figure, we conclude that,:

- the magnitude of conductance of the excitation branch is:

$$|Y_M| = \frac{I_{OC}}{V_{OC}}$$

- The power factor angle of magnetization branch is:

$$\theta_{OC} = \cos^{-1}\left(\frac{P_{OC}}{V_{OC}.I_{OC}}\right)$$

Notice that the power factor is always lagging for a real transformer, so the angle of the current always lags the angle of the voltage by θ degrees.

We know that the conductance of the excitation branch is:

$$Y_M = \frac{1}{R_C} - j\frac{1}{X_M} = G_C - jB_M$$

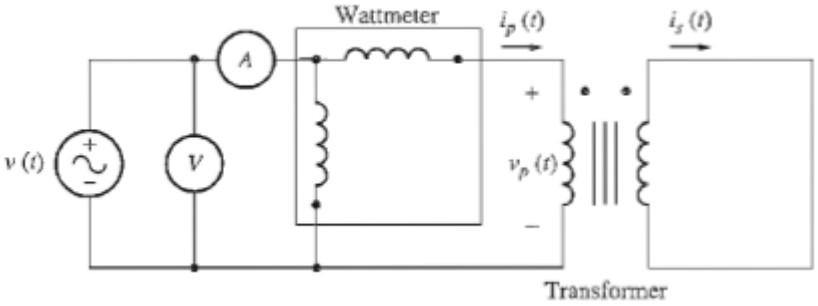
From this, we find:

$$\frac{1}{R_C} = G_C = \mathrm{Re}\{Y_M\} = |Y_M|.\cos\theta_{OC} = \frac{I_{OC}}{V_{OC}}\cos\theta_{OC}$$

And:

$$\frac{1}{X_M} = B_M = -\mathrm{Im}\{Y_M\} = -|Y_M|.\sin\theta_{OC} = -\frac{I_{OC}}{V_{OC}}\sin\theta_{OC}$$

The short-circuit test: In the short-circuit test, the low-voltage terminals of the transformer are short-circuited, and the high-voltage tenninals are connected to a variable voltage source, as shown below.



The input voltage is adjusted until the current in the short-circuited windings is equal to its rated value. The input voltage, current and power are again measured.

Since the input voltage is so low during the short-circuit test, negligible current flows through the excitation branch. If the excitation current is ignored, then all the voltage drop in the transfonner can be attributed to the series elements in the circuit.
The magnitude of the series impedances referred to the primary side of the transformer is:

$$|Z_{CS}| = \frac{V_{SC}}{I_{SC}}$$

- The power factor angle of circuit in the short-circuit test is:

$$\theta_{SC} = \cos^{-1}\left(\frac{P_{SC}}{V_{SC}.I_{SC}}\right)$$

Notice that the power factor is always lagging for a real transformer, so the angle of the current always lags the angle of the voltage by θ degrees.
We know that, in the short-circuit test:

$$Z_{CS} = Z_P + a^2 Z_S = \left(R_P + a^2 R_S\right) + j\left(X_P + a^2 X_S\right)$$

Thus:

$$R_P + a^2 R_S = \mathrm{Re}\{Z_{SC}\} = |Z_{SC}|.\cos\theta_{SC} = \frac{V_{SC}}{I_{SC}}\cos\theta_{SC} \qquad (1)$$

And:

$$X_P + a^2 X_S = \mathrm{Im}\{Z_{SC}\} = |Z_{SC}|.\sin\theta_{SC} = \frac{V_{SC}}{I_{SC}}\sin\theta_{SC} \qquad (2)$$

For some simplification, the resistance of the primary winding , R_P , can be estimated by measuring it by an ohmmeter while the seconary winding is open-circuited. After finding R_P , the resistance of the secondary winding refered to primary, $R_S^{'} = a^2 R_S$ can be found from (1), as the transformation ratio a is usually knowown, or easily can be found, the resistance R_S is found.

As regard to the reactances of the primary and secondary windings, and for practical transformers, X_P and $X_S^{'} = a^2 X_S$ are realively equal, as each of them can easily be found using (2).

Example:
Build the equivalent circuit of a $20kVA$, $8000/240V$, $60Hz$ transformer, knowing that the open-circuit test was performed on the secondary side of the transformer (to reduce the maximum voltage to be measured) and the short-circuit test were performed on the primary side of the transformer (to reduce the maximum current to be measured). The following data were taken:

Open-circuit test (on secondary)	**Short-circuit test (on primary)**
$V_{OC} = 240V$	$V_{SC} = 489V$
$I_{OC} = 7.133A$	$I_{SC} = 2.5A$
$P_{OC} = 400W$	$P_{SC} = 240W$

Solution:
The transformation ratio of the transformer is:

$$a = \frac{V_P}{V_S} = \frac{8000}{240} = 33.33$$

From the open-circuit test:
The power factor angle of magnetization branch is:

$$\theta_{OC} = \cos^{-1}\left(\frac{P_{OC}}{V_{OC}.I_{OC}}\right) = \cos^{-1}\left(\frac{400}{240x7.133}\right) = 76.5^0$$

The magnitude of conductance of the excitation branch is:

$$|Y_M| = \frac{I_{OC}}{V_{OC}} = \frac{7.133}{240} = 0.0297\Omega$$

$$Y_M = |Y_M| \angle \theta_{OC} = 0.0297 \angle -76.5^0 = 0.0297\cos(76.5^0) - j0.0297\sin(76.5^0)$$
$$= 0.00693 - j0.02888$$

But:

$$Y_M = \frac{1}{R_C} - j\frac{1}{X_M}$$

Thus:

$$\frac{1}{R_C} = 0.00693 \text{ and } R_C = \frac{1}{0.00693} = 144\Omega$$

$$\frac{1}{X_M} = 0.02888 \text{ and } X_M = \frac{1}{0.02888} = 34.63\Omega$$

From the short-circuit test:
- The power factor angle is:

$$\theta_{SC} = \cos^{-1}\left(\frac{240}{489x2.5}\right) = 78.7^0$$

The magnitude of the series impedances is:

$$|Z_{CS}| = \frac{V_{SC}}{I_{SC}} = \frac{489}{2.5} = 195.6\Omega$$

$$Z_{CS} = |Z_{CS}|\angle\theta_{SC} = 195.6\angle 78.7^0 = 195.6\cos(78.7^0) + j195.6\sin(78.7^0)$$

$$= 38.4 + j192 \quad \Omega$$

But:

$$Z_{CS} = Z_P + a^2 Z_S = (R_P + a^2 R_S) + j(X_P + a^2 X_S) = R_{eq} + jX_{eq}$$

Thus:

$$R_{eq} = 38.4\Omega \quad \text{and} \quad X_{eq} = 192\Omega$$

Therefore, the equivalent resistance and reactance referred to the high-voltage (primary) side are:

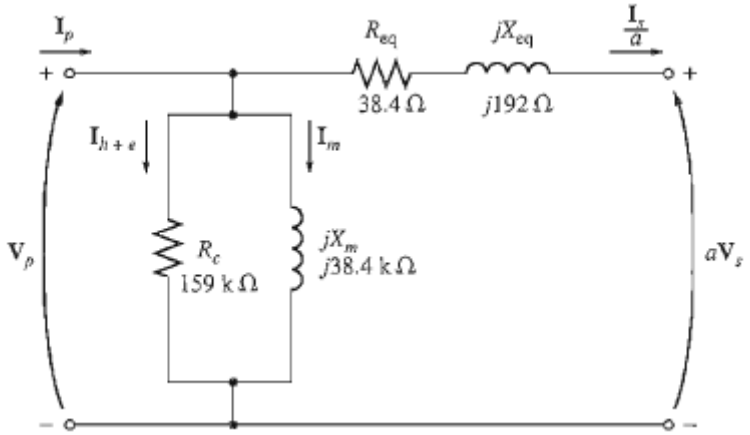
$$R_{eq} = 38.4\Omega \quad \text{and} \quad X_{eq} = 192\Omega$$

The resulting simplified equivalent circuit referred to the high-voltage (primary) side can be found by converting the excitation branch values to the high-voltage side.
Or:

$$R_{C,P} = a^2 R_{C,S} = (33.33)^2(144) = 159k\Omega$$

$$X_{M,P} = a^2 X_{M,S} = (33.33)^2(34.63) = 38.4k\Omega$$

The resulting equivalent circuit is shown in the figure below.



THE PER-UNIT SYSTEM OF MEASUREMENTS

Circuits containing transformers can be solved easily using the per-unit (pu) system of measurements. This method provides the following advantages:

1- per unit data provides relative magnitude information.
2- simplifies the analysis of circuits containing transformers. The per unit impedance of a transformer in three-phase system is the same, regardless of the type of the winding connections (star-delta, delta-star, stat-star, or delta-delta). The per unit impedance of a transformer is the same when referred to either the primary or the secondary side,
3- circuit parameter are in narrow ranges (around 1 p.u) and this makes error checking easier.
4- Manufacturers usually specify the impedance of an equipment in per unit on the base of its nameplate rating.
5- The per unit impedance values of various ratings of equipment lie in a narrow range, (whereas the actual ohmic values may vary widely). Therefore when

actual values are unknown, a good approximate (typical) value of similar equipment can be used.

This method is suitable for systems with different voltage levels (sections connected with transformers).
In this method base values are selected, and all the actual values are referred to their respective base values, as:

$$value\,in\ p.u = \frac{actual\,value}{base\,value}$$

normally two base values, base VA, $(VA)_B$ and the base voltage, V_B are chosen.
then for each section V_B is found from the transformers ratios. then the base current I_B and the base impedance Z_B are calculated for each section. the p.u values then are found.

For the case of single phase systems:

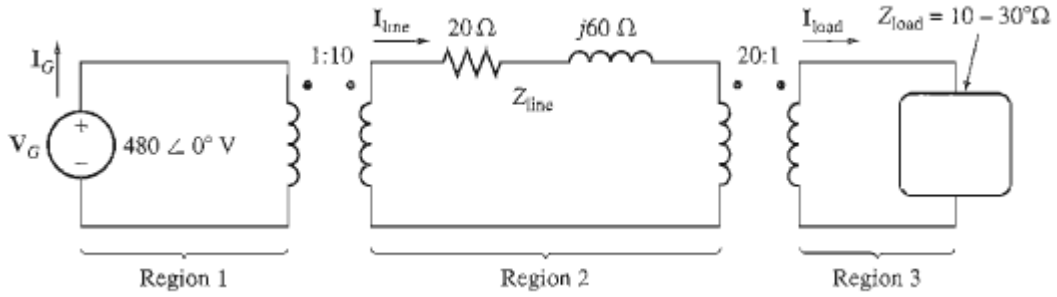
- select the base VA, S_{base}
- select V_{base} and calculate it for each section.
- the base current for each section is calculated as:

$$I_B = \frac{S_{base}}{V_{base}}$$

- the base impedance for each section is calculated as:
- $$Z_B = \frac{V_{base}}{I_{base}} = \frac{(V_{base})^2}{S_{base}}$$

Example:
A simple power system shown below contains a $480V$ generator connected to an ideal $1:10$ step-up transformer, a transmission line, an ideal $20:1$ step-down transformer, and a load. The impedance of the transmission line is $20 + j60\Omega$ and the impedance of the load is $10\angle 30^0\ \Omega$. The base values for this system are chosen to be $480V$ and $10kVA$ at the generator.

a) Find the base voltage, current, impedance, and apparent power at every point in the power system.
b) Convert this system to its per-unit equivalent circuit.
c) Find the power supplied to the load in this system.
d) Find the power lost in the transmission line.



Solution:
a)
In the generator region: $S_{base} = 10kVA$, $V_{base1} = 480V$, thus:

$$I_{base1} = \frac{S_{base}}{V_{base1}} = \frac{10000}{480} = 20.83A$$

$$Z_{base1} = \frac{V_{base1}}{I_{base1}} = \frac{480}{20.83} = 23.04\Omega$$

In the transmission line region: $S_{base} = 10kVA$, $V_{base2} = 4800V$, thus:

$$I_{base2} = \frac{S_{base}}{V_{base2}} = \frac{10000}{4800} = 2.083A$$

$$Z_{base2} = \frac{V_{base2}}{I_{base2}} = \frac{4800}{2.083} = 2304\Omega$$

In the load region: $S_{base} = 10kVA$, $V_{base3} = 240V$, thus:

$$I_{base3} = \frac{S_{base}}{V_{base3}} = \frac{10000}{240} = 41.67A$$

$$Z_{base3} = \frac{V_{base3}}{I_{base3}} = \frac{240}{41.67} = 5.76\Omega$$

b) the per-unit values of the circuit parameters are:
the generator voltage in p.u:

$$V_{G.pu} = \frac{V_{G,actual}}{V_{base1}} = \frac{480}{480} = 1.0\,p.u$$

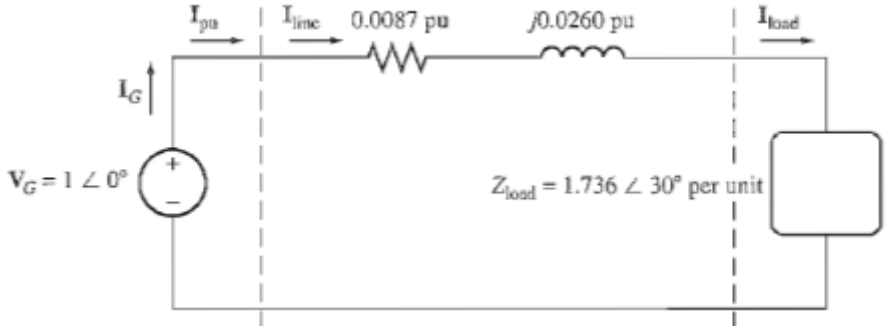
the impedance of the transmission line in p.u:

$$Z_{Line.pu} = \frac{Z_{Line,actual}}{Z_{base2}} = \frac{20 + j60}{2304} = 0.0087 + j0.026\,p.u$$

the impedance of the load in p.u:

$$Z_{Load.pu} = \frac{Z_{Load,actual}}{Z_{base3}} = \frac{10\angle 30^0}{5.76} = 1.736\angle 30^0\ p.u = 1.503 + j0.868 \quad p.u$$

The per-unit equivalent circuit of the power system is shown below.



c) From this circuit:

$$I_{G.pu} = I_{Line.pu} = I_{Load.pu}\frac{V_{G,pu}}{Z_{line.pu} + Z_{load.pu}} = \frac{1\angle 0^0}{0.0087 + j0.026 + 1.736\angle 30^0} = 0.569\angle -30.6^0\ p.u$$

The per-unit power of the load is:

$$P_{Load.pu} = \left(I_{Load.pu}\right)^2.R_{Load,p.u} = (0.569)^2(1.503) = 0.487\,p.u = 0.487x10000 = 4870W$$

d) The per-unit power loss in the transmission line is:

$$P_{Line.pu} = \left(I_{Line.pu}\right)^2 .R_{Line,p.u} = (0.569)^2 (0.0087) = 0.00282\,p.u = 0.00282x10000 = 28.2W$$

TRANSFORMER VOLTAGE REGULATION AND EFFICIENCY

The series impedances in the primary and secondary winding of the transformer are the reason that the output voltage of a transformer varies with the load even if the input voltage remains constant.

Full-load voltage regulation is a quantity that compares the output voltage of the transformer at no load with the output voltage at full load . It is defined as:

$$VR = \frac{V_{S,nl} - V_{S,fl}}{V_{S,fl}} x100\%$$

As at no load, $V_{S,nl} = \frac{V_P}{a}$, the voltage regulation can also be expressed as:

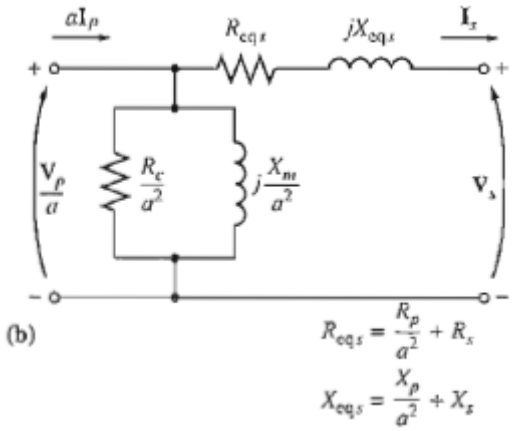
$$VR = \frac{\frac{V_P}{a} - V_{S,fl}}{V_{S,fl}} x100\% = \frac{V_P - a.V_{S,fl}}{a.V_{S,fl}} x100\%$$

The voltage regulation in per-unit can be expressed as:

$$VR = \frac{V_{P.pu} - V_{S,fl.pu}}{V_{S,fl.pu}} x100\%$$

For an ideal transformer, as the series impedances of the two transformer windings are zeros, thus: $V_{P.pu} = V_{S,fl.pu}$ and $VR = 0$.

Consider the simplified transformer equivalent circuit, shown below.



If the effects of the excitation branch on transformer voltage regulation is ignored, so only the series impedances need be considered. The phasor diagram of this equivalent circuit, at a lagging power factor, is shown below.

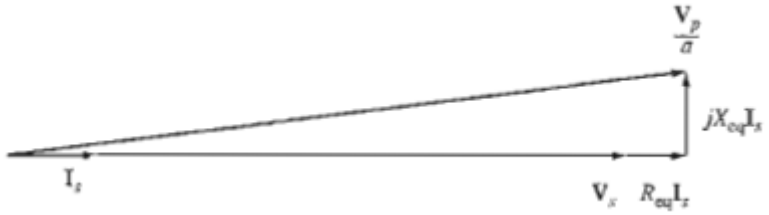


Note for equivalent circuit:

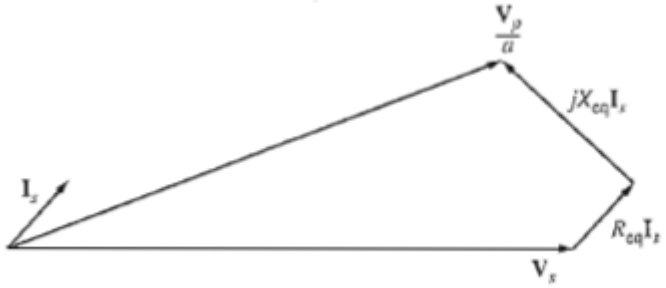
$$\frac{V_P}{a} = V_S + R_{eq} I_S + jX_{eq} I_S$$

From the phasor diagram, it is easy to see that, for lagging loads, $\frac{V_P}{a} > V_S$ and the voltage regulation of a transformer with lagging loads must be greater zero.

The phasor diagram at unity power factor is shown in the next Figure. Here again. the voltage at the secondary is lower than the voltage at the primary, so VR > O, but is smaller number than it was with a lagging current.



The phasor diagram of a transformer with a leading current is shown below.



Notice that the secondary voltage can actually be higher than the referred primary voltage, or the transformer actually has a negative voltage regulation.

Transformer Efficiency

The efficiency of a device is defined as:

$$\eta = \frac{P_{out}}{P_{in}} x100\% = \frac{P_{out}}{P_{out} + P_{loss}} x100\%$$

Transformer losses can be calculated using its equivalent circuit. There are three types of losses present in transformers:

1) Copper ($P_{Cu} = I^2 R$) losses. These losses are accounted for by the series resistance in the equivalent circuit.
2) Hysteresis losses.
3) Eddy current losses.

Notice that hysterisis and eddy current losses are included in resistor R_C , and their sum is called the core losses (P_{core})

The output power of the transformer is given as:

$$P_{out} = V_S . I_S . \cos\theta_S$$

Thus the efficiency of the transformer can be given as:

$$\eta = \frac{V_S . I_S . \cos\theta_S}{V_S . I_S . \cos\theta_S + P_{Cu} + P_{core}} x100\%$$

Example:

A $15kVA,\ 2300/230$ transformer is tested to build its equivalent circuit. The following are the results of the open-circuit and short-circuit tests of the transformer.

Open-circuit test (low-voltage side)	**Short-circuit test (high-voltage side)**
$V_{OC} = 230V$	$V_{SC} = 47V$
$I_{OC} = 2.1A$	$I_{SC} = 6.0A$
$P_{OC} = 50W$	$P_{SC} = 160W$

a) Find the equivalent circuit of this transformer referred to the high-voltage side.
b) Find the equivalent circuit of this transformer referred to the low-voltage side.
c) Calculate the full-load voltage regulation at 0.8 lagging power factor, 1.0 power factor, and at 0.8 leading power factor using the exact equation for V_P.
d) Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8 lagging, 1.0, and 0.8 leading.
e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Solution:
The turns ratio of this transformer is $a = 2300/230 = 10$
a)
From the open-circuit test data, the open-circuit impedance angle is:

$$\theta_{OC} = \cos^{-1}\left(\frac{P_{OC}}{V_{OC}.I_{OC}}\right) = \cos^{-1}\left(\frac{50}{230x2.1}\right) = 84^0$$

The magnitude of the excitation admittance is:

$$|Y_E| = \frac{I_{OC}}{V_{OC}} = \frac{2.1}{230} = 0.00913S$$

Or:

$$Y_E = |Y_E|\angle -\theta_{OC} = 0.00913\angle -84^0 = 0.00913\left(\cos 84^0 - j\sin 84^0\right) = 0.000954 - j0.00908$$

but:

$$Y_E = \frac{1}{R_C} - j\frac{1}{X_M}$$

Thus:

$$R_C = \frac{1}{0.000954} = 1050\Omega \text{ and } X_M = \frac{1}{0.00908}110\Omega$$

From the short-circuit test data, the short-circuit impedance angle is:

$$\theta_{SC} = \cos^{-1}\left(\frac{P_{SC}}{V_{SC}.I_{SC}}\right) = \cos^{-1}\left(\frac{160}{47x6}\right) = 55.4^0$$

The magnitude of the equivalent series impedance is:

$$|Z_{SE}| = \frac{V_{SC}}{I_{SC}} = \frac{47}{6} = 7.833\Omega$$

Or:

$$Z_{SE} = |Z_{SE}|\angle\theta_{SC} = 7.833\angle 55.4^0 = 7.833\left(\cos 55.4^0 + j\sin 55.4^0\right) = 4.45 + j6.45\Omega$$

But:

$$Z_{SE} = R_{eq,P} + jX_{eq,P}$$

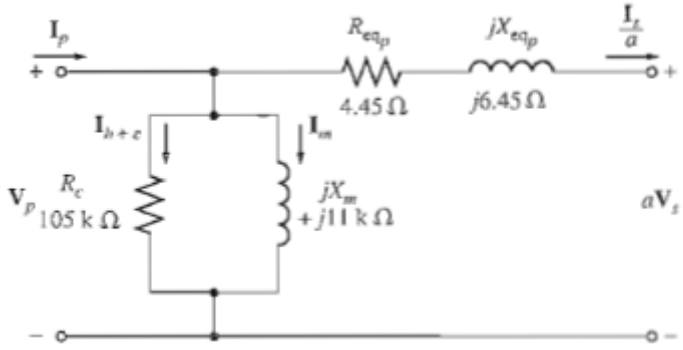
Or, the series elements referred to the primary side are:

$$R_{eq,P} = 4.45\Omega \qquad \text{and } X_{eq,P} = 6.45\Omega$$

The resulting simplified equivalent circuit referred to the primary side can be found by converting the excitation branch values to the primary side. Or:

$R_{C,P} = a^2 R_{C,S} = 10^2 x1050 = 105k\Omega$ and $X_{M,P} = a^2 X_{M,S} = 10^2 x110 = 11k\Omega$

This equivalent circuit is shown in the figure below:

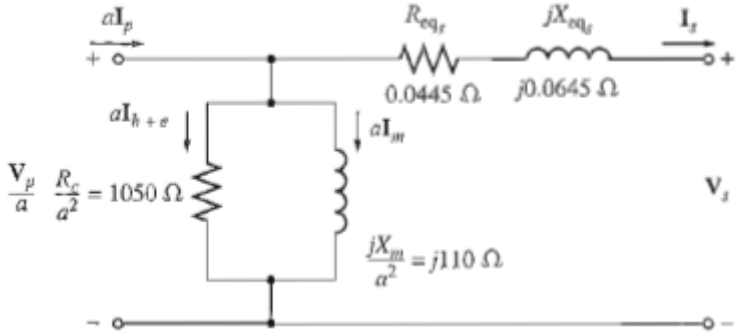


b)

To find the equivalent circuit referred to the low-voltage side, the impedance is devided by a^2, the resulting values are:

$R_C = 1050\Omega$, $X_M = 110\Omega$, $R_{eq} = 0.0445\Omega$ and $X_{eq} = 0.0645\Omega$

And the equivalent circuit is as below:



c)

The magnitude of the full-load current on the secondary side of this transformer is:

$$\left|I_{S,rated}\right| = \frac{S_{rated}}{V_{S,rated}} = \frac{15000}{230} = 65.2A$$

At PF=0.8 lagging, the power factor angle is: $\cos^{-1}(0.8) = -36.9^0$, thus:

$I_S = 65.2\angle -36.9^0 A$

From the previous equivalent circuit:

$$\frac{V_P}{a} = V_S + I_S\left(R_{eq,S} + jX_{eq,S}\right) = 230 + 65.2\angle -36.9^0(0.0445 + j0.0645)$$

$$= 234.85\angle 0.4^0$$

The resulting voltage regulation is:

$$VR = \frac{\frac{V_P}{a} - V_{S,fl}}{V_{S,fl}} = \frac{234.85 - 230}{230} x100\% = 2.1\%$$

At PF=1.0, the power factor angle is: $\cos^{-1}(1.0) = 0^0$, thus:

$I_S = 65.2\angle 0^0\, A$

And:

$$\frac{V_P}{a} = V_S + I_S\left(R_{eq,S} + jX_{eq,S}\right) = 230 + 65.2\angle 0^0\left(0.0445 + j0.0645\right)$$

$$= 232.94\angle 1.04^0$$

The resulting voltage regulation is:

$$VR = \frac{\frac{V_P}{a} - V_{S,fl}}{V_{S,fl}} = \frac{232.94 - 230}{230} x100\% = 1.28\%$$

At PF=0.8 leading, the power factor angle is: $\cos^{-1}(0.8) = 36.9^0$, thus:

$I_S = 65.2\angle 36.9^0\, A$

And:

$$\frac{V_P}{a} = V_S + I_S\left(R_{eq,S} + jX_{eq,S}\right) = 230 + 65.2\angle 36.9^0\left(0.0445 + j0.0645\right)$$

$$= 229.85\angle 1.27^0$$

The resulting voltage regulation is:

$$VR = \frac{\frac{V_P}{a} - V_{S,fl}}{V_{S,fl}} = \frac{229.85 - 230}{230} x100\% = -0.062\%$$

d)

To plot the voltage regulation as a function of load, the calculation made in (c) are repeated for many different loads, this can be done using a MATLAB program.

e)

The efficiency of the transformer is found after calculating the losses in the transformer:

- The copper losses are:

$P_{CU} = (I_S)^2 . R_{eq} = (65.2)^2 (0.0445) = 189W$

- The core losses are given by:

$$P_{core} = \frac{\left(\frac{V_P}{a}\right)^2}{R_C} . = \frac{(234.85)^2}{1050} = 52.5W$$

The output power of the transformer at this power factor is:

$P_{out} = V_S . I_S \cos\theta = (230)(62.5)(\cos 36.9^0) = 12000W$

And the efficiency of the transformer at this condition is:

$$\eta = \frac{P_{out}}{P_{out} + P_{Cu} + P_{core}} = \frac{12000}{12000 + 189 + 52.5} = 98.03\%$$

TRANSFORMER TAPS AND VOLTAGE REGULATION

In almost all real distribution transformers a series of taps in the windings are existing. These taps are switched using a manual or an automaic tap-changer to permit small changes in the turns ratio of the transformer. These taps in most of the cases are on the high voltage winding, because:

- the current to switched by the tap-changer is less
- the high voltage winding is usually the external one on the core, this makes installing the taps and the tap-changer easier.

A typical distribution transformet might have four taps in addition to the nominal setting with spacings of 2.5 percent of full-load voltage between them. Such an arrangement provides for adjustments up to 5 percent above or below the nominal voltage rating of the transformer.

Example:

A $500kVA$, $11000/400V$ distribution transformer has four 2.5 percent taps on its primary winding. What are the voltage ratios of this transformer at each tap setting?

Solution: The five possible voltage ratings of this transformer are

Nominal tap: $11000/400V$

$+5\%tap:\ 11550/400V$

$+2.5\%tap:\ 11275/400V$

$-2.5\%tap:10725/400V$

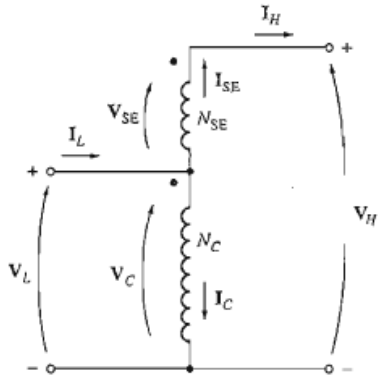
$-5\%tap:10450/400V$

Notice that the taps are on the high-voltage winding.

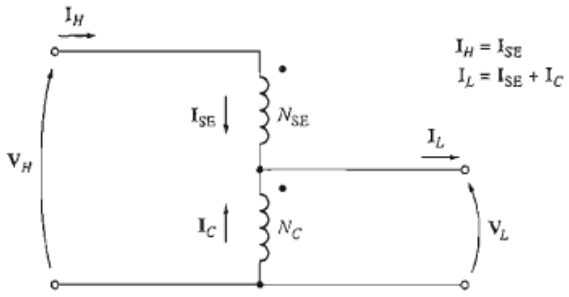
The taps on a transformer permit the transformer to be adjusted in the field to accommodate variations in local voltages. In order to keep the voltage on the load constant a tap changing under load (TCUL) transformer is used. In these transformer the tap is changing automatically.

THE AUTOTRANSFORMER

In the autotransformer the primary and the secondary windings are on the same coil, as shown below.



Step-up autotransformer

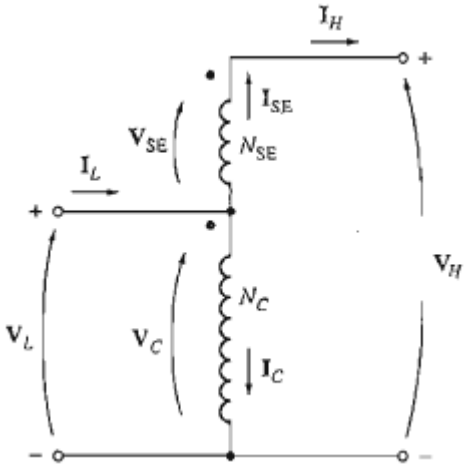


Step-down autotransformer

The voltage on the first winding and the voltage on the second winding is given by the turns ratio of the transformer.
Notice that, one transformer winding is called the common winding, because its voltage appears on both sides of the transformer. The smaller winding is called the series winding, because it is connected in series with the common winding.

Notice that the voltage on the common coil is called the common voltage V_C , and the current in that coil is called the common current I_C. The voltage on the series coil is called the series voltage V_{SE} , and the current in that coil is called the series current I_{SE} . The voltage and current on the low-voltage side of the transfonner are called V_L and I_L respectively, while the corresponding quantities on the high-voltage side are called V_H and I_H .
The primary side of the autotransformer (the side with power into it) can be either the high-voltage side or the low-voltage side, depending on whether the autotransformer is acting as a step-down or a step-up transformer.

From the figure of the step-up autotransformer, shown below.



$$\frac{V_C}{V_{SE}} = \frac{N_C}{N_{SE}}$$

$$N_C.I_C = N_{SE}.I_{SE}$$

Also:

$$V_L = V_C$$

$$V_H = V_C + V_{SE}$$

$$I_L = I_C + I_{SE}$$

$$I_H = I_{SE}$$

Voltage and Current Relationships in an Autotransformer

The relationship between V_H and V_L can be dertermined from the diagram of the step-up autotransformer preiously shown.

$$V_H = V_C + V_{SE}$$

But:

$\frac{V_C}{V_{SE}} = \frac{N_C}{N_{SE}}$, so:

$$V_H = V_C + \frac{N_{SE}}{N_C} V_C$$

But:

$$V_L = V_C$$

Thus:

$$V_H = V_L + \frac{N_{SE}}{N_C} V_L = \frac{N_C + N_{SE}}{N_C} V_L$$

Or:

$$\frac{V_L}{V_H} = \frac{N_C}{N_C + N_{SE}}$$

The current relationship between the two sides of the transformer can be found by noting that:

$$I_L = I_C + I_{SE} \qquad (1)$$

But:

$$I_C . N_C = I_{SE} . N_{SE}$$

Or:

$$I_C = I_{SE} . \frac{N_{SE}}{N_C} \qquad (2)$$

Substituting (2) in (1):

$$I_L = \frac{N_{SE}}{N_C} I_{SE} + I_{SE}$$

But:

$$I_H = I_{SE}$$

Thus:

$$I_L = \frac{N_{SE}}{N_C} I_H + I_H = \frac{N_C + N_{SE}}{N_C} I_H$$

Or:

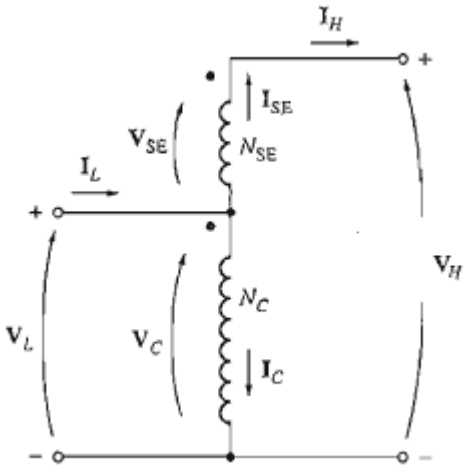
$$\frac{I_L}{I_H} = \frac{N_C + N_{SE}}{N_C}$$

And it's the reciprocal of the ration $\frac{V_L}{V_H}$.

The Apparent Power Rating Advantage of Autotransformers

It has to be noted that not all the power traveling from the primary to the secondary in the autotransformer goes through the windings. As a result, if a conventional transformer is reconnected as an autotransformer, it can handle much more power than it was originally rated for.

Consider the autotransformer circuit shown below.



The input apparent power to the autotransformer is given by:

$$S_{in} = V_L I_L$$

and the output apparent power is given by:

$$S_{out} = V_H I_H$$

From these:

$$\frac{S_{in}}{S_{out}} = \frac{V_L I_L}{V_H I_H} = \left(\frac{V_L}{V_H}\right)\left(\frac{I_L}{I_H}\right) = \frac{N_C}{N_C + N_{SE}} . \frac{N_C + N_{SE}}{N_C} = 1$$

Or:

$$S_{in} = S_{out} = S_{IO}$$

Here S_{IO} is the input and output apparent powers of the transformer.

However, the apparent power in the transformer windings is:

$$S_W = V_C . I_C = V_{SE} . I_{SE}$$

The relationship between the power going into the primary (and out the secondary) of the transformer and the power in the transformer 's actual windings can be found as follows:

$$S_W = V_C . I_C = V_L (I_L - I_H) = V_L . I_L - V_L I_H$$

As found before:

$$\frac{I_L}{I_H} = \frac{N_C + N_{SE}}{N_C}$$

Or:

$$I_H = \frac{N_C}{N_C + N_{SE}} I_L$$

Thus:

$$S_W = V_L . I_L - V_L I_L \frac{N_C}{N_C + N_{SE}} = V_L . I_L \frac{N_{SE}}{N_C + N_{SE}}$$

$$= S_{IO} \frac{N_{SE}}{N_C + N_{SE}}$$

Therefore, the ratio of the apparent power in the primary and secondary of the autotransformer to the apparent power actually traveling through its windings is:

$$\frac{S_{IO}}{S_W} = \frac{N_C + N_{SE}}{N_{SE}}$$

This equation shows the apparent power rating advantage of the autotransformer over the conventional transfonner. In this equation, S_{IO} is the apparent power entering the primary and leaving the secondary of the transformer, while S_W is the apparent power actually traveling through the transformer's windings (the rest passes from primary to secondary without being coupled through the transformer's windings). Note that the smaller the series winding, the greater the advantage.

For example, a $500kVA$ autotransformer connecting a $110kV$ system to a $138kV$ system would have an $\frac{N_C}{N_{SE}} = \frac{110}{28}$. Such an autotransformer would actually have windings rated at:

$$S_W = S_{IO}\frac{N_{SE}}{N_{SE}+N_C} = (5000)\frac{28}{28+110} = 1015kVA$$

The autotransfomler would have windings rated at only about $1015kVA$, while a conventional transformer doing the same job would need windings rated at $5000kVA$. The autotransformer could be 5 times smaller than the conventional transformer and also would be much less expensive. For this reason, it is very advantageous to build transformers between two nearly equal voltages as autotransformers.

It is common practice in power systems to use autotransfonners whenever two voltages fairly close to each other in level need to be transformed, because the closer the two voltages are, the greater the autotransformer power advantage becomes. They are also used as variable transformers, where the low-voltage tap moves up and down the winding. This is a very convenient way to get a variable ac voltage.

The principal disadvantage of autotransformers is that, unlike ordinary transformers, there is a direct physical connection between the primary and the secondary circuits, so the electrical isolation of the two sides is lost. If a particular application does not require electrical isolation, then the autotransformer is a convenient and inexpensive way to tie nearly equal voltages together.

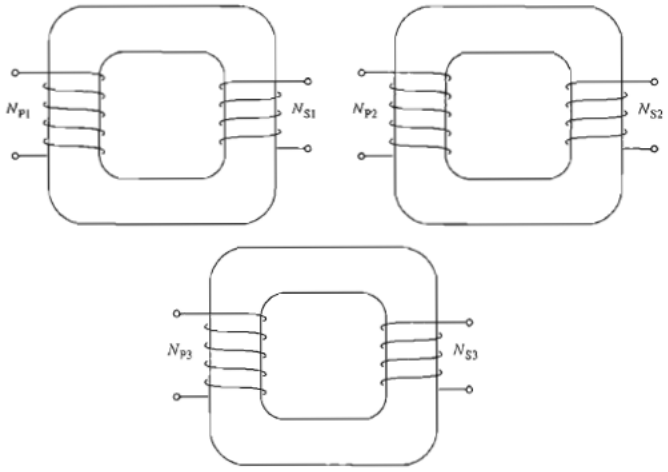
The Internal Impedance of an Autotransformer

Autotransformers have one additional disadvantage compared to conventional transformers. It turns out that, compared to a given transformer connected in the conventional manner, the effective per-unit impedance of an autotransformer is smaller by a factor equal to the reciprocal of the power advantage of the autotransformer connection.

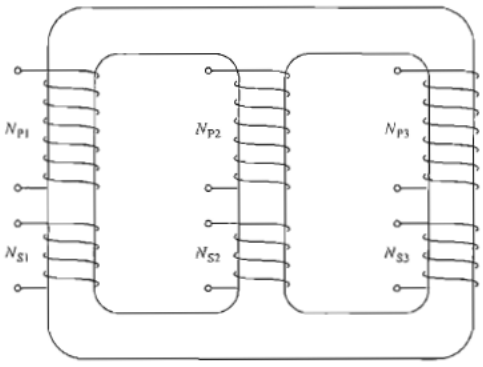
The reduced internal impedance of an autotransformer compared to a conventional two-winding transformer can be a serious problem in some applications where the series impedance is needed to limit current flows during power system faults (short circuits). The effect of the smaller internal impedance provided by an autotransformer must be taken into account in practical applications before autotransformers are selected.

THREE-PHASE TRANSFORMERS

Transformers for three-phase circuits can be constructed in one of two ways:
- three single-phase transformers connected in a three-phase bank.
- three sets of windings wrapped on a common core, as below.



three single-phase transformers



three sets of windings wrapped on a common core
single three-phase transformer

A single three-phase transformer is lighter, smaller, cheaper, and slightly more efficient, but using three separate single-phase transformers has the advantage that each unit in the bank could be replaced individually in the event of trouble. A utility would only need to stock a single spare single-phase transformer to back up all three phases, potentially saving money.

Three-Phase Transformer Connections

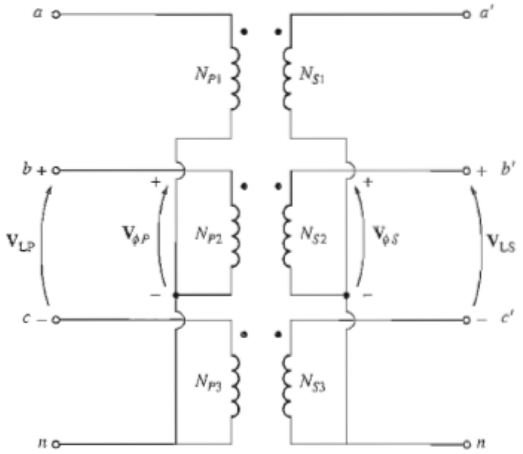
The primaries and secondaries of any three-phase transformer can be independently connected in either a Wye (Y) or a delta (Δ). This gives a total of four possible connections for a three-phase transformer bank:

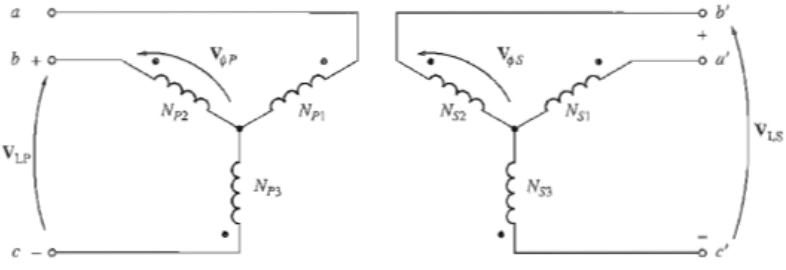
1) Wye- wye (Y - Y)
2) Wye-delta (Y - Δ)
3) Delta-wye (Δ -Y)
4) Delta-delta (Δ - Δ)

The impedance, voltage regulation, efficiency, and similar calculations for three-phase transformers are done on a per-phase basis.

WYE-WYE CONNECTION

This connection is shown below.





In this connection, the primary voltage on each phase of the transformer is given by $V_{\phi P} = \frac{V_{LP}}{\sqrt{3}}$. The primary-phase voltage is related to the secondary-phase voltage by the turns ratio of the transformer. The phase voltage on the secondary is then related to the line voltage on the secondary by $V_{LS} = \sqrt{3}V_{\phi S}$. Therefore, overall the voltage ratio on the transformer is:

$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{\sqrt{3}V_{\phi S}} = a$$

The Y-Y connection has two very serious problems:

1) If loads on the transformer circuit are unbalanced, then the voltages on the phases of the transformer can become severely unbalanced.
2) Third-harmonic voltages can be large.

Notice thet if a three-phase set of voltages is applied to the Y-Y transformer, the voltages in any phase will be 120^0 apart from the voltages in any other phase. However, the third-harmonic components of each of the three phase voltage will be in phase with each other, since there are three cycles in the third harmonic for each cycle of the fundamental frequency. There are always some third-harmonic components in transformer voltage because of the nonlinearity of the core, and these components add up. The result is a very large third-harmonic component of voltage on top of the fundamental voltage. This third-harmonic voltage can be larger than the fundamental voltage itself.

Both the unbalance problem and the third-harmonic problem can be solved using one of two techniques:

1) Solidly ground the neutrals of the transformers, especially the primary winding's neutral. This connection permits the additive third-harmonic components to cause a current flow in the neutral instead of building up large voltages. The neutral also provides a return path for any current imbalances in the load.

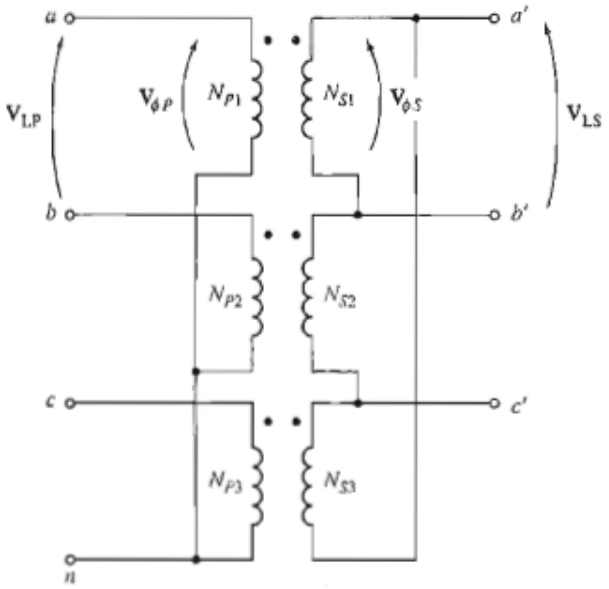
2) Add a third (tertiary) winding connected in Δ to the transformer bank. If a third Δ-connected winding is added to the transformer then the third-harmonic components of voltage in the Δ will add up, causing a circulating current flow within the winding. This suppresses the third-harmonic components of voltage in the same manner as grounding the transformer neutrals.

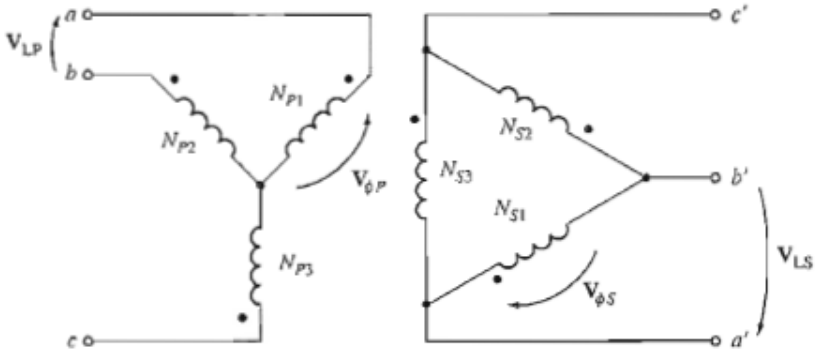
The Δ-connected tertiary windings need not even be brought out of the transformer case, but they often are used to supply lights and auxiliary power within the substation where it is located. The tertiary windings must be large enough to handle the circulating currents, so they are usually made about one-third the power rating of the two main windings.

One or the other of these correction techniques must be used any time a Y-Y transformer is installed. In practice, very few Y - Y transformers are used, since the same jobs can be done by one of the other types of three-phase transformers.

WYE-DELTA CONNECTION

This connection is shown in the following figure.





In this connection, the primary line voltage is related to the primary phase voltage by $V_{LP} = \sqrt{3}V_{\phi P}$, while the secondary line voltage is equal to the secondary phase voltage $V_{LS} = V_{\phi S}$. The voltage ratio of each phase is:

$$\frac{V_{\phi P}}{V_{\phi S}} = a$$

The relationship between the line voltage on the primary side of the bank and the line voltage on the secondary side of the bank is:

$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{V_{\phi S}} = \sqrt{3}a$$

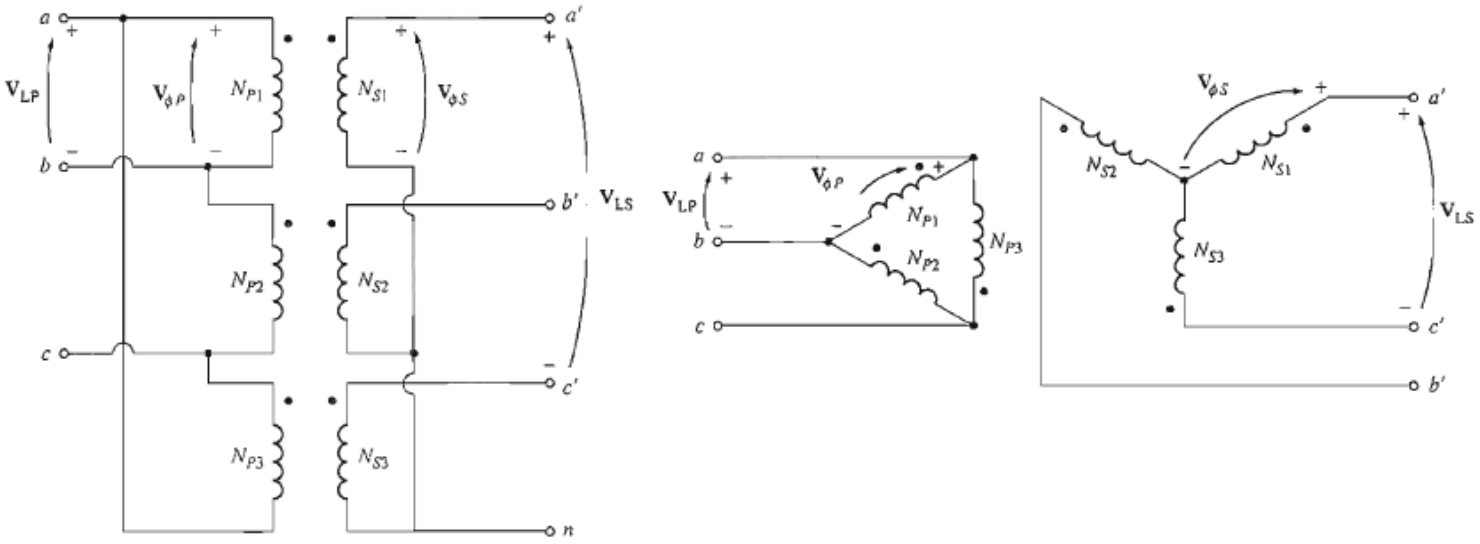
Notice that this connection has no problem with the third-harmonic components in its voltages, since they are consumed in a circulating current on the Δ side. This connection is also more stable with respect to unbalanced loads, since the Δ partially redistributes any imbalance that occurs.

This arrangement does have one problem, though. Because of the connection, the secondary voltage is shifted 30° relative to the primary voltage of the transformer. The fact that a phase shift has occurred can cause problems in paralleling the secondaries of two transformer banks together. The phase angles of transformer secondaries must be equal if they are to be paralleled, which means that attention must be paid to the direction of the 30° phase shift occuring in each transformer bank to be paralleled together.

Notice that, the connection shown in the previous figure will cause the secondary voltage to be lagging if the system phase sequence is *abc* (positive sequence). If the system phase sequence is *acb* (negative sequence), then the secondary voltage will be leading the primary voltage by 30°.

DELTA- WYE CONNECTION

This connection is shown in the following figure.



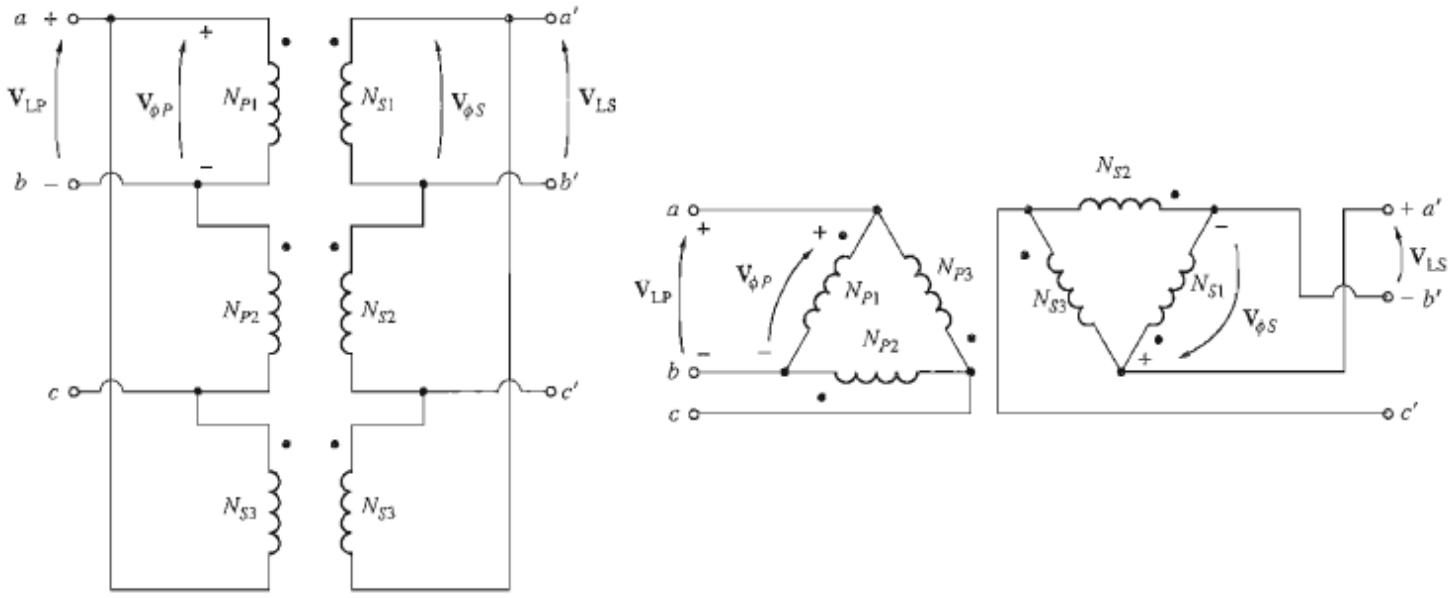
In this connection, the primary line voltage is equal to the primary-phase voltage $V_{LP} = V_{\phi P}$, while the secondary voltages are related by $V_{LS} = \sqrt{3}V_{\phi S}$. Therefore, the line-to-line voltage ratio of this transformer connection is:

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{\sqrt{3}V_{\phi S}} = \frac{1}{\sqrt{3}}\frac{V_{\phi P}}{V_{\phi S}} = \frac{a}{\sqrt{3}}$$

This connection has the same advantages and the same phase shift as the Y-Δ transformer. The connection shown in the previous figure makes the secondary voltage lag the primary voltage by 30^0, as before.

DELTA-DELTA CONNECTION.

This connection is shown in the figure below.



In this connection:

$V_{LP} = V_{\phi P}$ and $V_{LS} = V_{\phi S}$, thus:

$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{V_{\phi S}} = a$$

This transformer connection has no phase shift associated with it and no problems with unbalanced loads or harmonics.

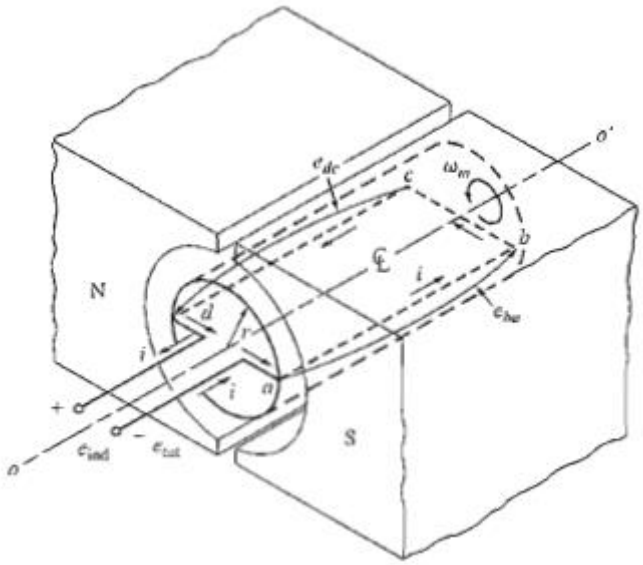
DC MACHINERY FUNDAMENTALS

DC machines are:
- generators that convert mechanical energy to dc electric energy, and
- motors that convert dc electric energy to mechanical energy.

Inside the DC machines, the voltages and the currents are AC, but they are converted to DC at the machine outputs by a mechanism called a commutator. Because of this commutator, dc machinery is also known as commutating machinery.

A SIMPLE ROTATING LOOP BETWEEN CURVED POLE FACES

The simplest possible rotating DC machine is shown in the figure below.



It consists of a single loop of wire rotating about a fixed axis. The rotating part of this machine is called the rotor, and the stationary part is called the stator. The magnetic field for this machine is supplied by the magnetic north and south poles.

Notice that the loop of rotor wire lies in a slot carved in a ferromagnetic core. The iron rotor, together with the curved shape of the pole faces, provides a constant-width air gap between the rotor and the stator.

As the magnetic flux takes the shortest path through the air, it is perpendicular to the rotor surface everywhere under the pole faces. Also, since the air gap is of uniform width, the reluctance is the same everywhere under the pole faces. The uniform reluctance means that the magnetic flux density is constant everywhere under the pole faces.

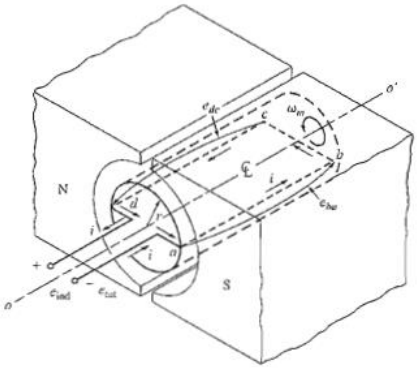
The Voltage Induced in a Rotating Loop

If the rotor of this machine is rotated, a voltage will be induced in the wire loop. The sides *ab* and *cd* are perpendicular to the plane of the page while the sides *bc* and *da* are parallel to the plane of the page. The magnetic field is constant and perpendicular to the surface of the rotor everywhere under the pole faces and zero beyond the edges of the poles.

The total voltage on the loop is the sum of the voltages induced in the four loop segments. The voltage on each segment is given by:

$$e_{ind} = (\vec{V} x \vec{B})\vec{l}$$

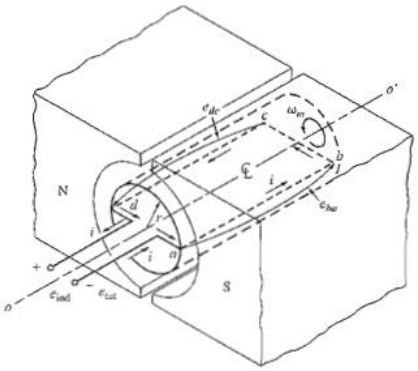
a) Segment *ab*: In this segment, the velocity of the wire is tangential to the path of rotation (for the figure it poins up). The magnetic field B points out perpendicular to the rotor surface (from left to right) everywhere under the pole face and is zero beyond the edges of the pole face. Under the pole face, velocity $\vec{V}$ is perpendicular to $\vec{B}$, and the quantity $\vec{V}X\vec{B}$ points into the page. Therefore, the induced voltage on the segment is:



$$e_{ba} = \left(\vec{V}x\vec{B}\right).\vec{l} = \begin{cases} VBl & positive \text{ int } o \quad page \quad under \quad the \quad pole \quad face \\ 0 & beyond \quad the \quad pole \quad edges \end{cases}$$

***b) Segment bc*:** In this segment, the quantity $\vec{V}X\vec{B}$ is either into or out of the page, while length $\vec{l}$ is in the plane of the page, so $\vec{V}X\vec{B}$ is perpendicular to $\vec{l}$. Therefore the voltage in segment be will be zero. $e_{cb} = 0$.

c) Segment cd: In this segment, the velocity of the wire is tangential to the path of rotation (for the figure it poins down). The magnetic field $\vec{B}$ points out perpendicular to the rotor surface (from left to right) everywhere under the pole face and is zero beyond the edges of the pole face. Under the pole face, velocity $\vec{V}$ is perpendicular to $\vec{B}$, and the quantity $\vec{V}X\vec{B}$ points out of the page. Therefore, the induced voltage on the segment is:



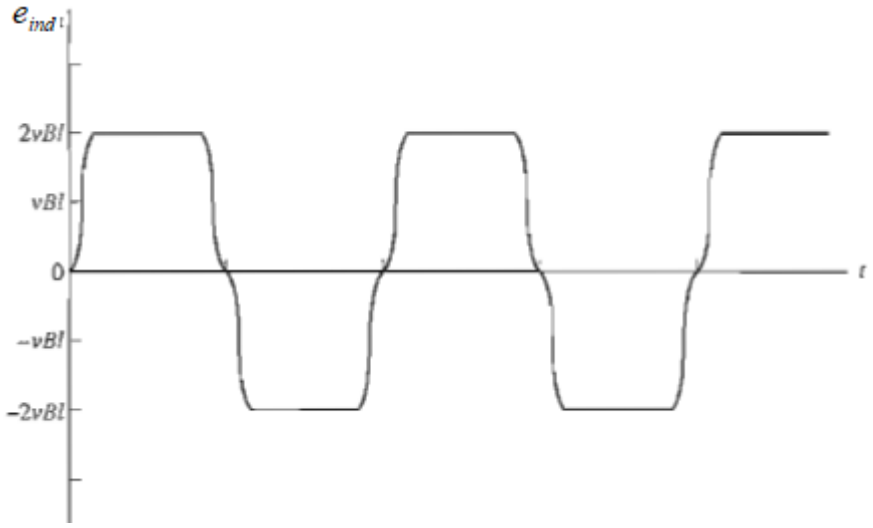
$$e_{dc} = \left(\vec{V}x\vec{B}\right).\vec{l} = \begin{cases} VBl & positive \quad out \quad of \quad the \, page \quad under \quad the \quad pole \quad face \\ 0 & beyond \quad the \quad pole \quad edges \end{cases}$$

d) Segment da*:** As in segment ***bc, the quantity $\vec{V}X\vec{B}$ is perpendicular to $\vec{l}$. Therefore the voltage in this segment be will be zero. $e_{da} = 0$.

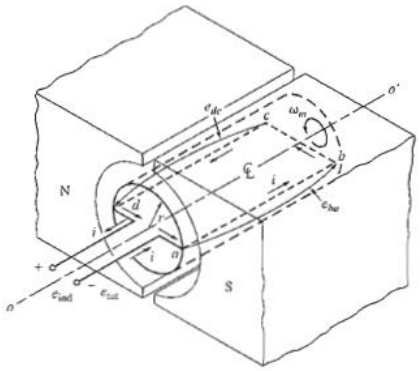
The total induced voltage on the loop is given by:

$$e_{ind} = e_{ba} + e_{cb} + e_{dc} + e_{ad} = \begin{cases} 2VBl & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases} \qquad (1)$$

When the loop rotates throug 180°, segment *ab* becomes under the north pole face instead of the south pole face. At that time, the direction of the voltage on the segment reverses, but its magnitude remains constant. The resulting voltage is shown as a function of time in the figure below.



Equation (1) can be expressed in another way.
Referring to the figure below.



The volocity of the loop wire can be expressed as:

$$V = r.\omega_m \qquad (2)$$

Where r is the radius from the axis of rotation out to the edge of the loop and ω_m is the angular velocity of the loop. Thus equation (1) can be written as:

$$e_{ind} = \begin{cases} 2r.\omega_m.B.l & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases} \qquad (3)$$

From the figure, it can be seen that the rotor surface is a cylinder, and the area of the rotor surface is $A = 2\pi rl$. Since there are two poles, the area of the rotor under each pole (ignoring the small gaps between poles) will be $A_P = \pi rl$.
Therefore,

$$e_{ind} = \begin{cases} \frac{2}{\pi} A_P.\omega_m.B & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases} \qquad (4)$$

Since the flux density B is constant everywhere in the air gap under the pole faces, the tota1 flux under each pole is:

$$\phi = B.A_P \qquad (5)$$

Thus:

$$e_{ind} = \begin{cases} \frac{2}{\pi}\phi.\omega_m. & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases} \qquad (6)$$

From this equation, it can be seen that the voltage induced in the machine winding will depend on the three factors:

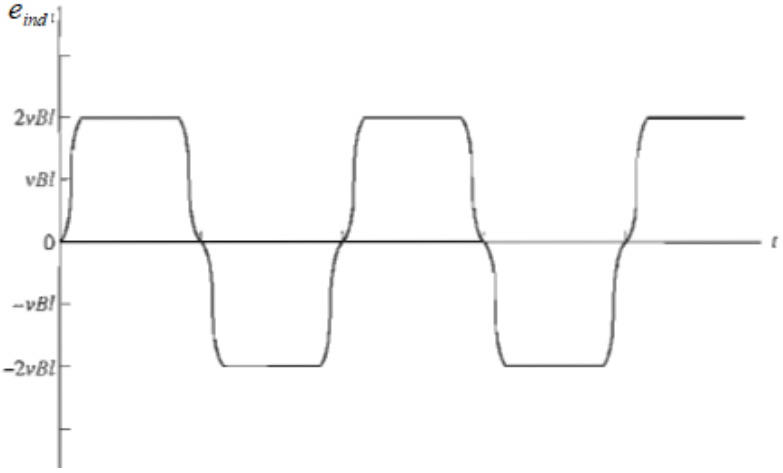
1) The flux in the machine, ϕ

2) The speed of rotation, ω_m

3) A constant representing the construction of the machine.

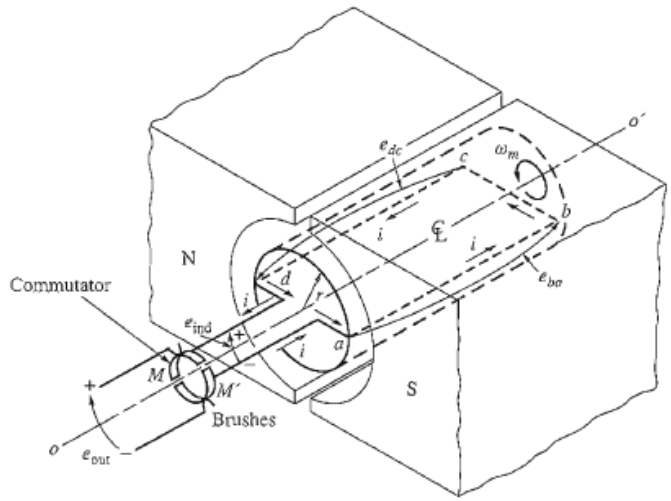
Getting DC Voltage Out of the Rotating Loop

The figure below shows the voltage induced in the machine winding.



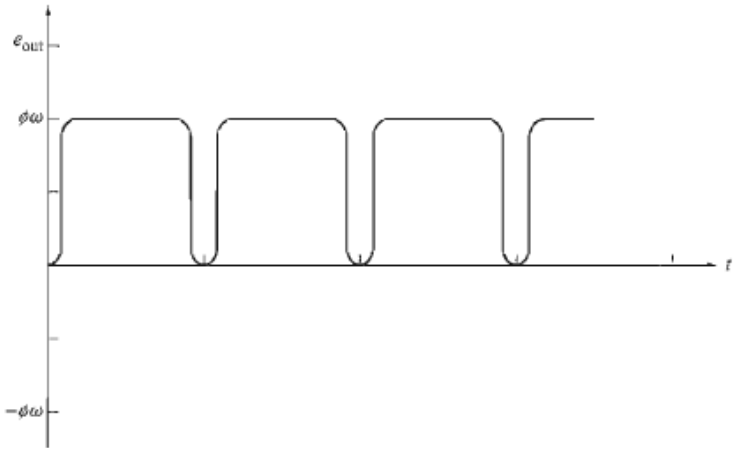
It can be seen that this voltage is alternately a constant positive value and a constant negative value.

In the DC machine two semicircular conducting segments with two fixed contacts are added to the end of the loop, as shown below.



The two fixed contacts are set up at an angle such that at the instant when the voltage in the loop is zero, the contacts short-circuit the two segments. In this fashion, every time the voltage of the loop switches direction, the contacts also switch connections,

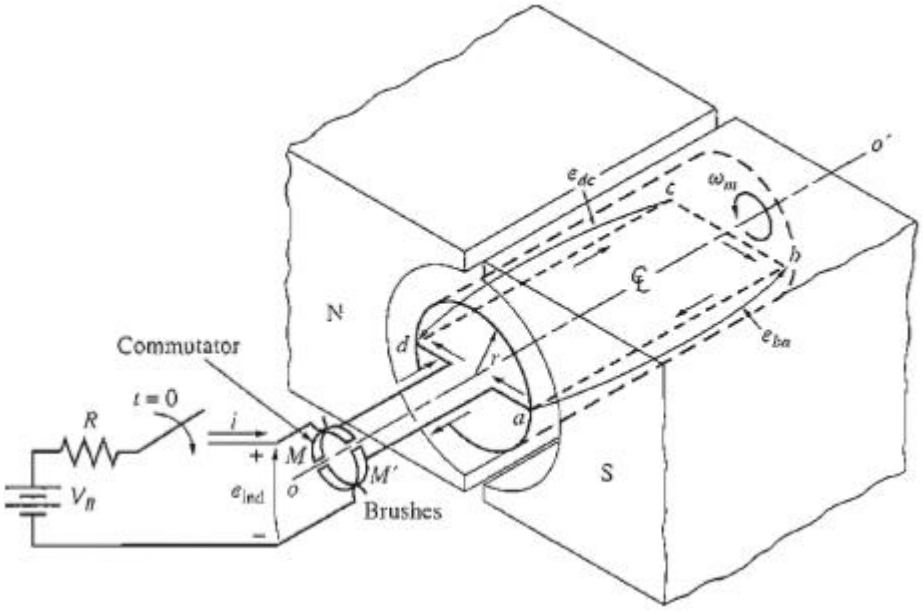
and the output voltage of the contacts is always built up in the same direction, as shown below.



This connection-switching process is known as commutation. The rotating semicircular segments are called commutator segments, and the fixed contacts are called brushes.

The Induced Torque in the Rotating Loop

If a battery is connected to the machine, as shown below, a current will flow in the loop conductor. The interaction of this current and the magnetic field, inside the machine, will produce a torque that causes rotation of the machine rotor.



As known, the force on a segment of the loop is given by:

$\vec{F} = i(\vec{l} x \vec{B})$

and the torque on the segment is given by:

$\vec{T} = \vec{r} X \vec{F} = rF \sin\theta$

where θ is the angle between $\vec{r}$ and $\vec{F}$. The torque is essentially zero whenever the loop is beyond the pole edges.

The forces on the loop segments are:

a) segment *ab*: In this segment, the current is directed out of the page. The magnetic field under the pole face is pointing radialIy from north pole to south pole. The force on this wire will be:

$\vec{F}_{ab} = i(\vec{l} x \vec{B}) = ilB$ tangent to the direction of the motion.

The torque on the rotor caused by this force is:

$\vec{T}_{ab} = rilB \sin 90^0 = rilB$ CCW (counter clockwise).

b) segment *bc*: In this segment, the current is directed from left to right. The magnetic field under the pole face is pointing radialIy from north pole to south pole. The force on this wire will be:

$\vec{F}_{bc} = i(\vec{l} x \vec{B}) = 0$ as $i\vec{l}$ is parallel to $\vec{B}$.

c) segment *cd*: In this segment, the current is directed into the page. The magnetic field under the pole face is pointing radialIy from north pole to south pole. The force on this wire will be:

$\vec{F}_{cd} = i(\vec{l} x \vec{B}) = ilB$ tangent to the direction of the motion.

The torque on the rotor caused by this force is:

$\vec{T}_{cd} = rilB \sin 90^0 = rilB$ CCW (counter clockwise).

b) segment *da*: In this segment, the current is directed from right to left. The magnetic field under the pole face is pointing radialIy from north pole to south pole. The force on this wire will be:

$\vec{F}_{da} = i\left(\vec{l} x \vec{B}\right) = 0$ as $i\vec{l}$ is parallel to $\vec{B}$.

The resulting torque on the loop is given by:

$$\vec{T} = \vec{T}_{ab} + \vec{T}_{bc} + \vec{T}_{cd} + \vec{T}_{da} = \begin{cases} 2rilB & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases}$$

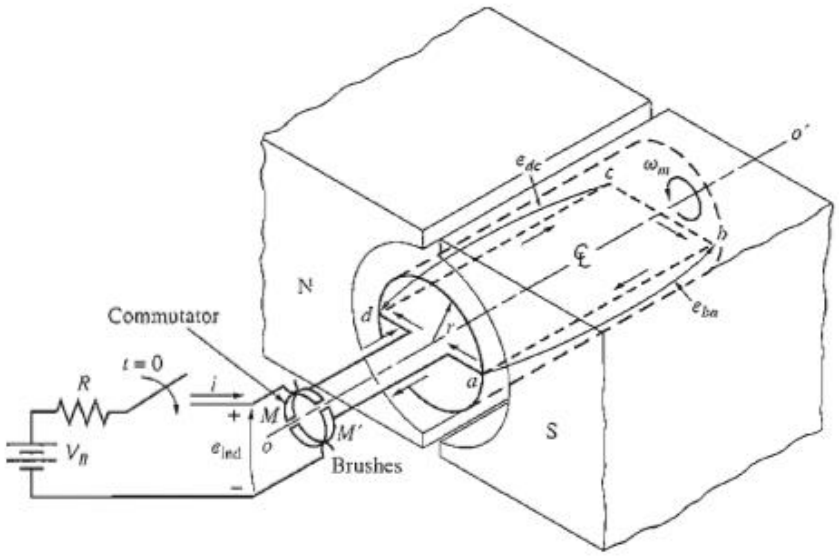
As $A_P = \pi r l$ and $\phi = A_P . B$, thus:

$$\vec{T} = \begin{cases} \frac{2}{\pi}\phi i & under \quad the \quad pole \quad faces \\ 0 & beyond \quad the \quad pole \quad edges \end{cases}$$

It can be seen that, the torque produced in the machine is the product of the flux in the machine and the current in the machine, times some quantity representing the mechanical construction of the machine (the percentage of the rotor covered by pole faces).

In general, the torque in any real machine will depend on the same three factors:
1) The flux in the machine
2) The current in the machine
3) A constant representing the construction of the machine.

Example:
The figure below shows a simple DC machine connected to a battery and a resistor through a switch. The resistor shown models the total resistance of the battery and the wire in the machine. The physical dimensions and characteristics of this machine are:
$r = 0.5m$, $l = 1.0m$, $R = 0.3\Omega$, $B = 0.25T$, $V_B = 120V$



a) What happens when the switch is closed?.

b) What is the machine's maximum starting current?. What is its steady-state angular velocity at no load.

c) Suppose a load is attached to the loop, and the resulting load torque is $10N.m$. What would the new steady-state speed be? How much power is supplied by the shaft of the machine? How much power is being supplied by the battery? Is this machine a motor or a generator?

d) Suppose the machine is again unloaded, and a torque of $7.5N.m$ is applied to the shaft in the direction of rotation. What is the new steady-state speed? Is this machine now a motor or a generator?

e) Suppose the machine is running unloaded. What would the final steady-state speed of the rotor be if the flux density were reduced to $0.2T$?

Solution:

a) When the switch in is closed, a current will flow in the loop. Since the loop is initially stationary, $e_{ind} = 0$. Therefore, the current will be given by:

$$i = \frac{V_B - e_{ind}}{R} = \frac{120 - 0}{0.3} = 400A$$

This current flows through the rotor loop, producing a torque:

$$T = \frac{2}{\pi}\phi.i \qquad \text{CCW}$$

This induced torque produces an angular acceleration in a counterclockwise direction, so the rotor of the machine begins to turn. But as the rotor begins to turn, an induced voltage is produced in the motor, given by:

$$e_{ind} = \frac{2}{\pi}\phi.\omega_m$$

so the current falls. As the current falls, the torque decreases, and the machine winds up.

In steady state: $e_{ind} = V_B$, the current $i = \frac{V_B - e_{ind}}{R} = 0$ and the torque $T = \frac{2}{\pi}\phi.i = 0$,

This is the same as the starting behavior of the linear dc machine, discussed before.

b) At starting conditions, the machine's currenl is:

$$i = \frac{V_B - e_{ind}}{R} = \frac{120 - 0}{0.3} = 400A$$

At no-load steady-state conditions, the induced torque is zero. But as $T = \frac{2}{\pi}\phi.i$, this implies that current i. The fact that $i = 0$ means that $V_B = e_{ind}$.

Therefore, the speed of the rotor is found from:

$$e_{ind} = \frac{2}{\pi}\phi.\omega_m = V_B$$

Thus:

$$\omega_m = \frac{\pi.V_B}{2\phi} \qquad (1)$$

Where:

$$\phi = B.A_P = B.\pi.r.l \qquad (2)$$

Substituting (2) in (1), we have:

$$\omega_m = \frac{\pi.V_B}{2\pi rlB} = \frac{V_B}{2rlB} = \frac{120}{2(0.5)(1.0)(0.25)} = 480 rad/s$$

c) If a load torque of $10N.m$ is applied to the shaft of the machine, it will begin to slow down. But as ω_m decreases, e_{ind} decreases and the rotor current increases. As the rotor current increases, the torque produced increases too, until $T_{ind} = T_{Load} = 10N.m$. At steady state:

$T_{ind} = \frac{2}{\pi}\phi.i$, as $\phi = B.A_P = B.\pi.r.l$, thus:

$$T_{ind} = \frac{2}{\pi} B.\pi rl.i = 2B.rl.i$$

From which:

$$i = \frac{T_{ind}}{2Brl} = \frac{10}{2(0.25)(0.5)(1.0)} = 40A$$

and:

$$e_{ind} = V_B - i.R = 120 - 40(0.3) = 108V$$

But:

$e_{ind} = \frac{2}{\pi}\phi.\omega_m$, as $\phi = B.A_P = B.\pi.r.l$, thus:

$$e_{ind} = \frac{2}{\pi} B\pi rl.\omega_m = 2Brl.\omega_m$$

Or:

$$\omega_m = \frac{e_{ind}}{2Brl} = \frac{108}{2(0.25)(0.5)(1.0)} = 432 rad/s$$

The power supplied from the shaft is:

$$P_{mech} = T.\omega = 10(432) = 4320W$$

The power out of the battery is:

$$P_B = V_B.i = 120(40) = 4800W$$

The difference is the power losses in the resistor, or:

$$P_{loss} = Ri^2 = 0.3(40)^2 = 480W$$

Note that:

$$P_B - P_{mech} = 4800 - 4320 = 480W$$

The machine is operating as a motor, converting electric power to mechanical power.

d) If a torque is applied in the direction of motion, the rotor accelerates. As the speed increases, the internal voltage e_{ind} increases and exceeds V_B. The current flows into the battery and the machine is now a generator. This current causes an induced torque opposite to the direction of motion. The induced torque opposes the external applied torque.

At steady state:

$$T_{app} = T_{ind} = 2B.rl.i$$

From which:

$$i = \frac{T_{app}}{2B.rl} = \frac{7.5}{2(0.25)(0.5)(1.0)} = 30A$$

The induced voltage is:

$$e_{ind} = V_B + Ri = 120 + 0.3(30) = 129V$$

The speed of the shaft is:

$$\omega_m = \frac{e_{ind}}{2Brl} = \frac{129}{2(0.25)(0.5)(1.0)} = 516 rad/s$$

e) Since the machine is unloaded thus the machine have a zero torque, and then a zero current, or:

$$e_{ind} = V_B = 120V$$

And the new speed of the machine is:

$$\omega_m = \frac{e_{ind}}{2Brl} = \frac{120}{2(0.2)(0.5)(1.0)} = 600 rad/s$$

Notice that when the flux in the machine is decreased, its speed increases. This is the same behavior seen in the linear machine and the same way that real de motors behave.

COMMUTATION IN A SIMPLE FOUR-LOOP DC MACHINE

Commutation is the process of converting the ac voltages and currents in the rotor of a dc machine to dc voltages and currents at its terminals.

Here, the technique of commutation of a four-loop two-pole machine will be explained as an example.

A simple four-loop, two-pole dc machine is shown in the figure below.

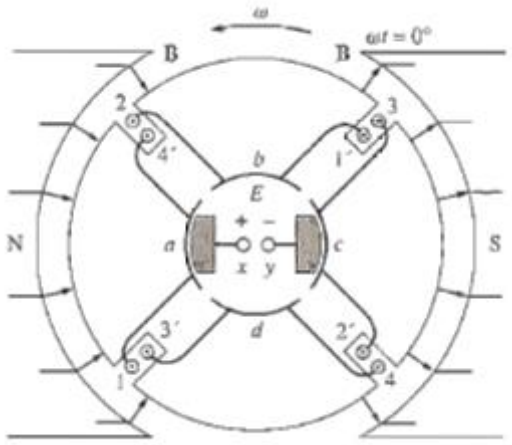


Fig. (1)

This machine has four complete loops buried in slots carved in the laminated steel of its rotor. The pole faces of the machine are curved to provide a uniform air-gap width and to give a uniform flux density everywhere under the faces.

The winding's connections to the machine's commutator are shown in following figure.

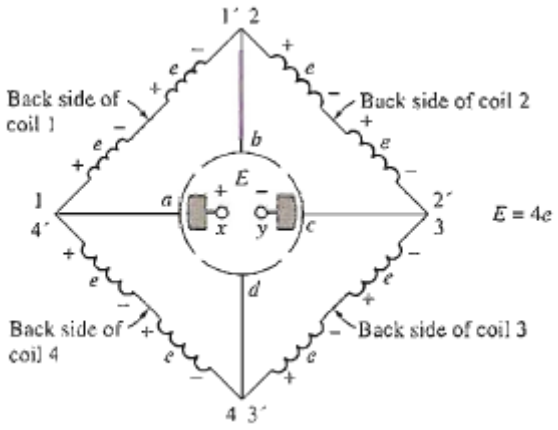


Fig. (2)

Notice that loop 1 stretches between commutator segments a and b, loop 2 stretches between segments b and c, and so forth around the rotor.

Notice that figure (1) is drawn at the instant of time when 1,2,3', and 4' ends of the loops are under the north pole face, while the 1',2',3, and 4 ends of the loops are under the south pole face, The voltage in each of the 1, 2, 3', and 4' ends of the loops is given by:

$e_{ind} = (\vec{V} X \vec{B})\vec{l} = VBl$ positive out of the page.

The voltage in each of the 1',2', 3, and 4 ends of the loops is given by:

$e_{ind} = (\vec{V} X \vec{B})\vec{l} = VBl$ positive into the page.

The overall result is shown in Figure (2). In Figure (2), each coil represents one side (or conductor) of a loop. If the induced voltage on anyone side of a loop is e, then the total voltage at the brushes of the machine is:

$E = 4e$ at $\omega t = 0$.

Notice also that there are two parallel paths for current through the machine.

As known, the rotor of the machine continues to rotate. The following figure shows the machine at time $\omega t = 45^0$.

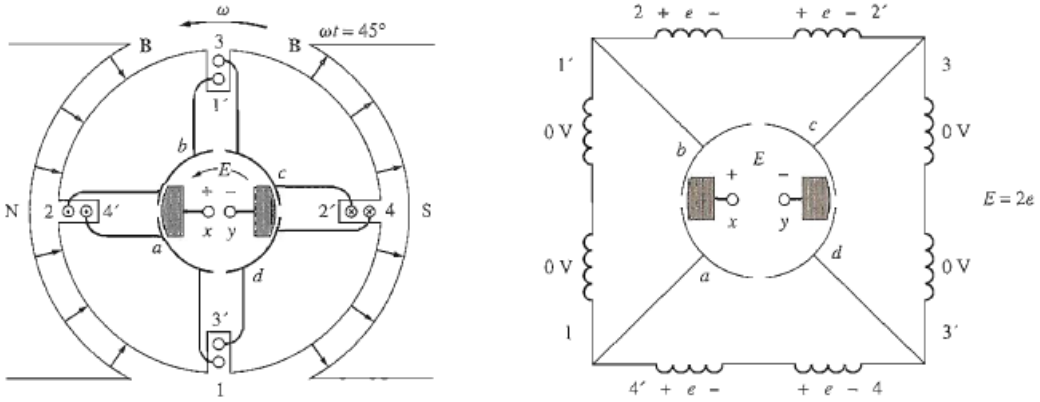


Fig. (3)

At that time, loops 1 and 3 have rotated into the gap between the poles, so the voltage across each of them is zero. Notice that at this instant the brushes of the machine are shorting out commutator segments *ab* and *cd*. This bappens just at the time when the loops between these segments have $0V$ across them, so shorting out the segments creates no problem. At this time, only loops 2 and 4 are under the pole faces, so the terminal voltage E is given by:

$E = 2e$ at $\omega t = 45^0$.

When the rotor continues to turn through another 45°, the resulting situation is shown in figure (4a).

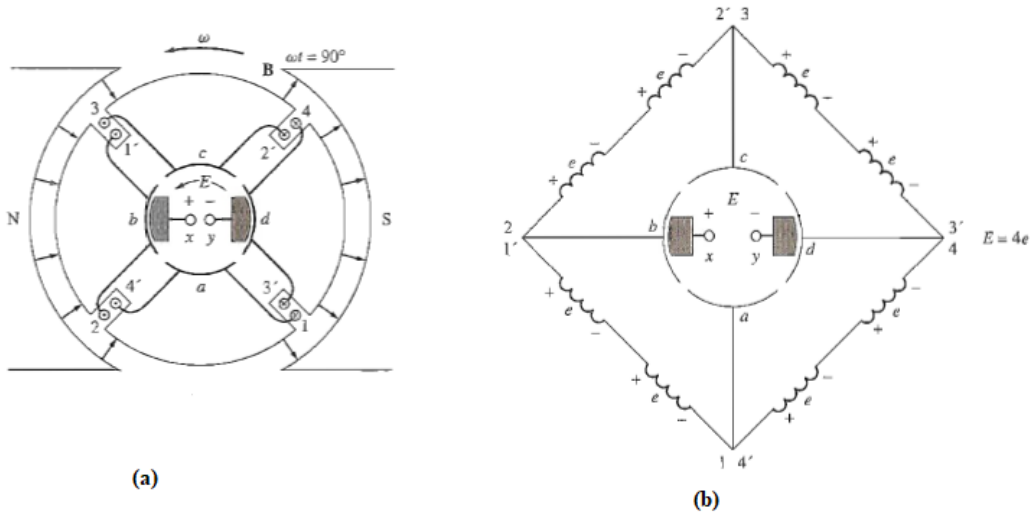


Fig. (4)

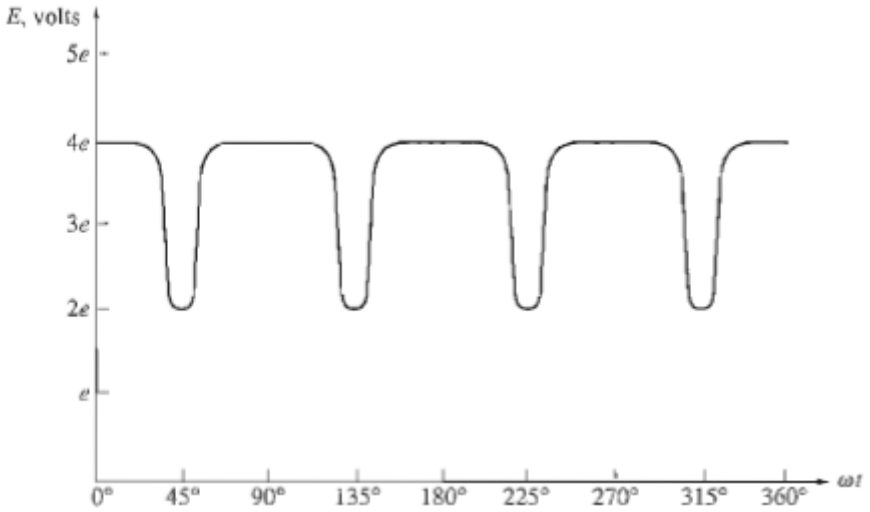
Here, the 1',2,3, and 4' ends of the loops are under the north pole face, and the 1,2',3', and 4 ends of the loops are under the south pole face. The voltages are still built up out of the page for the ends under the north pole face and into the page for the ends

under the south pole face. The resulting voltage diagram is shown in figure (4b). There are now four voltage-carrying ends in each parallel path through the machine, so the terminal voltage E is given by:

$E = 4e$ at $\omega t = 90^0$.

Comparing figures (1) and (4), it can be seen that the voltages on loops 1 and 3 have reversed between the two pictures, but since their connections have also reversed, the total voltage is still being built up in the same direction as before. This fact is at the heart of every commutation scheme. Whenever the voltage reverses in a loop, the connections of the loop are also switched, and the total voltage is still built up in the original direction.

The terminal voltage of this machine as a function of time is shown in the figure below.



It can be seen that this machine output voltage waveform is a better approximation to a constant dc level than the single rotating loop, discussed before. It has to be mentioned that as the number of loops on the rotor increases, the approximation to a perfect dc voltage continues to get better and better.

Also from the previous discussion, commutation ca be defined as the process of switching the loop connections on the rotor of the dc machine just as the voltage in the loop switches polarity, in order to maintain a constant dc output voltage.

It has to be mentioned that the rotating segments to which the loops are attached are called commutator segments, and the stationary pieces that ride on top of the moving segments are called brushes. The commutator segments in real machines are typically made of copper bars. The brushes are made of a mixture containing graphite, so that they cause very little friction as they rub over the rotating commutator segments.

COMMUTATION AND ARMATURE CONSTRUCTION IN REAL DC MACHINES

The rotor of the DC machine is also called the armature. The rotor (armature) coils are inserted into the armature slots (channels). The ends of the rotor coils are connected to the commutator segments. Each coil consists of a number of turns (loops) of wire, each turn taped and insulated from the other turns and from the rotor slot. Each side of a turn is ca11ed a conductor. The number of conductors on a machine's armature is given by:

$$Z = 2C.N_C$$

Where:

Z : is the number of conductors on the rotor.

C : is the number of coils on the rotor.

N_C : is the number of turns per coil.

Normally, a coil spans 180 electrical degrees. This means that when one side is under the center of a given magnetic pole, the other side is under the center of a pole of opposite polarity.

Notice that, the physical poles may not be located 180 mechanical degrees apart. The relationship between the electrical angle and mechanical angle in a given machine is given by:

$$\theta_e = \frac{P}{2}\theta_m$$

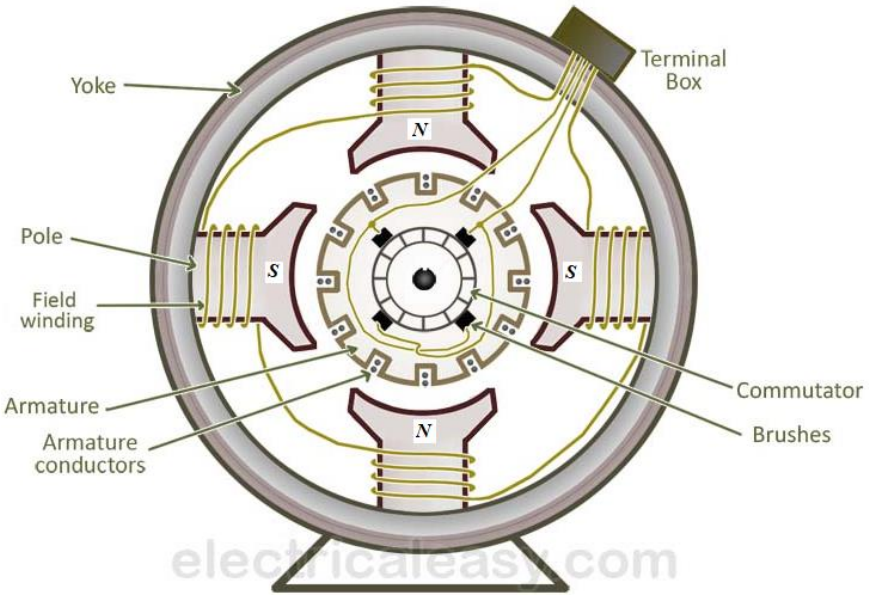
Here:

θ_e : the angle in electrical degrees.

θ_m : the angle in mechanical degrees.

P : the number of machine poles.

Example: the figure below shows a 4-pole DC machine. Note that for this machine 180 electrical degrees is equal to 90 mechanical degrees.



Notice that, if a coil spans 180 electrical degrees, the voltages in the conductors on either side of the coil will be exactly the same in magnitude and opposite in direction at all times. Such a coil is called a full-pitch coil.
Sometimes a coil is built that spans less than 180 electrical degrees. Such a coil is called a fractional-pitch coil, and a rotor winding wound with fractional-pitch coils is called a chorded winding. The amount of chording in a winding is described by a pitch-factor p , which is defined as:

$$p = \frac{electrical \quad angle \quad of \quad coil}{180^0} x100\%$$

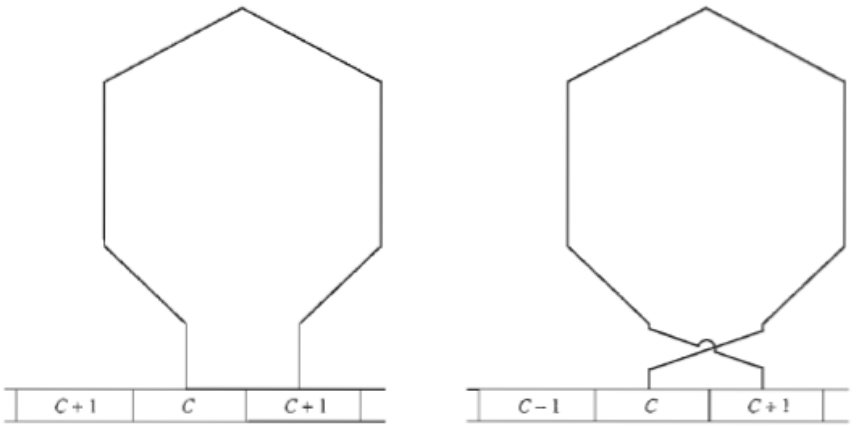
Sometimes a small amount of chording will be used in dc rotor windings to improve commutation.

Notice also that, most rotor windings are two-layer windings, meaning that sides from two different coils are inserted into each slot. One side of each coil will be at the bottom of its slot, and the other side will be at the top of its slot.

Connections to the Commutator Segments

There are a number of ways in which armature windings are connected to commutator segments. These connection methods result in different advantages and disadvantages.

The distance (in number of segments) between the commutator segments to which the two ends of a coil are connected is called the commutator pitch y_C . If the end of a coil is connected to a commutator segment ahead of the one its beginning is connected to, the winding is called a progressive winding, but if the end of a coil is connected to a commutator segment behind the one its beginning is connected to, the winding is called a retrogressive winding (see the figure below).

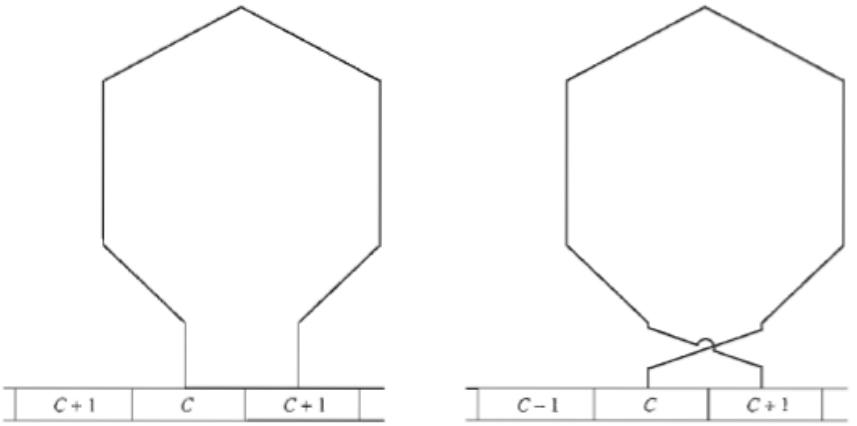


Rotor (armature) windings are further classified according to the plex of their windings. A simplex rotor winding is a single, complete, closed winding wound on a rotor. A duplex rotor winding is a rotor with two complete and independent sets of rotor windings. If a rotor has a duplex winding, then each of the windings will be associated with every other commutator segment: One winding will be connected to segments 1, 3, 5, etc., and the other winding will be connected to segments 2, 4, 6, etc. Similarly, a triplex winding will have three complete and independent sets of windings, each winding connected to every third commutator segment on the rotor. Collectively, all armatures with more than one set of windings are said to have multiplex windings.

Armature windings are also classified according to the sequence of their connections to the commutator segments. There are two basic sequences of armature winding connections. These are called lap windings and wave windings. In addition, there is a third type of winding, called a frog-leg winding, which combines lap and wave windings on a single rotor.

The Lap Winding

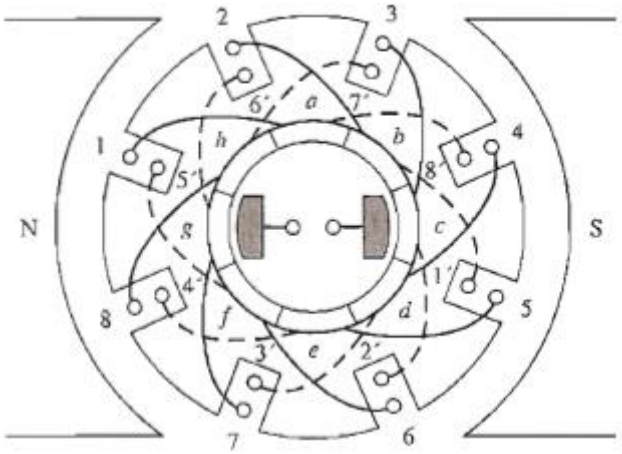
This winding is the simplest type of winding construction. A simplex lap winding is a rotor winding consisting of coils containing one or more turns of wire with the two ends of each coil coming out at adjacent commutator segments, see the figure below.



Progressive rotor winding Retrogressive rotor winding

Notice that, if the end of the coil is connected to the segment after the segment that the beginning of the coil is connected to, the winding is a progressive lap winding and $y_C = 1$; if the end of the coil is connected to the segment before the segment that the beginning of the coil is connected to, the winding is a retrogressive lap winding and $y_C = -1$.

A simple two-pole machine with lap windings is shown in the figure below.

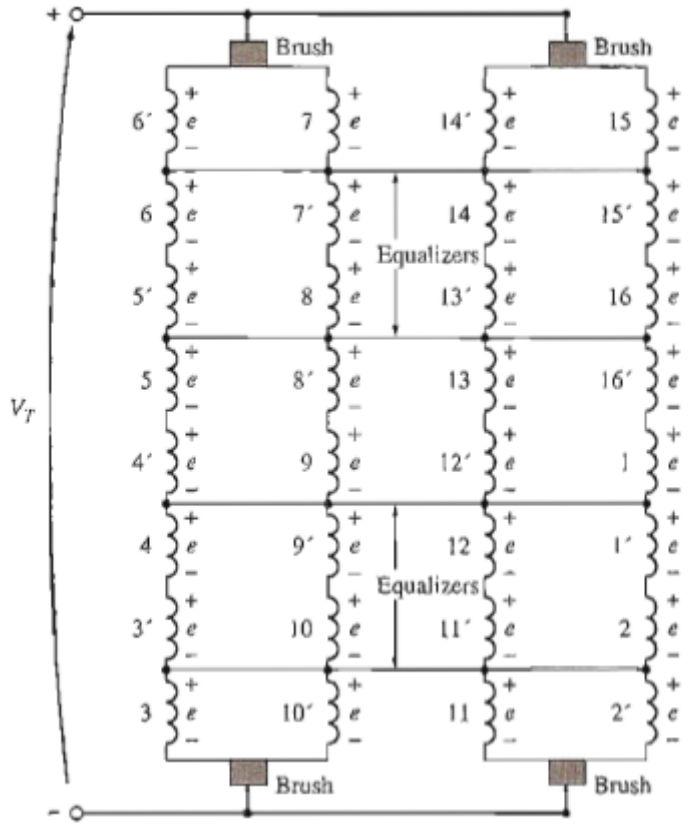


An important feature of simplex lap windings is that there are as many parallel current paths through the machine as there are poles on the machine.

If C is the number of coils and commutator segments present in the rotor and P is the number of poles on the machine, then there will be C/P coils in each of the P parallel current paths through the machine. The fact that there are P current paths also requires that there be as many brushes on the machine as there are poles in order to tap all the current paths. The fact that there are many current paths in a multipole

machine makes the lap winding an ideal choice for fairly low-voltage, high-current machines, since the high currents required can be split among the several different current paths. This current splitting permits the size of individual rotor conductors to remain reasonable even when the total current becomes extremely large.

The fact that there are many parallel paths through a multipole lap-wound machine can lead to a problem of circulating current in the parallel paths. The problem of circulating currents within the parallel paths of a machine can be reduced by equalizer rings or equalizing windings. Equalizers are bars located on the rotor of a lap-wound dc machine that short together points at the same voltage level in the different parallel paths. The effect of this shorting is to cause any circulating currents that occur to flow inside the small sections of windings thus shorted together and to prevent this circulating current from flowing through the brushes of the machine. An equalizer for a four-pole machine is shown in the figure below.



If a lap winding is duplex, then there are two completely independent windings wrapped on the rotor, and every other commutator segment is tied to one of the sets. Therefore, an individual coil ends on the second commutator segment down from where it started, and $y_C = \pm 2$ (depending on whether the winding is progressive or retrogressive), Since each set of windings has as many current paths as the machine has poles, there are twice as many current paths as the machine has poles in a duplex lap winding.

In general. for an m-plex lap winding, the commutator pitch y_C is:

$y_C = \pm m$

and the number of current paths in a machine is:

$a = m.P$

Where:

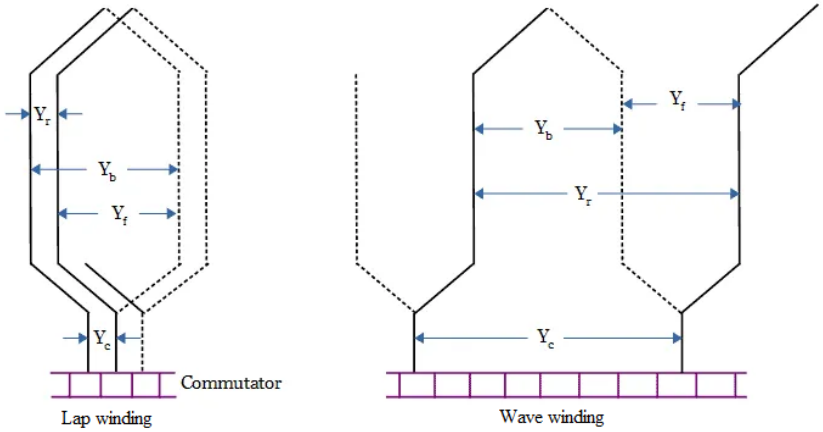
a : the number of current paths in the rotor

m : the plex of the windings (1, 2, 3, etc.)

P : the number of poles on the machine.

The Wave Winding

The series or wave winding is an alternative way to connect the rotor coils to the commutator segments. The figure below shows a lap and wave winding, for comparison.



Notice that there are two coils in series between the adjacent commutator segments. Furthermore, since each pair of coils between adjacent segments has a side under each pole face, all output voltages are the sum of the effects of every pole, and there can be no voltage imbalances.

Notice also that the lead from the second coil may be connected to the segment either ahead of or behind the segment at which the first coil begins. If the second coil is connected to the segment ahead of the first coil, the winding is progressive; if it is connected to the segment behind the first coil, it is retrogressive.

In general, if there are P poles on the machine, then there are P/2 coils in series between adjacent commutator segments.

In a simplex wave winding, there are only two current paths. There are C/2 or one-half of the windings in each current path. The brushes in such a machine will be located a full pole pitch apart from each other.

The general expression for commutator pitch in any simplex wave winding is:

$$y_C = \frac{2(C \pm 1)}{P}$$

where C is the number of coils on the rotor and P is the number of poles on the machine. The plus sign is associated with progressive windings, and the minus sign is associated with retrogressive windings.

Since there are only two current paths through a simplex wave-wound rotor, only two brushes are needed to draw off the current. This is because the segments undergoing commutation connect the points with equal voltage under all the pole faces. More

brushes can be added at points 180 electrical degrees apart if desired, since they are at the same potential and are connected together by the wires undergoing commutation in the machine. Extra brushes are usually added to a wave-wound machine, even though they are not necessary, because they reduce the amount of current that must be drawn through a given brush set.

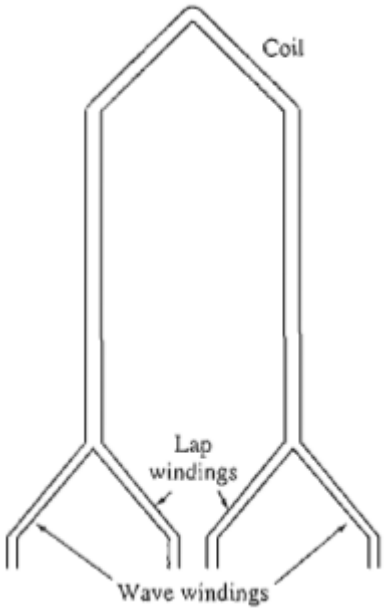
Wave windings are well suited to building higher-voltage DC machines, since the number of coils in series between commutator segments permits a high voltage to be built up more easily than with lap windings.

A multiplex wave winding is a winding with multiple independent sets of wave windings on the rotor. These extra sets of windings have two current paths each, so the number of current paths on a multiplex wave winding is:

$$a = 2m$$

The Frog-Leg Winding

The frog -leg winding or self-equalizing winding gets its name from the shape of its coils, as shown below.



This winding consists of a lap winding and a wave winding combined.

The equalizers in an ordinary lap winding are connected at points of equal voltage on the windings. Wave windings reach between points of essentially equal voltage under successive pole faces of the same polarity, which are the same locations that equalizers tie together. A frog-leg or self-equalizing winding combines a lap winding with a wave winding, so that the wave windings can function as equalizers for the lap winding.

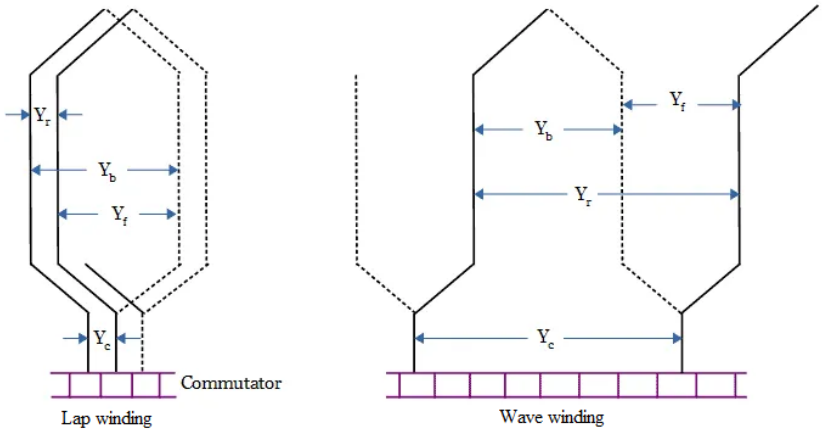
The number of current paths present in a frog-leg winding is:

$$a = 2P.m_{lap}$$

where P is the number of poles on the machine and m_{lap} is the plex of the lap winding.

Winding Parameters

The figure below shows a lap winding and a wave winding with the parameters describing them.



In this figure:

y_C : the commutator pith. It is the distance between the commutator segments to which the two ends of a coil are connected. From the figure, you can observe, for lap winding, commutator pitch (y_C) is the difference of back pitch(y_b) and front pitch(y_f). For wave winding, it is the sum of the back pitch and front pitch.

y_b : the bach pitch. It is the distance at which a coil advances on the back of the armature. It can also be defined as the distance between the first and the last conductors of a coil. It is the same as coil span.

y_f : the front pitch. It is the distance between the second conductor of one coil and the first conductor of the next coil.

y_r : the resultant pitch. It is the distance between the beginning of one coil and the beginning of the next coil to which it is connected.

y_P : the pole pith. Pole pitch is the peripheral distance between two adjacent poles. It is measured in terms of armature slots or armature conductors. It can also be defined as the number of armature slots(or armature conductors) per pole. Let there be 48 conductors and 4 poles in a dc generator, then the pole pitch is 48/4 = 12.

P : the number of machine poles

Z : the number of winding conductors

PROBLEMS WITH COMMUTATION IN REAL MACHINES

Two factors are affecting the commutation in real DC machines, these are:
1) armature reaction
2) $L\frac{di}{dt}$ voltages.

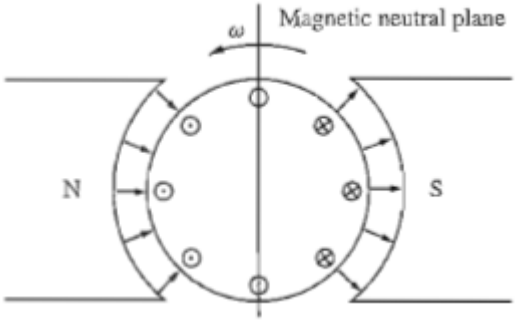
Armature Reaction

If the magnetic field windings of a dc machine are connected to a power supply and the rotor of the machine is turned by an external source of mechanical power, then a voltage will be induced in the conductors of the rotor. This voltage will be rectified into a dc output by the action of the machine's commutator.

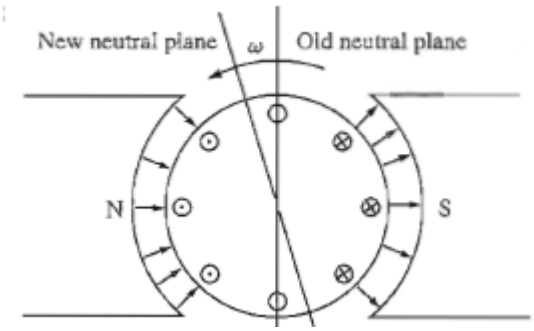
If a load is connected to the terminals of the machine a current will flow in its armature windings. This current flow will produce a magnetic fie1d of its own, which will distort the original magnetic field from the machine's poles. This distortion of the flux in a machine is called armature reaction. It causes two serious problems in real dc machines. These are:

a) Neutral-plane shift. The magnetic neutral plane is defined as the plane within the machine where the velocity of the rotor wires is exactly parallel to the magnetic flux lines, so that e_{ind} in the conductors in the plane is exactly zero.

The figure below show a two pole machine. Notice that when the machine is not loaded, the neutral plane in this machine is exactly vertical.



If a load is connected to this machine, a current will flow in its armature winding. This armature current will produce its own magnetic field that distorting the resultantant magnetic field inside the machine and causing the neutral plane to be shifted as shown below.



Notice that for the loaded generator, the magnetic neutral plane is shifted in the direction of rotation. If the machine is operating as a motor, the current in its rotor would be reversed and the magnetic neutral plane would shift in the opposite

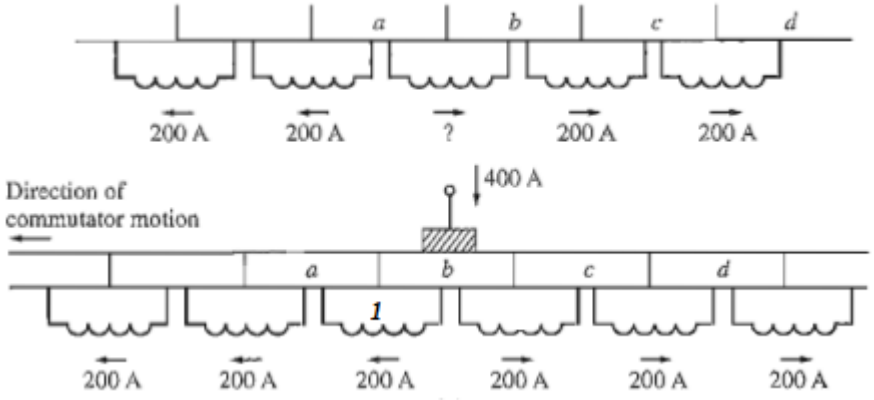
direction. Or, in general, the neutral plane shifts in the direction of motion for a generator and opposite to the direction of motion for a motor. Notice also that the amount of this shift depends on the machine loading.

The effect of neutral plane shift on machine commutation can be understood taking into consideration that the brushes are shortening the commutator segments when the voltage between then is zero. Because of neutral plane shift, the position of brushes at which the voltage between the segments under them is zero will be shifted. And if the position of the brushes is not changed a voltage will appear between the segments under the brushes causing arcing and sparking between the brushes and the commutator segments.

Notice that position of the neutral plane depends on the machine loading, so this problem cannot be fixed completely. For example, if the brushes are placed over the full-load neutral plane, they would spark at no load.

b) Flux weakening. When the dc mahine is loaded, and because of the magnetic flux generated by the armatue current, the net flux with the machine will be decreased. Flux weakening causes problems in both generators and motors. In generators, the effect of flux weakening is simply to reduce the voltage supplied by the generator for any given load. In motors, when the flux is decreased, its speed increases. But increasing the speed of a motor can increase its load, resulting in more flux weakening. It is possible for some shunt dc motors to reach a runaway condition as a result of flux weakening, where the speed of the motor just keeps increasing until the machine is disconnected from the power line or until it destroys itself.

***Ldi/dt* voltages.**
This occurs in commutator segments being shorted out by the brushes and sometimes is called inductive kick.
To understand this problem, look at the figure below.



This figure represents a series of commutator segments and the conductors connected between them. Assuming that the current in the brush is 400 A. This current will be distributed between the two paths, resulting in 200A in each path. Notice that when a commutator segment is shorted out, the current flow through that commutator segment will reverse.

مثلاً في الشكل أعلاه، وعندما تكون الفحمه في وضع b، فان التيار في ملف 1 هو باتجاة اليسار، وعند دوران الاله وفي لحظة انتقال الفحمة الى وضع a سيتغير اتجام التيار في ملف 1 ليصبح الى اليمين.

Assuming that the machine is turning at 800 r/min and that there are 50 commutator segments (a reasonable number for a typical motor), the number of segments passed under a brush in one second is: $\frac{800x50}{60} = 666.67$ segment/sec.

And the time for passing one segment is:

$$t = \frac{1}{666.67} = 0.0015s$$

Therefore, the rate of change in current with respect to time in the shorted loop must average:

$$\frac{di}{dt} \approx \frac{\Delta i}{\Delta t} = \frac{200 - (-200)}{0.0015} = \frac{400}{0.0015} = 266666.7 A/s$$

With even a tiny inductance in the loop, a very significant inductive voltage kick $v = L\frac{di}{dt}$ will be induced in the shorted commutator segment. This high voltage naturally causes sparking at the brushes of the machine, resulting in the same arcing problems that the neutral-plane shift causes.

Solutions to the Problems with Commutation

Three methods can be used to partially or completely correct the problems of armature reaction $L\frac{di}{dt}$ voltages, these are:

1) Brush shifting
2) Commutating poles or interpoles
3) Compensating windings

BRUSH SHIFTING.

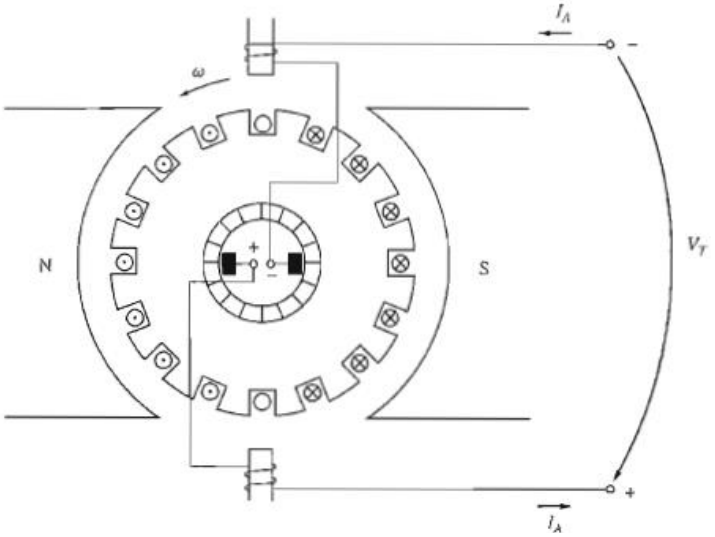
It is the simplest attempt to stop the sparking at the brushes caused by the neutral-plane shifts and $L\frac{di}{dt}$ effects. The problem with this solution is movement of the neutral plane with loading. Even more the shift direction reverses when the machine goes from motor operation to generator operation. Therefore, someone had to adjust the brushes every time the load on the machine changed.

A partial solution to this problem will be to fix the brushes in a compromise position (say, one that caused no sparking at two-thirds of full load). In this case, the motor sparked at no load and somewhat at full load, but if it spent most of its life operating at about two-thirds of full load, then sparking was minimized. Of course, such a machine could not be used as a generator at all because the sparking would have been horrible.

It has to be noticed that the brush-shifting approach to controlling sparking was already obsolete. Today, brush shifting is only used in very small machines that always run as motors. This is done because better solutions to the problem are simply not economical in such small machines.

COMMUTATING POLES OR INTERPOLES.

In this method, small poles, called commutating poles or interpoles, are placed midway between the main poles, see the figure below.



These commutating poles are located directly over the conductors being commutated. By providing a flux from the commutating poles, the voltage in the coils undergoing commutation can be canceled and no sparking at the brushes will occur.

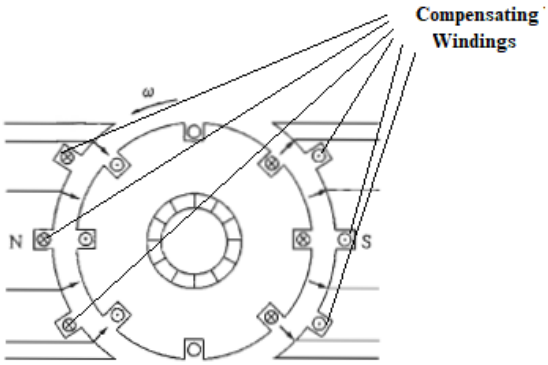
It has to be mentioned that, the commutating poles are so small and do not affect the operation of the machine, they affect only the few conductors about to undergo commutation.
Notice that to compensate for the effect of the load on the shifting of the neutral plane, these commutating poles are supplied by the armature current, as seen in the previous figure.

Note also that interpoles work for both motor and generator operation, since when the machine changes from motor to generator, the current both in its rotor and in its interpoles reverses direction. Therefore, the voltage effects from them still cancel.

The use of commutating poles or interpoles is very common, because they correct the sparking problems of dc machines at a fairly low cost. They are almost always found in any dc machine of 1 hp or larger.

COMPENSATING WINDINGS.

This method is used to cancel armature reaction and thus eliminate both neutral-plane shift and flux weakening. In this method, compensating windings are placed in slots carved in the faces of the poles, see the figure below.



Note that the compensating windings wires are parallel to the rotor conductors, to cancel the distorting effect of armature reaction. These compensating windings are connected in series with the rotor windings, so that whenever the load changes in the rotor, the current in the compensating windings changes, too.

The major disadvantage of compensating windings is that they are expensive and any motor that uses them must also have interpoles, since compensating windings do not cancel $L\frac{di}{dt}$ effects. The interpoles do not have to be as strong, though, since they are canceling only $L\frac{di}{dt}$ voltages in the windings and not the voltages due to neutral-plane shifting.

THE INTERNAL GENERATED VOLTAGE AND INDUCED TORQUE EQUATIONS OF REAL DC MACHINES

The induced machine voltage

The induced voltage in the DC machine depends on three factors:
1) The flux ϕ in the machine
2) The speed ω_m of the machine's rotor
3) A constant depending on the construction of the machine.

The voltage out of the armature of a real machine is equal to the number of conductors per path times the voltage on each conductor. The voltage in any single conductor under the pole faces was found as:

$$e_{ind} = vBl \qquad (1)$$

The number of conductors per path is $\frac{Z}{a}$, where Z is the total number of conductors and *a* is the number of parallel paths. Thus the voltage out of the armature of the machine is:

$$E_A = \frac{ZvBl}{a} \qquad (2)$$

The velocity of each conductor in the rotor can be expressed as $v = r.\omega_m$, where $r.$ is the radius of the rotor. So:

$$E_A = \frac{Zr\omega_m Bl}{a} \qquad (3)$$

Multiply and divide (3) by $\frac{2\pi}{P}$, we have:

$$E_A = \frac{2\pi Zr\omega_m Bl.P}{2\pi Pa} = \left(\frac{Z.P}{2\pi a}\right)\left(\frac{2\pi rlB.}{P}\right).\omega_m \qquad (4)$$

The flux of a pole is equal to the flux density under the pole times the pole's area:

$$\phi = B.A_P \qquad (5)$$

Where A_P is the portion of the area associated with each pole and is equal to the total area of the armature cylindrical surface, A, divided by the number of poles P, or:

$$A_P = \frac{A}{P} = \frac{2\pi r.l}{P} \qquad (6)$$

Thus, the flux of a pole is:

$$\phi = \frac{2\pi r.lB}{P} \qquad (7)$$

From (7) and (4), we find:

$$E_A = \left(\frac{Z.P}{2\pi a}\right).\phi.\omega_m \qquad (8)$$

Or:

$$E_A = K.\phi.\omega_m \qquad (9)$$

Where:

$$K = \frac{Z.P}{2\pi a} \qquad (10)$$

It is common to express the speed of a machine in revolutions per minute instead of radians per second. The conversion from revolutions per minute to radians per second is:

$$\omega_m = \frac{2\pi}{60} n_m \qquad (11)$$

Substituting (11) in (9), we find:

$$E_A = K.\phi.\frac{2\pi}{60} n_m = \left(\frac{2\pi K}{60}\right).\phi.n_m = K'.\phi.n_m \quad (12)$$

Where:

$$K' = \frac{2\pi K}{60} \qquad (13)$$

Substituting (10) in (13), we find:

$$K' = \frac{2\pi.Z.P}{60(2\pi).a} = \frac{.Z.P}{60.a}$$

The induced machine torque

The torque in any dc machine depends on three factors:

1) The flux ϕ in the machine

2) The armature (or rotor) current I_A in the machine

3) A constant depending on the construction of the machine.

The torque on the armature of a real machine is equal to the number of conductors Z times the torque on each conductor. The torque in any single conductor under the pole faces was previously shown to be:

$$T_{cond} = r.I_{cond}.l.B \qquad (1)$$

If there are a parallel paths in the machine, then the total armature current I_A is split among the a current paths, so the current in a single conductor is given by:

$$I_{cond} = \frac{I_A}{a} \qquad (2)$$

Substituting (2) in (1), we find:

$$T_{cond} = \frac{r.I_A.l.B}{a}$$

Since there are Z conductors, the total induced torque in a the machine rotor is:

$$T_{ind} = \frac{Z.r.l.B..I_A}{a} \qquad (3)$$

Multiply and divide (3) by $\frac{2\pi}{P}$, we have:

$$T_{ind} = \frac{2\pi.P.Z.r.l.B..I_A}{2\pi.a.P} = \left(\frac{Z.P}{2\pi.a}\right).\left(\frac{2\pi.r.l.B}{P}\right)..I_A \qquad (4)$$

The flux per pole in the machine can be expressed as:

$$\phi = \frac{2\pi.r.l.B}{P} \qquad (5)$$

From (5) and (4), we find:

$$T_{ind} = \left(\frac{Z.P}{2\pi.a}\right).\phi..I_A = K.\phi..I_A \qquad (6)$$

Where:

$$K = \frac{Z.P}{2\pi.a} \qquad (7)$$

Notice that, both the internal generated voltage and the induced torque equations just derived are only approximations, because not all the conductors in the machine are under the pole faces at any given time and also because the surfaces of each pole do not cover an entire $\frac{1}{P}$ of the rotor's surface. To achieve greater accuracy. the number of conductors under the pole faces could be used instead of the total number of conductors on the rotor.

Example 1: A duplex lap-wound armature is used in a six-pole dc machine with six brush sets, each spanning two commutator segments. There are 72 coils on the armature, each containing 12 turns. The flux per pole in the machine is 0.039 Wb, and the machine spins at 400 r/min.

a) How many current paths are there in this machine?

b) What is its induced voltage E_A?

Solution:

a) The number of current paths in this machine is:

$$a = m.P = 2(6) = 12$$

b) The induced voltage in the machine is:

$$E_A = K'.\phi.n_m$$

And:

$$K' = \frac{Z.P}{60a}$$

The number of conductors in this machine is:

$$Z = 2C.N_C = 2(72)(12) = 1728$$

Thus:

$$K' = \frac{Z.P}{60a} = \frac{1728(6)}{60(12)} = 14.4$$

And:

$$E_A = K'.\phi.n_m = 14.4(0.039)(400) = 224.6V$$

Example 2: A 12-pole dc generator has a simplex wave-wound armature containing 144 coils of 10 turns each. The resistance of each turn is 0.011Ω. Its flux per pole is 0.05 Wb, and it is turning at a speed of 200 r/min.

a) How many current paths are there in this machine?

b) What is the induced armature voltage of this machine?

c) What is the effective armature resistance of this machine?

d) If a $1k\Omega$ resistor is connected to the terminals of this generator, what is the resulting induced counter torque on the shaft of the machine? (Ignore the internal armature resistance of the machine.)

Solution:

a) There are a = 2m = 2 current paths in this winding.

b) There are $Z = 2C.N_C = 2(144)(10) = 2880$ conductors on this generator's rotor. Therefore,

$$K' = \frac{Z.P}{60a} = \frac{2880(12)}{60(2)} = 288$$

Therefore, the induced voltage is:

$$E_A = K'.\phi.n_m = 288(0.05)(200) = 2880V$$

c) There are two parallel paths through the rotor of this machine, each one consisting of $\frac{Z}{2} = \frac{2880}{2} = 1440$ conductors, or 720 turns. Therefore, the resistance in each current path is:

$$R(per \quad path) = (720)(0.011) = 7.92\Omega$$

Since there are two parallel paths, the effective armature resistance is:

$$R_A = \frac{7.92}{2} = 3.96\Omega$$

d) If a 1000Ω load is connected to the terminals of the generator, and if R_A is ignored, then a current of $I = \frac{2880V}{1000\Omega} = 2.88A$ flows.

The constant K is given by:

$$K = \frac{Z.P}{2\pi.a} = \frac{2880(12)}{2\pi(2)} = 2750.2$$

Therefore, the countertorque on the shaft of the generator is:

$$T_{ind} = K.\phi.I_A = 2750.2(0.05)(2.88) = 396N.m$$

POWER FLOW AND LOSSES IN DC MACHINES

The efficiency of a dc machine is defined by the equation:

$$\eta = \frac{P_{out}}{P_{in}} x100\%$$

The difference between the input power and the output power of a machine is the losses that occur inside it. Therefore:

$$\eta = \frac{P_{in} - P_{loss}}{P_{in}} x100\%$$

The Losses in DC Machines:

The losses that occur in dc machines can be divided into five basic categories:

1) Electrical or copper losses (I^2R losses)
2) Brush losses
3) Core losses
4) Mechanical losses
5) Stray load losses

ELECTRICAL OR COPPER LOSSES:

Copper losses are the losses that occur in the armature and field windings of the machine. The copper losses for the armature and field windings are given by:

Armature losses: $P_A = I_A^2.R_A$

Field losses: $P_F = I_F^2.R_F$

Where:

P_A : armature loss

P_F : field circuit loss

I_A: armature current

I_F : field current

R_A: armature resistance

R_F: field resistance

The resistances used in these calculations are usually the windings resistances at normal operating temperature.

BRUSH LOSSES:

The brush drop loss is the power lost across the contact potential at the brushes of the machine. It is given by the equation:

$$P_{BD} = V_{BD}.I_A$$

Where:

P_{BD}: brush drop loss

V_{BD}: brush voltage drop

I_A : armature current.

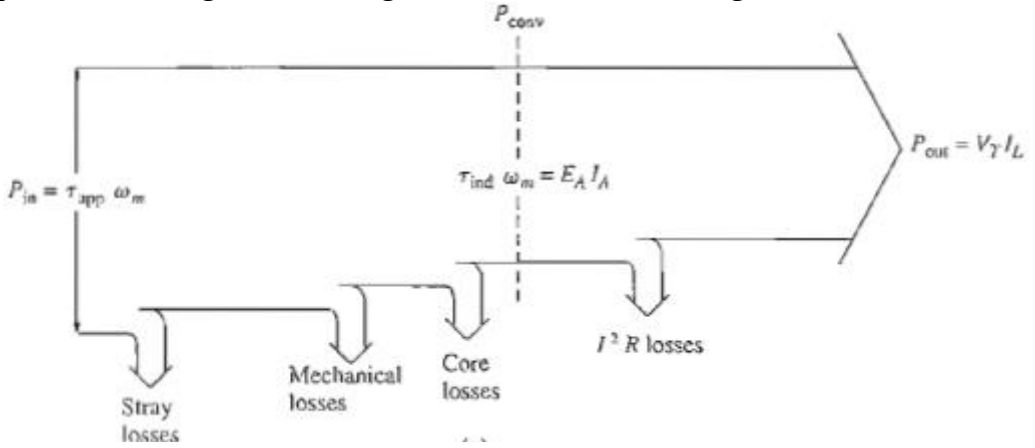
The reason that the brush losses are calculated in this manner is that the voltage drop across a set of brushes is approximately constant over a large range of armature currents. Unless otherwise specified, the brush voltage drop is usual1y assumed to be about 2 V.

CORE LOSSES:
The core losses are the hysteresis losses and eddy current losses occurring in the metal of the motor. These losses vary as the square of the flux density (B^2) and for the rotor as the 1.5th power of the speed of rotation ($n^{1.5}$).

MECHANICAL LOSSES:
The mechanical losses in a dc machine are the losses associated with mechanical effects. There are two basic types of mechanical losses: friction and windage. Friction losses are losses caused by the friction of the bearings in the machine, while windage losses are caused by the friction between the moving parts of the machine and the air inside the motor's casing. These losses vary as the cube of the speed of rotation of the machine.

STRAY LOSSES (OR MISCELLANEOUS LOSSES):
Stray losses are losses that cannot be placed in one of the previous categories. No matter how carefully losses are accounted for, some always escape inclusion in one of the above categories. All such losses are lumped into stray losses. For most machines, stray losses are taken by convention to be 1 percent of full load.

The Power-Flow Diagram
A power-f1ow diagram for a dc generator is shown in the figure below.



In this figure, mechanical power is input into the machine, and then the stray losses, mechanical losses, and core losses are subtracted. After they have been subtracted, the remaining power is ideally converted from mechanical to electrical form at the point labeled P_{conv}. The mechanical power that is converted is given by:

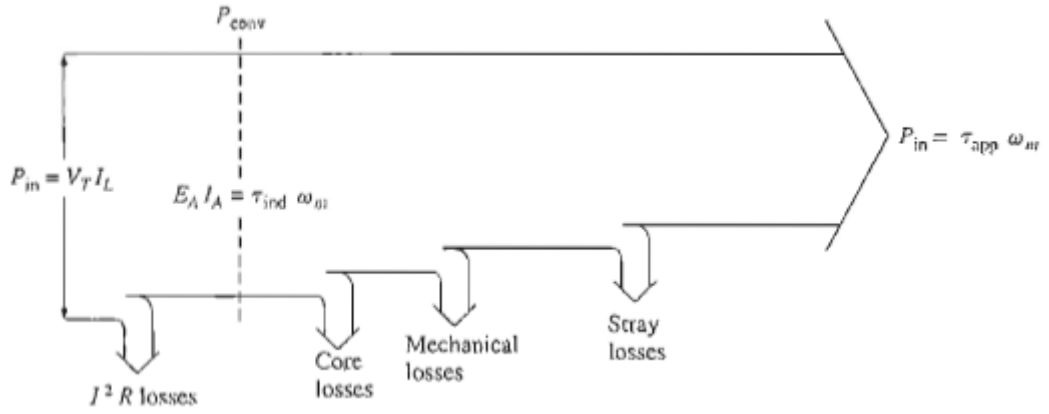
$$P_{conv} = T_{ind}.\omega_m$$

and the resulting electric power produced is given by:

$$P_{conv} = E_A.I_A$$

However, this is not the power that appears at the machine's termina1s. Before the terminals are reached, the electrical I^2R losses and the brush losses must be subtracted.

In the case of dc motors, this power-flow diagram is simply reversed. The power-flow diagram for a motor is shown in the figure below.


P_{conv}
$P_{in} = V_T I_L$
$E_A I_A = \tau_{ind}\ \omega_m$
$P_{in} = \tau_{app}\ \omega_m$
$I^2 R$ losses
Core losses
Mechanical losses
Stray losses

DC MOTORS AND GENERATORS

DC motors are generally used in applications where wide variations in speed are needed.

DC motors are often compared by their speed regulations. The speed regulation (SR) of a motor is defined by:

$$SR = \frac{\omega_{m.nl} - \omega_{m,fl}}{\omega_{m,fl}} x100\%$$

Or:

$$SR = \frac{n_{m.nl} - n_{m,fl}}{n_{m,fl}} x100\%$$

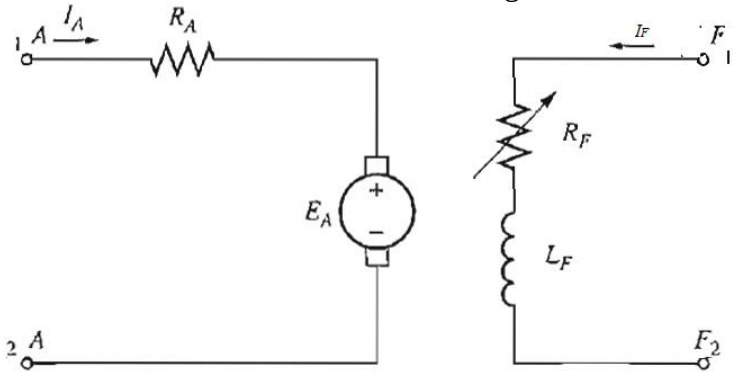
The motor speed regulation is a rough measure of the shape of a motor's torque- speed characteristic. A positive speed regulation means that a motor's speed drops with increasing load. A negative speed regulation means a motor's speed increases with increasing load. The magnitude of the speed regulation indicates how steep the slope of the torque-speed curve is.

There are five major types of dc motors in general use:

1) The separately excited dc motor
2) The shunt dc motor
3) The series dc motor
4) The compounded dc motor
5) The permanent-magnet dc motor

THE EQUIVALENT CIRCUIT OF THE DC MOTOR

The equivalent circuit of a dc motor is shown in the figure below.



In this figure, the armature circuit is represented by an ideal voltage source E_A and a resistor R_A. The field coils, which produce the magnetic flux in the generator, are represented by inductor L_F and resistor R_F. This resistor is represented adjustable to include an external variable resistor used to control the amount of current in the field circuit.

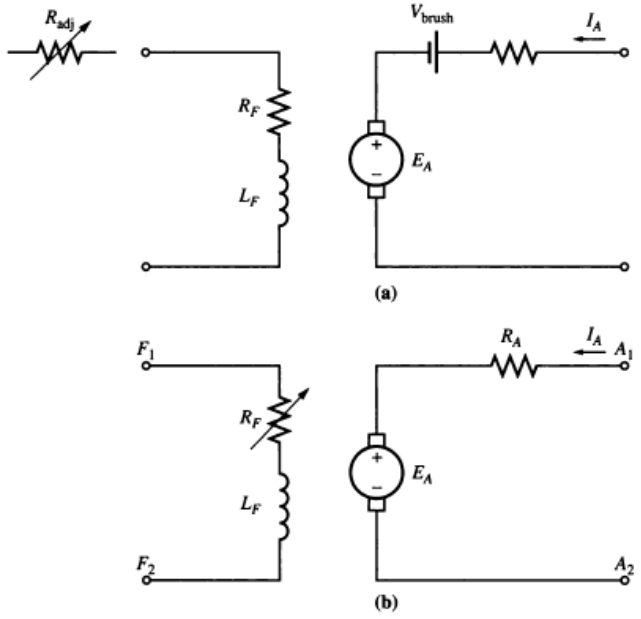
The internal generated voltage in the dc machine is given by:

$$E_A = K\phi\omega_m \qquad (1)$$

The induced torque developed by the machine is given by:

$$T_{ind} = K\phi.I_A \qquad (2)$$

1. The Equivalent Circuit of a DC Motor



(a) The equivalent circuit

(b) A simplified equivalent circuit eliminating the brush voltage drop and combining R_{adj} with the field resistance.

FIGURE 8–2
(a) The equivalent circuit of a dc motor. (b) A simplified equivalent circuit eliminating the brush voltage drop and combining R_{adj} with the field resistance.

In this figure, the armature circuit is represented by an ideal voltage source E_A and a resistor R_A. This representation is really the Thevenin equivalent of the entire rotor structure, including rotor coils, interpoles and compensating windings, if present.

The brush voltage drop is represented by a small battery V_{brush} opposing the direction of current flow in the machine.

The field coils, which produce the magnetic flux in the motor are represented by inductor L_F and resistor R_F. The separate resistor R_{adj} represents an external variable resistor used to control the amount of current in the field circuit.

Some of the few variations and simplifications:

i- The brush drop voltage is often only a very tiny fraction of the generated voltage in the machine. Thus, in cases where it is not too critical, the brush drop voltage may be left out or included in the R_A.

ii- The internal resistance of the field coils is sometimes lumped together with the variable resistor and the total is called R_F.

iii- Some generators have more than one field coil, all of which appear on the equivalent circuit.

The internal generated voltage is given by:

$$E_A = K\phi\omega$$

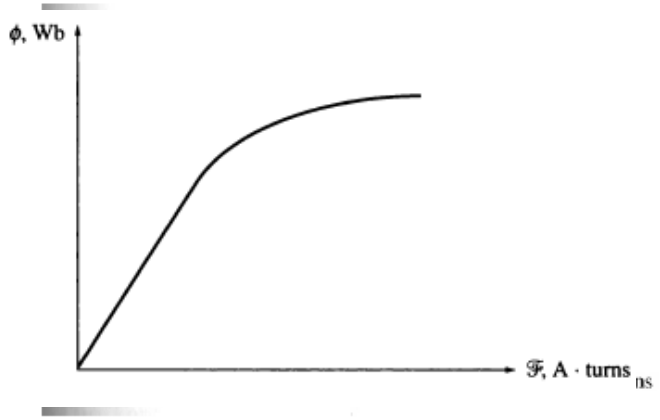
and the torque induced is

$$\tau_{ind} = K\phi I_A$$

2. The Magnetization Curve of a DC Machine

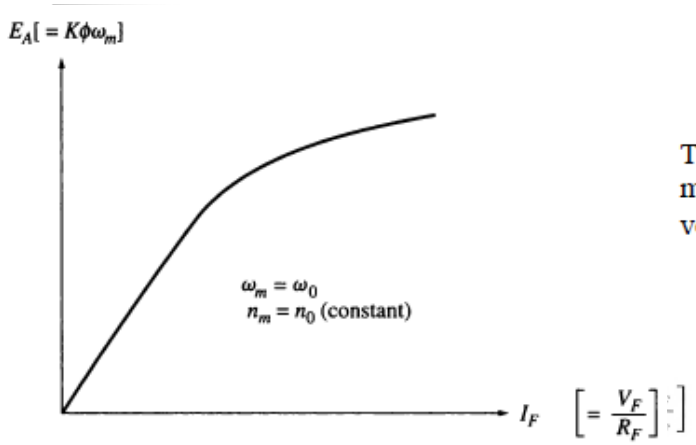
E_A is directly proportional to flux and the speed of rotation of the machine. How is the E_A related to the field current in the machine?

The field current in a dc machine produces a field mmf given by $F=N_F I_F$. This mmf produces a flux in the machine in accordance with its magnetization curve, shown below:



The magnetization curve of a ferromagnetic material

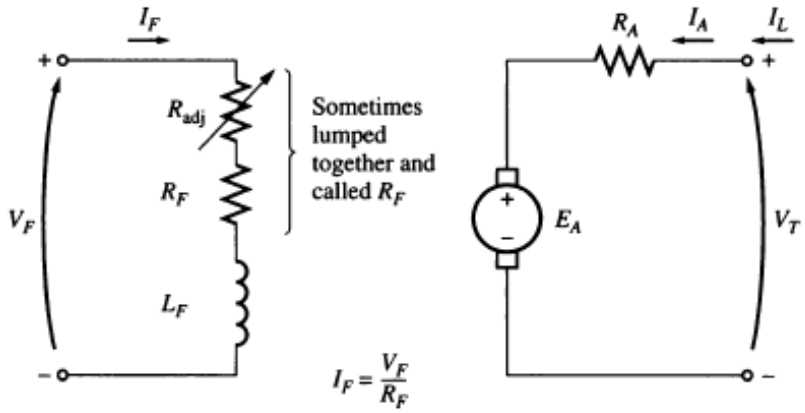
Since the field current is directly proportional to the mmf and since E_A is directly proportional to flux, it is customary to present the magnetization curve as a plot of E_A vs field current for a given speed ω_o.



The magnetization curve of a dc machine expressed as a plot of E_A versus I_F for a fixed speed ω_o.

NOTE: Most machines are designed to operate near the saturation point on the magnetization curve. This implies that a fairly large increase in field current is often necessary to get a small increase in E_A when operation is near full load.

3. Separately Excited and Shunt DC Motors

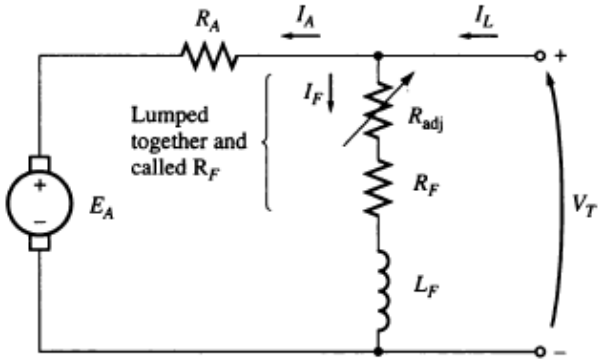


$$I_F = \frac{V_F}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A$$

The equivalent circuit of a separately excited dc motor



$$I_F = \frac{V_T}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A + I_F$$

The equivalent circuit of a shunt dc motor

A separately excited dc motor is a motor whose field circuit is supplied from a separate constant-voltage power supply, while a shunt dc motor is a motor whose field circuit gets its power directly across the armature terminals of the motor.

When the supply voltage to a motor is assumed constant, there is no practical difference in behaviour between these two machines. Unless otherwise specified, whenever the behaviour of a shunt motor is described, the separately excited motor is included too.

The KVL equation for the armature circuit is:

$$V_T = E_A + I_A R_A$$

The Terminal Characteristics of a Shunt DC Motor

A terminal characteristic of a machine is a plot of the machine's output quantities versus each other. For a motor, the output quantities are shaft torque and speed, so the terminal characteristic of a motor is a plot of its output torque versus speed.

How does a shunt dc motor respond to a load?

Suppose that the load on the shaft of a shunt motor is increased. Then the load torque τ_{load} will exceed the induced torque τ_{ind} in the machine, and the motor will start to slow down. When the motor slows down, its internal generated voltage drops ($E_A = K\phi\omega\downarrow$), so the armature current in the motor $I_A = (V_T - E_A\downarrow)/R_A$ increases. As the armature current rises, the induced torque in the motor increases ($\tau_{ind} = K\phi I_A\uparrow$), and finally the induced torque will equal the load torque at a lower mechanical speed of rotation.

The output characteristic of a shunt dc motor can be derived from the induced voltage and torque equations of the motor plus the KVL.

KVL → $V_T = E_A + I_A R_A$

The induced voltage $E_A = K\phi\omega$, so

$$V_T = K\phi\omega + I_A R_A$$

Since $\tau_{ind} = K\phi I_A$, current I_A can be expressed as:

$$I_A = \frac{\tau_{ind}}{K\phi}$$

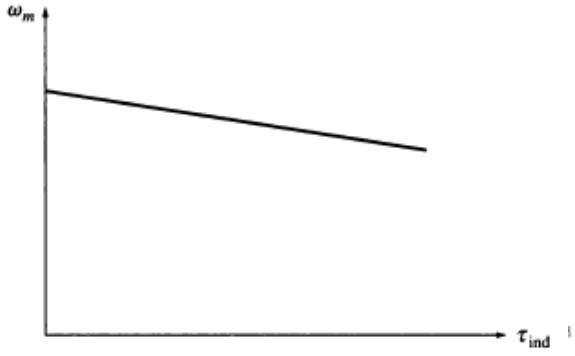
Combining the V_T and I_A equations:

$$V_T = K\phi\omega + \frac{\tau_{ind}}{K\phi} R_A$$

Finally, solving for the motor's speed:

$$\omega = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2}\tau_{ind}$$

This equation is just a straight line with a negative slope. The resulting torque-speed characteristic of a shunt dc motor is shown here:

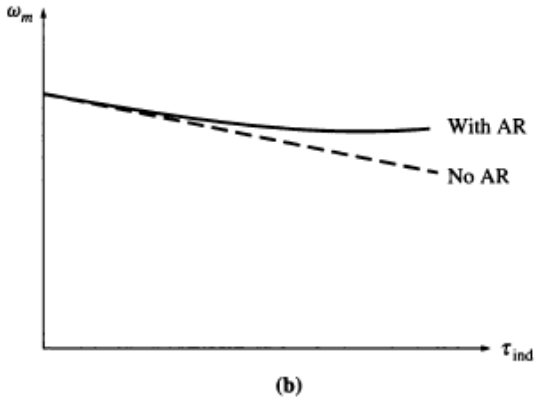


(a)

Torque-speed characteristic of a shunt or separately excited dc motor with compensating windings to eliminate armature reaction

It is important to realize that, in order for the speed of the motor to vary linearly with torque, the other terms in this expression must be constant as the load changes. The terminal voltage supplied by the dc power source is assumed to be constant - if it is not constant, then the voltage variations will affect the shape of the torque-speed curve.

Another effect internal to the motor that can also affect the shape of the torque-speed curve is armature reaction. If a motor has armature reaction, then as its load increases, the flux-weakening effects reduce its flux. From the motor speed equation above, the effect of reduction in flux is to increase the motor's speed at any given load over the speed it would run at without armature reaction. The torque-speed characteristic of a shunt motor with armature reaction is shown below:



(b)

Torque-speed characteristic of the motor with armature reaction present.

If a motor has compensating windings, there will be no flux weakening problems and the flux in the motor will be constant.

If a shunt dc motor has compensating windings so that flux is constant regardless of load, and the motor's speed and armature current are known at any one value of load, then it is possible to calculate its speed at any other value of load, as long as the armature current at that load is known or can be determined.

Example 8–1. A 50-hp, 250-V, 1200 r/min dc shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_{adj} + R_F$ of 50 Ω, which produces a *no-load* speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding (see Figure 8–7).

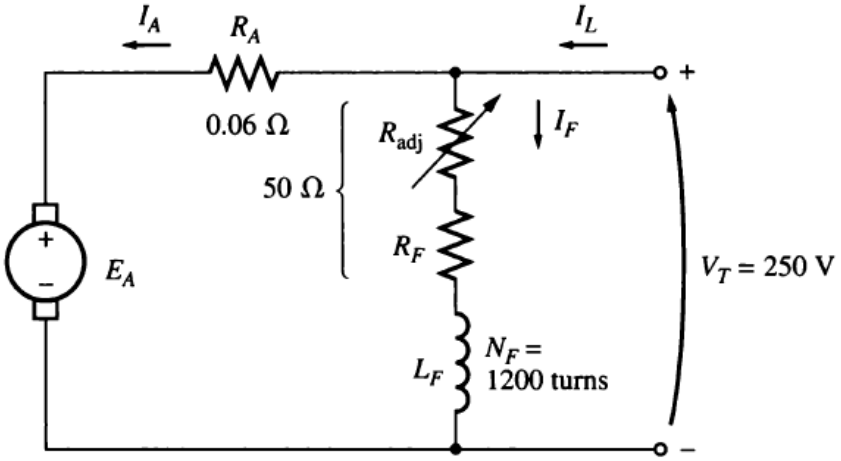


FIGURE 8–7
The shunt motor in Example 8–1.

(a) Find the speed of this motor when its input current is 100 A.
(b) Find the speed of this motor when its input current is 200 A.
(c) Find the speed of this motor when its input current is 300 A.
(d) Plot the torque–speed characteristic of this motor.

Solution

The internal generated voltage of a dc machine with its speed expressed in revolutions per minute is given by

$$E_A = K'\phi n_m \tag{7–41}$$

Since the field current in the machine is constant (because V_T and the field resistance are both constant), and since there are no armature reaction effects, *the flux in this motor is constant*. The relationship between the speeds and internal generated voltages of the motor at two different load conditions is thus

$$\frac{E_{A2}}{E_{A1}} = \frac{K'\phi n_{m2}}{K'\phi n_{m1}} \tag{8–8}$$

The constant K' cancels, since it is a constant for any given machine, and the flux ϕ cancels as described above. Therefore,

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} \tag{8–9}$$

At no load, the armature current is zero, so $E_{A1} = V_T = 250$ V, while the speed $n_{m1} = 1200$ r/min. If we can calculate the internal generated voltage at any other load, it will be possible to determine the motor speed at that load from Equation (8–9).

(a) If $I_L = 100$ A, then the armature current in the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$
$$= 100 \text{ A} - \frac{250 \text{ V}}{50\ \Omega} = 95 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$
$$= 250 \text{ V} - (95 \text{ A})(0.06\ \Omega) = 244.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{244.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1173 \text{ r/min}$$

(b) If $I_L = 200$ A, then the armature current in the motor is

$$I_A = 200 \text{ A} - \frac{250 \text{ V}}{50\ \Omega} = 195 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$

$$= 250 \text{ V} - (195 \text{ A})(0.06\ \Omega) = 238.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{238.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1144 \text{ r/min}$$

(c) If $I_L = 300$ A, then the armature current in the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$

$$= 300 \text{ A} - \frac{250 \text{ V}}{50\ \Omega} = 295 \text{ A}$$

Therefore, E_A at this load will be

$$E_A = V_T - I_A R_A$$

$$= 250 \text{ V} - (295 \text{ A})(0.06\ \Omega) = 232.3 \text{ V}$$

The resulting speed of the motor is

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{232.3 \text{ V}}{250 \text{ V}} 1200 \text{ r/min} = 1115 \text{ r/min}$$

(d) To plot the output characteristic of this motor, it is necessary to find the torque corresponding to each value of speed. At no load, the induced torque τ_{ind} is clearly zero. The induced torque for any other load can be found from the fact that power converted in a dc motor is

$$\boxed{P_{\text{conv}} = E_A I_A = \tau_{\text{ind}} \omega_m} \qquad (7\text{–}55, 7\text{–}56)$$

From this equation, the induced torque in a motor is

$$\tau_{\text{ind}} = \frac{E_A I_A}{\omega_m} \qquad (8\text{–}10)$$

Therefore, the induced torque when $I_L = 100$ A is

$$\tau_{\text{ind}} = \frac{(244.3 \text{ V})(95 \text{ A})}{(1173 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 190 \text{ N} \cdot \text{m}$$

The induced torque when $I_L = 200$ A is

$$\tau_{\text{ind}} = \frac{(238.3 \text{ V})(95 \text{ A})}{(1144 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 388 \text{ N} \cdot \text{m}$$

The induced torque when $I_L = 300$ A is

$$\tau_{\text{ind}} = \frac{(232.3 \text{ V})(295 \text{ A})}{(1115 \text{ r/min})(1 \text{ min/60s})(2\pi \text{ rad/r})} = 587 \text{ N} \cdot \text{m}$$

The resulting torque–speed characteristic for this motor is plotted in Figure 8–8.

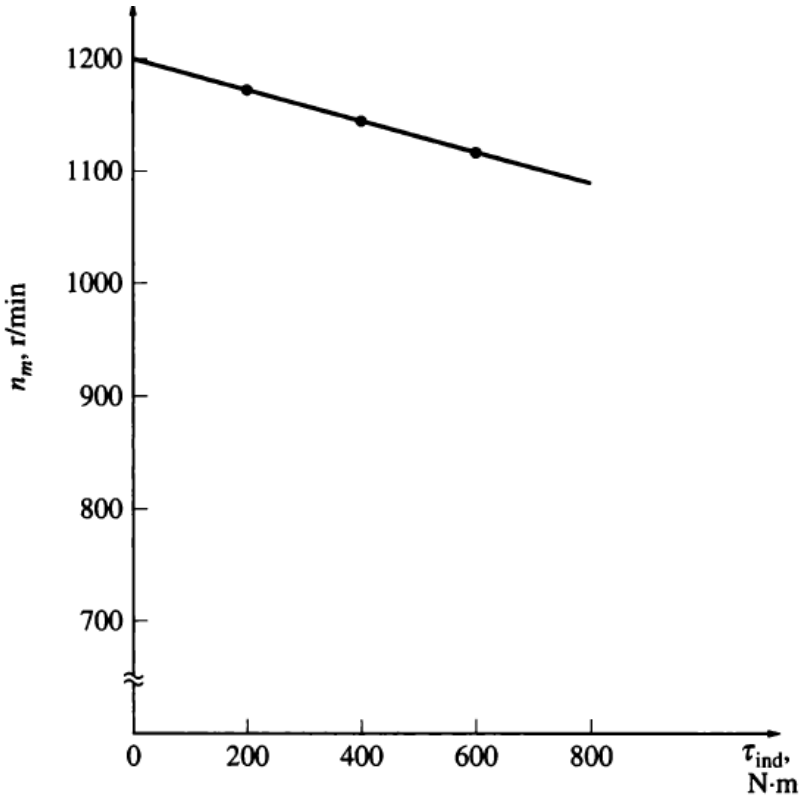


FIGURE 8–8

Nonlinear Analysis of a Shunt DC Motor

The flux and hence the internal generated voltage E_A of a dc machine is a non linear function of its magnetomotive force. Therefore, anything that changes the mmf in a machine will have a non linear effect on the E_A of the machine. Since the change in E_A cannot be calculated analytically, the magnetization curve of the machine must be used. Two principal contributors to the mmf in the machine are its field current and its armature reaction, if present.

Since the magnetization curve is a plot of E_A vs I_F for a given speed ω_o , the effect of changing a machine's field current can be determined directly from its magnetization curve.

If a machine has armature reaction, its flux will be reduced with each increase in load. The total mmf in a shunt dc motor is the field circuit mmf less the mmf due to armature reaction (AR):

$$F_{net} = N_F I_F - F_{AR}$$

Since the magnetization curves are expressed as plots of E_A vs field current, it is customary to define an equivalent field current that would produce the same output voltage as the combination of all the mmf in the machine. The resulting voltage E_A can then be determined by locating that equivalent field current on the magnetization curve. The equivalent field current:

$$I_F^* = I_F - \frac{F_{AR}}{N_F}$$

One other effect must be considered when non linear analysis is used to determine E_A of a dc motor. The magnetization curves for a machine are drawn for a particular speed, usually the rated speed of the machine. How can the effects of a given field current be determined if the motor is turning at other than rated speed?

The equation for the induced voltage in a dc machine when speed is expressed as rev/min: $E_A = K'\phi n$

For a given effective field current, the flux in the machine is fixed, so the E_A is related to speed by:

$$\frac{E_A}{E_{A0}} = \frac{n}{n_0}$$

where E_{A0} and n_0 represent the reference values of voltages and speed respectively. If the reference conditions are known from the magnetization curve and the actual E_A is known, then it is possible to determine the actual speed n.

Example 8–2. A 50-hp, 250-V, 1200 r/min dc shunt motor *without* compensating windings has an armature resistance (including the brushes and interpoles) of 0.06 Ω. Its field circuit has a total resistance $R_F + R_{adj}$ of 50 Ω, which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding, and the armature reaction produces a demagnetizing magnetomotive force of 840 A • turns at a load current of 200 A. The magnetization curve of this machine is shown in Figure 8–9.

(a) Find the speed of this motor when its input current is 200 A.

(b) This motor is essentially identical to the one in Example 8–1 except for the absence of compensating windings. How does its speed compare to that of the previous motor at a load current of 200 A?

(c) Calculate and plot the torque–speed characteristic for this motor.

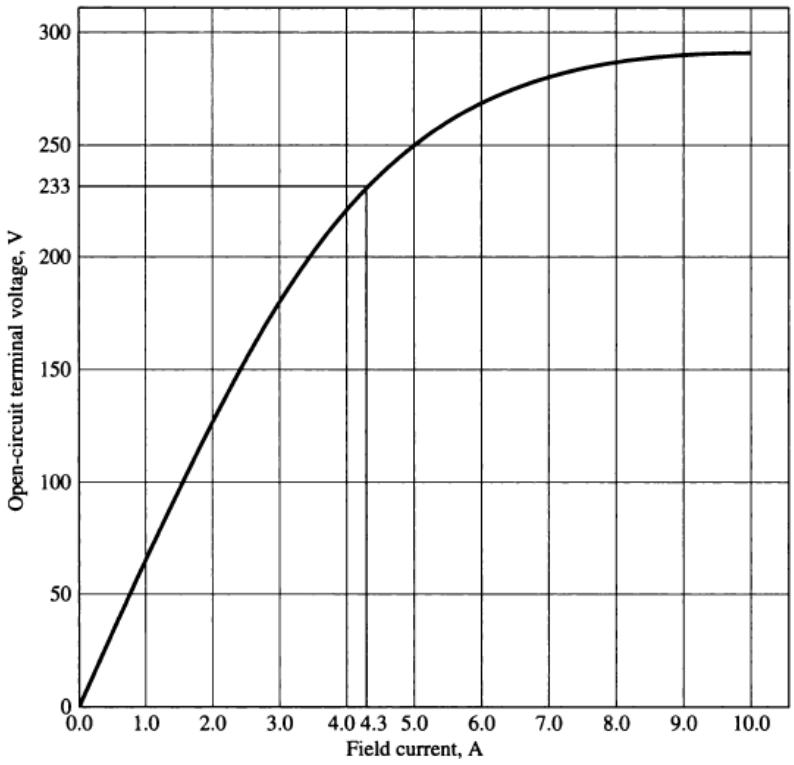


FIGURE 8–9

Solution

(a) If $I_L = 200$ A, then the armature current of the motor is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_F}$$

$$= 200 \text{ A} - \frac{250 \text{ V}}{50 \ \Omega} = 195 \text{ A}$$

Therefore, the internal generated voltage of the machine is

$$E_A = V_T - I_A R_A$$

$$= 250 \text{ V} - (195 \text{ A})(0.06 \ \Omega) = 238.3 \text{ V}$$

At $I_L = 200$ A, the demagnetizing magnetomotive force due to armature reaction is 840 A • turns, so the effective shunt field current of the motor is

$$I_F^* = I_F - \frac{\mathcal{F}_{AR}}{N_F} \tag{8–12}$$

$$= 5.0 \text{ A} - \frac{840 \text{ A} \cdot \text{turns}}{1200 \text{ turns}} = 4.3 \text{ A}$$

From the magnetization curve, this effective field current would produce an internal generated voltage E_{A0} of 233 V at a speed n_0 of 1200 r/min.

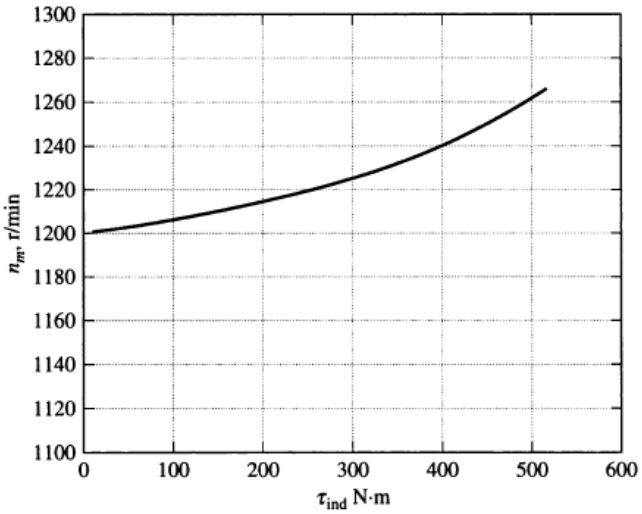
We know that the internal generated voltage E_{A0} would be 233 V at a speed of 1200 r/min. Since the actual internal generated voltage E_A is 238.3 V, the actual operating speed of the motor must be

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \tag{8–13}$$

$$n_m = \frac{E_A}{E_{A0}} n_0 = \frac{238.3 \text{ V}}{233 \text{ V}} (1200 \text{ r/min}) = 1227 \text{ r/min}$$

(b) At 200 A of load in Example 8–1, the motor's speed was n_m = 1144 r/min. In this example, the motor's speed is 1227 r/min. *Notice that the speed of the motor with armature reaction is higher than the speed of the motor with no armature reaction.* This relative increase in speed is due to the flux weakening in the machine with armature reaction.

(c) To derive the torque–speed characteristic of this motor, we must calculate the torque and speed for many different conditions of load. Unfortunately, the demagnetizing armature reaction magnetomotive force is only given for one condition of load (200 A). Since no additional information is available, we will assume that the strength of $\mathcal{F}_{AR}$ varies linearly with load current.



Speed Control of Shunt DC Motors

Two common methods (as already been seen in Chapter 1 simple linear machine):

i- Adjusting the field resistance R_F (and thus the field flux)

ii- Adjusting the terminal voltage applied to the armature.

Less common method:

iii- Inserting a resistor in series with the armature circuit.

Changing the Field Resistance

If the field resistance increases, then the field current decreases ($I_F\downarrow = V_T/R_F\uparrow$), and as the field current decreases, the flux decreases as well. A decrease in flux causes an instantaneous decrease in the internal generated voltage $E_A\downarrow$ ($=K\phi\downarrow\omega$), which causes a large increase in the machine's armature current since,

$$I_A\uparrow = \frac{V_T - E_A\downarrow}{R_A}$$

The induced torque in a motor is given by $\tau_{ind} = K\phi I_A$. Since the flux in this machine decreases while the current I_A increases, which way does the induced torque change?

Look at this example:

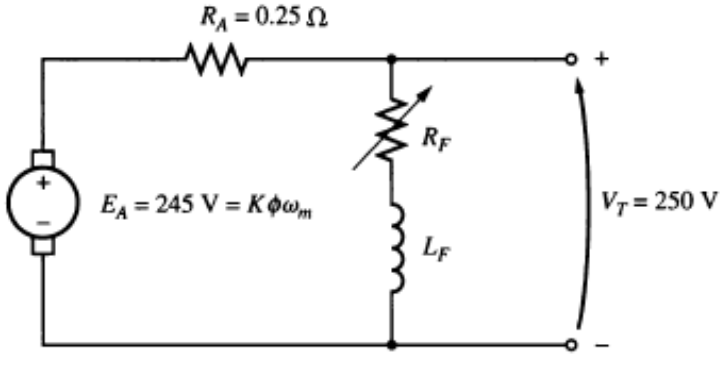


Figure above shows a shunt dc motor with an internal resistance of 0.25Ω. It is currently operating with a terminal voltage of 250V and an internal generated voltage of 245V. Therefore, the armature current flow is $I_A = (250V-245V)/0.25\Omega = 20A$.

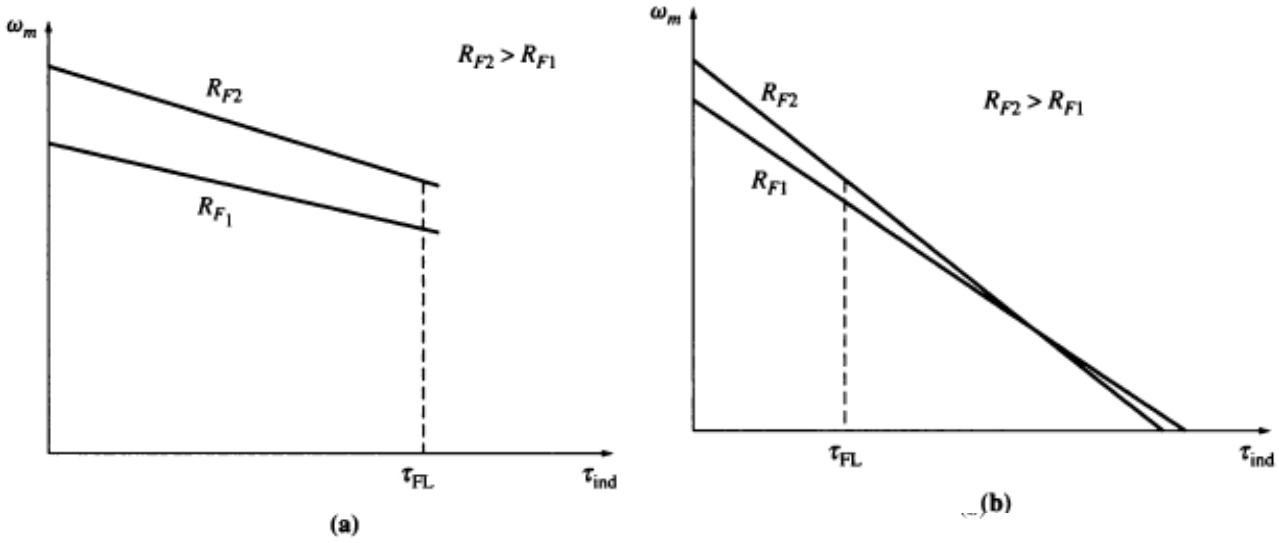
What happens in this motor if there is a 1% decrease in flux? If the flux decrease by 1%, then E_A must decrease by 1% too, because $E_A = K\phi\omega$. Therefore, E_A will drop to:
$E_{A2} = 0.99\ E_{A1} = 0.99\ (245) = 242.55V$

The armature current must then rise to
$I_A = (250-242.55)/0.25 = 29.8\ A$

Thus, a 1% decrease in flux produced a 49% increase in armature current.

So, to get back to the original discussion, the increase in current predominates over the decrease in flux. so, $\tau_{ind} > \tau_{load}$, the motor **speeds up**.

However, as the motor speeds up, E_A rises, causing I_A to fall. Thus, induced torque τ_{ind} drops too, and finally τ_{ind} equals τ_{load} at a higher steady-sate speed than originally.



The effect of field resistance R_F speed control on a shunt motor's torque-speed characteristics.
(a) over the motor's normal operating range
(b) over the entire range from no load to stall conditions

WARNING:

The effect of increasing the R_F is shown in figure (b) above. Notice that as the flux in the machine decreases, the no-load speed of the motor increases, while the slope of the torque-speed curve becomes steeper.

This figure shows the terminal characteristic of the motor over the full range from no-load to stall conditions. It is apparent that at very slow speeds an increase in field resistance will actually decrease the speed of the motor. This effect occurs because, at very low speeds, the increase in armature current caused by the decrease in E_A is no longer large enough to compensate for the decrease in flux in the induced torque equation. With the flux decrease actually larger than the armature current increase, the induced torque decreases, and the motor slows down.

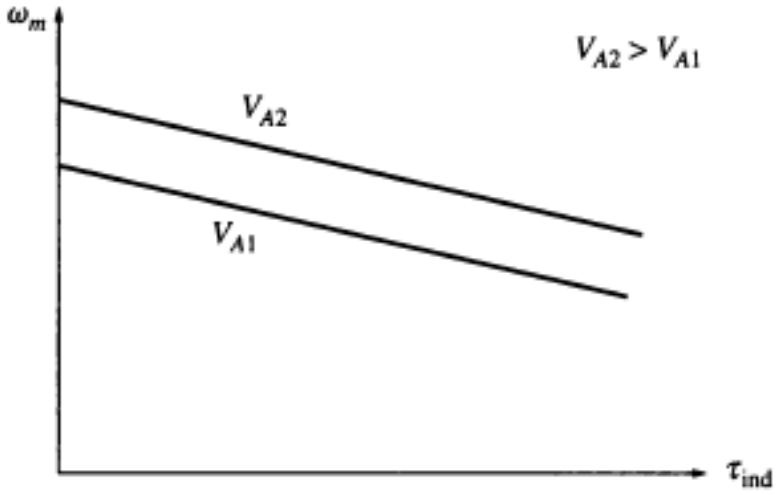
Some small dc motors used for control purposes actually operates at speeds close to stall conditions. For these motors, an increase in R_F might have no effect, or it might even decrease the speed of the motor. Since the results are not predictable, field resistance speed control should not be used in these types of dc motors. Instead, the armature voltage method should be employed.

Changing the Armature Voltage

This method involves changing the voltage applied to the armature of the motor without changing the voltage applied to the field.

If the voltage V_A is increased, then the I_A must rise [$I_A = (V_A \uparrow - E_A)/R_A$]. As I_A increases, the induced torque $\tau_{ind} = K\phi I_A \uparrow$ increases, making $\tau_{ind} > \tau_{load}$, and the speed of the motor increases.

But, as the speed increases, the E_A ($=K\phi\omega\uparrow$) increases, causing the armature current to decrease. This decrease in I_A decreases the induced torque, causing $\tau_{ind} = \tau_{load}$ at a higher rotational speed.



Inserting a Resistor in Series with the Armature Circuit

If a resistor is inserted in series with the armature circuit, the effect is to drastically increase the slope of the motor's torque-speed characteristic, making it operate more slowly if loaded. This fact can be seen from the speed equation:

$$\omega = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2}\tau_{ind}$$

The insertion of a resistor is a very wasteful method of speed control, since the losses in the inserted resistor are very large. For this reason, it is rarely used.

Safe Ranges of Operation for the 2 common methods

Field Resistance Control

The lower the field current in a shunt (or separately excited) dc motor, the faster it turns: and the higher the field current, the slower it turns. Since an increase in field current causes decrease in speed, there is always a minimum achievable speed by field circuit control. This minimum speed occurs when the motor's field circuit has the maximum permissible current flowing through it.

If a motor is operating at its rated terminal voltage, power and field current, then it will be running at rated speed, also known as base speed. Field resistance control can control the speed of the motor for speeds above base speed but not for speeds below base speed. To achieve a speed slower than base speed by field circuit control would require excessive field current, possibly burning up the field windings.

Armature Voltage Control

The lower the armature voltage on a separately excited dc motor, the slower it turns, and the higher the armature voltage, the faster it turns. Since an increase in armature voltage causes an increase in speed, there is always a maximum achievable speed by armature voltage control. This maximum speed occurs when the motor's armature voltage reaches its maximum permissible level.

If a motor is operating at its rated terminal voltage, power and field current, then it will be running at rated speed, also known as base speed. Armature voltage control can control the speed of the motor for speeds below base speed but not for speeds above base speed. To achieve a speed faster than base speed by armature voltage control would require excessive armature voltage, possibly damaging the armature circuit.

→ These two techniques of speed control are obviously complementary. Armature voltage control works well for speeds below base speed, and field resistance control works well for speeds above base speed.

There is a significant difference in the torque and power limits on the machine under these two types of speed control. The limiting factor in either case is the heating of the armature conductors, which places an upper limit on the magnitude of the armature current I_A.

For armature voltage control, the **flux in the motor is constant**, so the maximum torque in the motor is $\tau_{max} = K\phi I_{A,max}$

This maximum torque is constant regardless of the speed of the rotation of the motor. Since the power out of the motor is given by $P=\tau\omega$, the maximum power is $P_{max} = \tau_{max}\omega$. Thus, the **max power out is directly proportional to its operating speed under armature voltage control**.

On the other hand, when field resistance control is used, the flux does change. In this form of control, a speed increase is caused by a decrease in the machine's flux. In order for the armature current limit is not exceeded, the induced torque limit must decrease as the speed of the motor increases. Since the power out of the motor is given by $P=\tau\omega$ and the torque limit decreases as the speed of the motor increases, **the max power out of a dc motor under field current control is constant, while the maximum torque varies as the reciprocal of the motor's speed**.

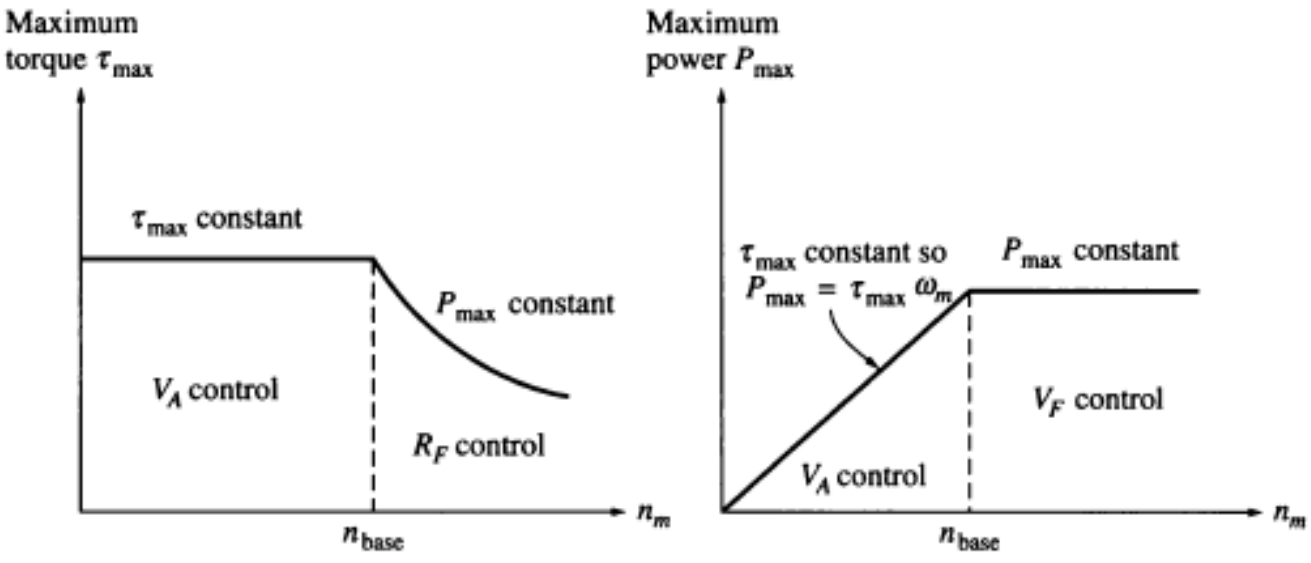


FIGURE 8–16
Power and torque limits as a function of speed for a shunt motor under armature volt and field resistance control.

Example 8–3. Figure 8–17a shows a 100-hp, 250-V, 1200 r/min shunt dc motor with an armature resistance of 0.03 Ω and a field resistance of 41.67 Ω. The motor has compensating windings, so armature reaction can be ignored. Mechanical and core losses may be assumed to be negligible for the purposes of this problem. The motor is assumed to be driving a load with a line current of 126 A and an initial speed of 1103 r/min. To simplify the problem, assume that the amount of armature current drawn by the motor remains constant.

(a) If the machine's magnetization curve is shown in Figure 8–9, what is the motor's speed if the field resistance is raised to 50 Ω?

(b) Calculate and plot the speed of this motor as a function of the field resistance R_F assuming a constant-current load.

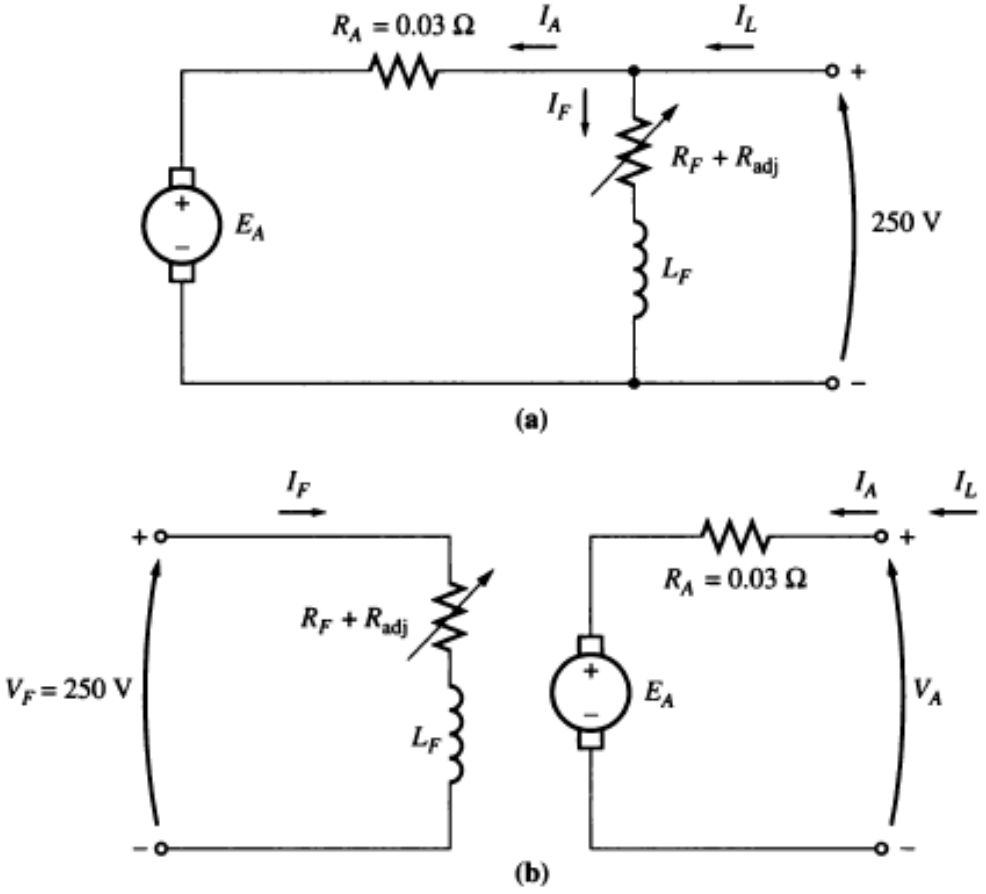


FIGURE 8–17
(a) The shunt motor in Example 8–3. (b) The separately excited dc motor in Example 8–4.

Solution

(a) The motor has an initial line current of 126 A, so the initial armature current is

$$I_{A1} = I_{L1} - I_{F1} = 126\text{ A} - \frac{150\text{ V}}{41.67\ \Omega} = 120\text{ A}$$

Therefore, the internal generated voltage is

$$\begin{aligned} E_{A1} &= V_T - I_{A1}R_A = 250\text{ V} - (120\text{ A})(0.03\ \Omega) \\ &= 246.4\text{ V} \end{aligned}$$

After the field resistance is increased to 50 Ω, the field current will become

$$I_{F2} = \frac{V_T}{R_F} = \frac{250\text{ V}}{50\ \Omega} = 5\text{ A}$$

The ratio of the internal generated voltage at one speed to the internal generated voltage at another speed is given by the ratio of Equation (7–41) at the two speeds:

$$\frac{E_{A2}}{E_{A1}} = \frac{K'\phi_2 n_{m2}}{K'\phi_1 n_{m1}} \tag{8–16}$$

Because the armature current is assumed constant, $E_{A1} = E_{A2}$, and this equation reduces to

$$1 = \frac{\phi_2 n_{m2}}{\phi_1 n_{m1}}$$

or

$$n_{m2} = \frac{\phi_1}{\phi_2} n_{m1} \tag{8–17}$$

A magnetization curve is a plot of E_A versus I_F for a given speed. Since the values of E_A on the curve are directly proportional to the flux, the ratio of the internal generated voltages read off the curve is equal to the ratio of the fluxes within the machine. At $I_F = 5$ A, $E_{A0} = 250$ V, while at $I_F = 6$ A, $E_{A0} = 268$ V. Therefore, the ratio of fluxes is given by

$$\frac{\phi_1}{\phi_2} = \frac{268\text{ V}}{250\text{ V}} = 1.076$$

and the new speed of the motor is

$$n_{m2} = \frac{\phi_1}{\phi_2} n_{m1} = (1.076)(1103\text{ r/min}) = 1187\text{ r/min}$$

Example 8–4. The motor in Example 8–3 is now connected separately excited, as shown in Figure 8–17b. The motor is initially running with $V_A = 250$ V, $I_A = 120$ A, and $n = 1103$ r/min, while supplying a constant-torque load. What will the speed of this motor be if V_A is reduced to 200 V?

Solution

The motor has an initial line current of 120 A and an armature voltage V_A of 250 V, so the internal generated voltage E_A is

$$E_A = V_T - I_A R_A = 250\text{ V} - (120\text{ A})(0.03\ \Omega) = 246.4\text{ V}$$

By applying Equation (8–16) and realizing that the flux ϕ is constant, the motor's speed can be expressed as

$$\begin{aligned} \frac{E_{A2}}{E_{A1}} &= \frac{K'\phi_2 n_{m2}}{K'\phi_1 n_{m1}} \qquad (8–16) \\ &= \frac{n_{m2}}{n_{m1}} \\ n_{m2} &= \frac{E_{A2}}{E_{A1}} n_{m1} \end{aligned}$$

To find E_{A2} use Kirchhoff's voltage law:

$$E_{A2} = V_T - I_{A2}R_A$$

Since the torque is constant and the flux is constant, I_A is constant. This yields a voltage of

$$E_{A2} = 200\text{ V} - (120\text{ A})(0.03\ \Omega) = 196.4\text{ V}$$

The final speed of the motor is thus

$$n_{m2} = \frac{E_{A2}}{E_{A1}} n_{m1} = \frac{196.4\text{ V}}{246.4\text{ V}}\,1103\text{ r/min} = 879\text{ r/min}$$

The Effect of an Open Field Circuit

As the field resistance increased, the speed of the motor increased with it. What would happen if this effect were taken to the extreme, if the field resistance really increased? What would happen if the field circuit were actually opened while the motor is running?

From the previous discussion, the flux in the machine will drop, and E_A will drop as well. This would cause a really large increase in the armature current, and the resulting induced torque would be quite a bit higher than the load torque of the motor. Therefore, the motor's speed starts to rise and just keeps going up.

In ordinary shunt dc motors operating with light fields, if the armature reaction effects are severe enough, the effect of speed rising can take place. If the armature reaction on a dc motor is severe, an increase in load can weaken its flux enough to actually cause the motor's speed to rise. However, most loads have torque-speed curves whose torque increases with speed, so the increased speed increases its load, which increases the armature reaction, weakening the flux again. The weaker flux causes a further increase in speed, further increase the load, etc. etc. until the motor overspeeds. This condition is known as **runaway**.

4. The Permanent-Magnet DC Motor

A permanent magnet dc motor (PMDC) is a dc motor whose poles are made of permanent magnets. PMDC motor offer a number of benefits compared with shunt dc motors in some applications.

Advantage: Since the motors do not require an external field circuit, they do not have the field circuit copper losses. Because no field windings are required, they can be smaller than corresponding shunt dc motors.

Disadvantages:
Permanent magnets cannot produce as high flux density as an externally supplied shunt field., so a PMDC motor will have a lower induced torque per ampere of armature current than a shunt motor of the same size. Also, PMDC motors run the risk of demagnetization.

A PMDC motor is basically the same machine as a shunt dc motor, except that the flux of a PMDC motor is fixed. Therefore, it is not possible to control the speed of the PMDC motor by varying the field current or flux. The only methods of speed control available for a PMDC motor are armature voltage control and armature resistance control.

5. The Series DC Motor

A series DC motor is a dc motor whose field windings consist of relatively few turns connected in series with the armature circuit.

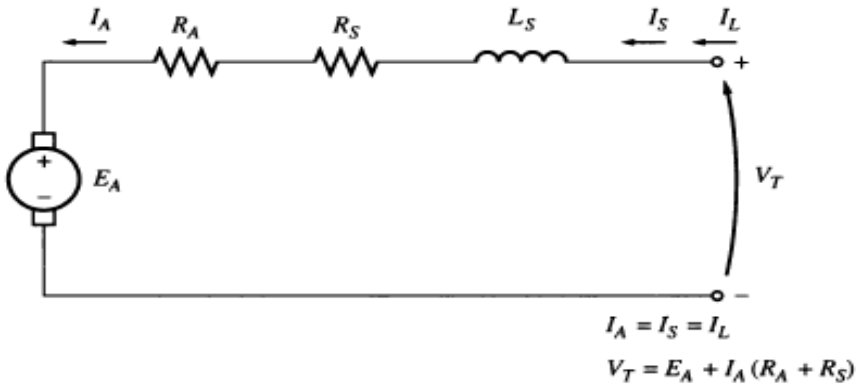


FIGURE 8–20
The equivalent circuit of a series dc motor.

The KVL for this motor is $V_T = E_A + I_A (R_A + R_S)$

Induced Torque in a Series DC Motor

The basic behaviour of a series dc motor is due to the fact that the flux is directly proportional to the armature current, at least until saturation is reached. As the load on the motor increases, its flux increases too. As seen earlier, an increase in flux in the motor causes a decrease in its speed. The result is that a series dc motor has a sharply drooping torque-speed characteristic.

The induced torque is $\tau_{ind} = K\phi I_A$

The flux in this machine is directly proportional to its armature current (at least until metal saturates). Therefore, the flux in the machine can be given by $\phi = cI_A$ where c is a constant of proportionality. Thus, $\tau_{ind} = K\phi I_A = KcI_A^2$

Series dc motors are therefore used in applications requiring very high torques. Example: starter motors in cars, elevator motors, tractor motors etc.

The Terminal Characteristic of a Series DC Motor

The assumption of a linear magnetization curve implies that the flux in the motor will be given by $\phi = cI_A$. This equation will be used to derive the torque-speed characteristic curve for the series motor.

Derivation of the torque-speed characteristic:

1. $V_T = E_A + I_A (R_A + R_S)$

2. $I_A = \sqrt{\frac{\tau_{ind}}{Kc}}$

3. Also, $E_A = K\phi\omega$, thus substituting this in the KVL gives:

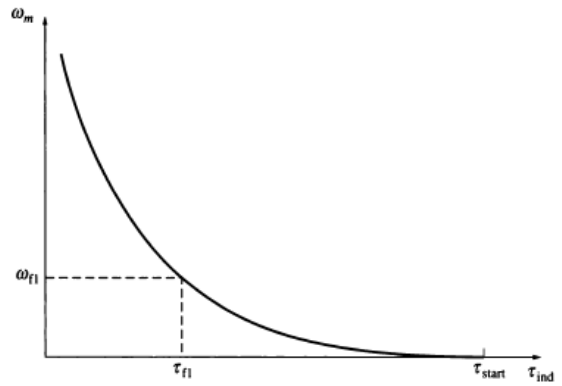
$$V_T = K\phi\omega + \sqrt{\frac{\tau_{ind}}{Kc}}(R_A + R_S)$$

4. If the flux can be eliminated from this expression, it will directly relate the torque of a motor to its speed. Notice that $I_A = \phi/c$ and $\tau_{ind} = (K/c)\phi^2$. Thus,

$$\phi = \sqrt{\frac{c}{K}}\sqrt{\tau_{ind}}$$

5. Substituting the flux equation into equation in part 3, results in:

$$\omega = \frac{V_T}{\sqrt{Kc}}\frac{1}{\sqrt{\tau_{ind}}} - \frac{R_A + R_S}{Kc}$$



The torque-speed characteristic of a series dc motor

Speed Control of Series DC Motors.

Unlike with the shunt dc motor, there is only one efficient way to change the speed of a series dc motor. That method is to change the terminal voltage of the motor. If terminal voltage is increased, the speed will increase for any given torque.

Example 8–5. Figure 8–20 shows a 250-V series dc motor with compensating windings, and a total series resistance $R_A + R_S$ of 0.08 Ω. The series field consists of 25 turns per pole, with the magnetization curve shown in Figure 8–22.

(a) Find the speed and induced torque of this motor for when its armature current is 50 A.

(b) Calculate and plot the torque–speed characteristic for this motor.

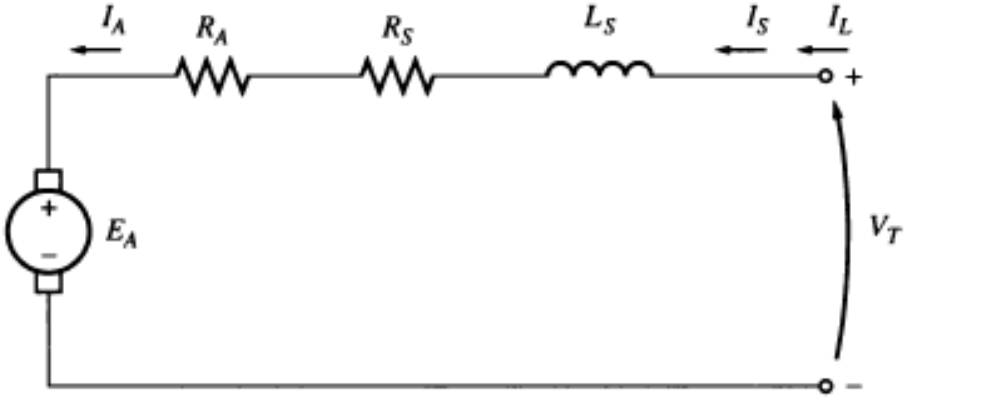


FIGURE 8–20
The equivalent circuit of a series dc motor.

Solution

(a) To analyze the behavior of a series motor with saturation, pick points along the operating curve and find the torque and speed for each point. Notice that the magnetization curve is given in units of magnetomotive force (ampere-turns) versus E_A for a speed of 1200 r/min, so calculated E_A values must be compared to the equivalent values at 1200 r/min to determine the actual motor speed.

For $I_A = 50$ A,

$$E_A = V_T - I_A(R_A + R_S) = 250 \text{ V} - (50\text{A})(0.08\ \Omega) = 246 \text{ V}$$

Since $I_A = I_F = 50$ A, the magnetomotive force is

$$\mathcal{F} = NI = (25 \text{ turns})(50 \text{ A}) = 1250 \text{ A} \cdot \text{turns}$$

From the magnetization curve at $\mathcal{F} = 1250$ A • turns, $E_{A0} = 80$ V. To get the correct speed of the motor, remember that, from Equation (8–13),

$$n_m = \frac{E_A}{E_{A0}} n_0$$

$$= \frac{246 \text{ V}}{80 \text{ V}} 120 \text{ r/min} = 3690 \text{ r/min}$$

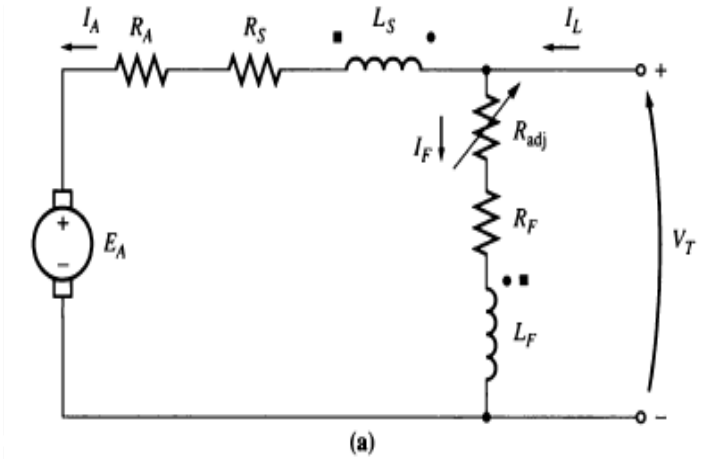
To find the induced torque supplied by the motor at that speed, recall that $P_{conv} = E_A I_A = \tau_{ind}\omega_m$. Therefore,

$$\tau_{ind} = \frac{E_A I_A}{\omega_m}$$

$$= \frac{(246 \text{ V})(50 \text{ A})}{(3690 \text{ r/min})(1 \text{ min}/60 \text{ s})(2\pi \text{ rad/r})} = 31.8 \text{ N} \cdot \text{m}$$

6. The Compounded DC Motor

A compounded dc motor is a motor with both a shunt and a series field. This is shown below:



- Cumulatively compounded
- Differentially compounded

The equivalent circuit of compounded dc motor:
(a) long-shunt connection.
(b) short-shunt connection.

The KVL for a compounded dc motor is:

$$V_T = E_A + I_A (R_A + R_S)$$

and the currents are:

$$I_A = I_L - I_F$$
$$I_F = V_T/R_F$$

The net mmf and the effective shunt field current are:

$$F_{net} = F_F \pm F_{SE} - F_{AR}$$

$$I_F^* = I_F \pm (N_{SE}/N_F)\, I_A - F_{AR}/N_F$$

+ve sign associated with a cumulatively compounded motor
-ve sign associated with a differentially compounded motor

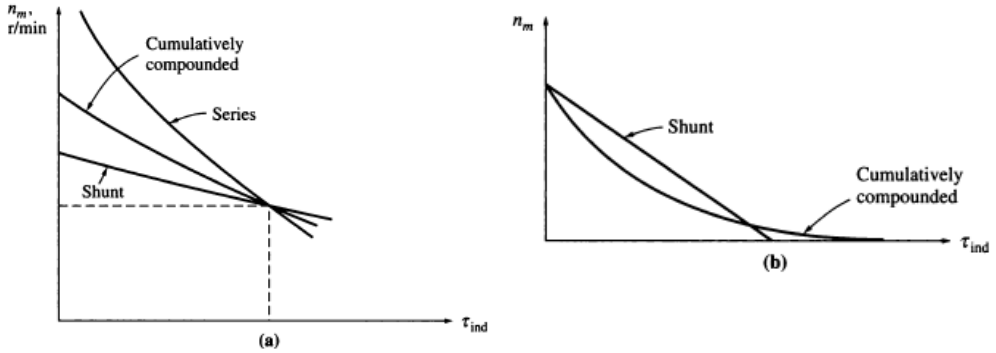
The Torque-Speed Characteristic of a Cumulatively Compounded DC Motor (CC)

There is a component of flux which is constant and another component which is proportional to its armature current (and thus to its load). Thus, CC motor has a higher starting torque than a shunt motor (whose flux is constant) but a lower starting torque than a series motor (whose entire flux is proportional to armature current).

The CC motor combines the best features of both the shunt and series motors. Like a series motor, it has extra torque for starting; like a shunt, it does not overspeed at no load.

At light loads, the series field has a very small effect, so the motor behaves approximately as a shunt dc motor. As the load gets very large, the series flux becomes quite important and the torque-speed curve begins to look like a series motor's characteristic.

A comparison of the torque-speed characteristics of each of these types of machines is shown below:



(a) The torque–speed characteristic of a cumulatively compounded dc motor compared to series and shunt motors with the same full-load rating. (b) The torque–speed characteristic of a cumulatively compounded dc motor compared to a shunt motor with the same no-load speed.

The Torque-Speed Characteristic of a Differentially Compounded DC Motor

In a differentially compounded dc motor, the shunt mmf and series mmf subtract from each other. This means that as the load on the motor increases, I_A increases and the flux in the motor decreases. But as the flux decreases, the speed of the motor increases. This speed increase causes another increase in load, which further increases I_A, further decreasing the flux, and increasing the speed again. The result is that a differentially compounded motor is unstable and tends to runaway.

This instability is much worse than that of a shunt motor with armature reaction. It is so bad that a differentially compounded motor is unsuitable for any application.

Differentially compounded motor is also impossible to start. At starting conditions, the armature current and the series field current are very high. Since the series flux subtracts from the shunt flux, the series field can actually reverse the magnetic polarity of the machine's poles. The motor will typically remain still or turn slowly in the wrong direction while burning up, because of the excessive armature current. When this type of motor is to be started, its series field must be short-circuited, so that it behaves as an ordinary shunt motor during the starting period.

Speed Control in the Cumulatively Compounded DC Motor

Same as for a shunt motor.